Title of Work: Virtual Chemistry Lab for Higher Secondary Science Students using html, CSS and JavaScript.

Description: This work focuses on enhancing students' understanding and visualization of chemistry concepts and processes. It addresses the limitations of existing tools that are either too advanced, not user-friendly, or not aligned with standard curricula by developing a virtual chemistry lab that is interactive, practical, and aligned with educational standards.

The description of the modules is given below:

Sr. No	**Module Name**	**Description**
1	index.html	This HTML module serves as the **landing page** for the *Virtual Chemistry Lab*. It provides a visually engaging interface featuring animated particles, a live digital clock, and interactive buttons that link to core lab functionalities such as performing virtual experiments and accessing the periodic table. Built with Tailwind CSS and custom styling, it offers users an immersive and user-friendly starting point for exploring chemistry concepts.
2	periodic_table.html	This module dynamically generates an interactive periodic table using HTML, CSS, and JavaScript. It visually represents chemical elements with color-coded categories (e.g., metals, nonmetals, noble gases) in a responsive grid layout. Each element tile displays its symbol and number, and users can click on a tile to view detailed information about the element. The design includes a modern dark-themed aesthetic and hover effects for better user experience.
3	experiment.html	This module presents a visually appealing, interactive landing page that allows users to select between different types of chemistry experiments—Organic, Physical, or Inorganic. It features a modern glassmorphism design with animated glowing orbs, styled using Tailwind CSS, and includes navigation buttons that direct users to specific experiment pages.
4	organic.html	This module provides an interactive virtual lab interface for Organic Chemistry, allowing users to explore various topics such as IUPAC nomenclature, reaction mechanisms, and stereochemistry through a sidebar menu. It features a structured layout with a main content area, an informative right panel, and

		navigation links for immersive learning and simulation tools.
5	IUPAC_nomenclature.html	This module provides an interactive learning platform for **IUPAC nomenclature**, featuring a 3D molecule viewer, step-by-step naming guides, practice exercises with instant feedback, and reference tables for functional groups, substituents, and naming rules to master systematic chemical naming conventions.
6	reaction_mechanisms.html	This module provides an **interactive learning platform for organic reaction mechanisms**, featuring 3D molecular visualization, step-by-step mechanism breakdowns, example reactions, and quizzes to test understanding across key processes like SN1/SN2, elimination, addition, and pericyclic reactions.
7	kinetics_energy.html	This module provides an **interactive learning platform for organic reaction kinetics**, enabling users to visualize energy diagrams, analyze reaction rates, explore catalysis effects, compare mechanisms (SN1/SN2, E1/E2), and study temperature dependencies through adjustable parameters, dynamic charts, and real-time simulations.
8	stereochemistry.html	This module provides an interactive learning platform for stereochemical concepts, featuring 3D molecular visualization, step-by-step stereochemistry breakdowns, for example molecular structures, and tools to test understanding across key processes like R/S configuration, stereoisomerism, optical activity, and CIP rules.
9	iconic_compounds.html	This module allows users to interactively explore and visualize the 3D structures of various organic molecules using a 3D molecular viewer. By selecting a molecule (like methane, ethanol, or benzene) from a set of buttons, users can load and view the molecular structure in real-time. The page also offers various display options, such as changing the visualization style (stick, sphere, or line), toggling labels, and enabling molecular rotation. Additionally, users can view detailed information about each molecule, including its chemical formula and a brief description. This interactive tool is useful for learning about molecular structures and their properties.

10	physical.html	This module features three interactive sections: **Electrochemistry** calculators for cell potential, Faraday's law, and Gibbs free energy; **Crystal Structures** visualization using 3D models (FCC, BCC, HCP) rendered with Three.js; and **Quantum Chemistry** tools for basic Schrödinger equation solutions and orbital visualization, all designed to simulate key chemistry concepts through dynamic calculations and visualizations.
11	inorganic.html	This module creates a simple layout for an Inorganic Chemistry Virtual Lab webpage. It includes a navigation bar, a sidebar featuring various inorganic chemistry topics such as Periodic Table & Trends, Coordination Chemistry, Crystal Field Theory, Acids, Bases & Salts, Qualitative Analysis, Metal Extraction & Metallurgy, and Redox Reactions & Electrochemistry. The main content area provides an interactive lab space for users to explore these topics, and a right panel with additional features like visualizers, stepwise coordination complex formation, electrochemical cell simulations, and quizzes. The page also contains a back button to navigate to a previous page and a footer. The layout is responsive, using CSS Flexbox, and aims to offer users an interactive and educational experience.

The remaining document contains the source code for these modules.

- **Source Codes:**

Module 1
index.html –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Virtual Chemistry Lab</title>

  <!-- Tailwind CSS CDN -->
  <script src="https://cdn.tailwindcss.com"></script>
  <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600;700&display=swap" rel="stylesheet">

  <style>
   body {
     font-family: 'Poppins', sans-serif;
     overflow: hidden;
     background: linear-gradient(to bottom right, #e0f7ff, #d7e3f4); /* Light, subtle gradient */
```

```
    color: #1a202c; /* Dark text for contrast */
  }

  .glass {
    background: rgba(255, 255, 255, 0.08);
    backdrop-filter: blur(12px);
    border: 1px solid rgba(255, 255, 255, 0.15);
    border-radius: 1rem;
    box-shadow: 0 10px 40px rgba(0, 0, 0, 0.2);
  }

  .button-hover:hover {
    transform: scale(1.04);
    box-shadow: 0 0 15px rgba(255, 255, 255, 0.4);
  }

  /* Beaker wiggle */
  .wiggle:hover {
    animation: wiggle 0.8s ease-in-out infinite;
  }

  @keyframes wiggle {
    0% { transform: rotate(0deg); }
    25% { transform: rotate(3deg); }
    50% { transform: rotate(-3deg); }
    75% { transform: rotate(2deg); }
    100% { transform: rotate(0deg); }
  }

  /* Bubbles under button */
  .bubbles {
    position: absolute;
    bottom: -10px;
    left: 50%;
    transform: translateX(-50%);
    width: 100%;
    height: 20px;
    z-index: -1;
  }

  .bubbles span {
    position: absolute;
    width: 8px;
    height: 8px;
    background: white;
    border-radius: 50%;
    animation: rise 4s linear infinite;
    opacity: 0.2;
  }

  @keyframes rise {
    0% { transform: translateY(0) scale(1); opacity: 0.3; }
    50% { opacity: 1; }
    100% { transform: translateY(-50px) scale(0.6); opacity: 0; }
  }

  .bubbles span:nth-child(1) { left: 20%; animation-delay: 0s; }
  .bubbles span:nth-child(2) { left: 40%; animation-delay: 0.6s; }
  .bubbles span:nth-child(3) { left: 60%; animation-delay: 1.2s; }
  .bubbles span:nth-child(4) { left: 80%; animation-delay: 1.8s; }
  .bubbles span:nth-child(5) { left: 30%; animation-delay: 2.4s; }

  /* Floating particles */
.particle {
  position: absolute;
  width: 10px; /* slightly larger */
  height: 10px;
```

```
  background-color: white;
  border-radius: 50%;
  opacity: 0.25; /* more visible */
  animation: float 12s ease-in-out infinite;
  box-shadow: 0 0 6px rgba(255, 255, 255, 0.4); /* subtle glow */
}

   @keyframes float {
    0% { transform: translateY(0) translateX(0); }
    50% { transform: translateY(-40px) translateX(20px); }
    100% { transform: translateY(0) translateX(0); }
   }

   /* Clock */
   #clock {
    position: absolute;
    top: 1.5rem;
    right: 2rem;
    font-size: 3.1rem;
    font-weight: 600;
    text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);
    color: #e2e8f0;
   }
  </style>
</head>
<body class="min-h-screen flex items-center justify-center bg-gradient-to-br from-gray-900 via-slate-800 to-slate-900 text-white
px-4 relative">

  <!-- ⏰ Clock -->
  <div id="clock"></div>

  <!-- 🧫 Chemistry Particles -->
  <div class="particle" style="top: 20%; left: 10%;"></div>
  <div class="particle" style="top: 40%; left: 80%;"></div>
  <div class="particle" style="top: 70%; left: 50%;"></div>
  <div class="particle" style="top: 10%; left: 60%;"></div>
  <div class="particle" style="top: 80%; left: 20%;"></div>

  <!-- 🩹 Main Content Card -->
  <div class="glass text-center px-8 py-10 md:px-14 md:py-12 max-w-2xl w-full z-10 relative">
   <h1 class="text-4xl md:text-5xl font-bold mb-4" style="text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);">
     <span class="inline-block wiggle">🩹</span> Welcome to the <br/> Virtual Chemistry Lab
   </h1>
   <p class="text-lg text-gray-300 mb-8">
    Explore and perform virtual chemistry experiments.
   </p>

   <!-- Buttons -->
   <div class="flex flex-wrap justify-center gap-6 relative">
    <div class="relative">
     <button onclick="location.href='experiment.html'"
       class="bg-green-500 hover:bg-green-600 text-white px-6 py-3 rounded-full text-lg font-semibold transition-all duration-300
ease-in-out button-hover flex items-center gap-2">
       😳 Try Experiment
     </button>
     <div class="bubbles">
      <span></span><span></span><span></span><span></span><span></span>
     </div>
    </div>

    <div class="relative">
     <button onclick="location.href='periodic_table.html'"
       class="bg-blue-500 hover:bg-blue-600 text-white px-6 py-3 rounded-full text-lg font-semibold transition-all duration-300
ease-in-out button-hover flex items-center gap-2">
       💉 View Periodic Table
     </button>
```

```
      <div class="bubbles">
        <span></span><span></span><span></span><span></span><span></span><span></span>
      </div>
    </div>
  </div>
 </div>

 <!-- ⏰ Live Clock Script -->
 <script>
  function updateClock() {
    const now = new Date();
    const hours = now.getHours().toString().padStart(2, '0');
    const minutes = now.getMinutes().toString().padStart(2, '0');
    const seconds = now.getSeconds().toString().padStart(2, '0');
    document.getElementById('clock').textContent = `${hours}:${minutes}:${seconds}`;
  }
  setInterval(updateClock, 1000);
  updateClock();
 </script>
</body>
</html>
```

Module 2
periodic_table.html –

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Periodic Table</title>
   <style>
     body {
          background: linear-gradient(135deg, #88bbd7, #17559c);
          color: white;
          font-family: Arial, sans-serif;
          text-align: center;
          margin: 0;
          padding: 20px;
          min-height: 100vh;
      }

       h1 {
   font-size: 4rem; /* Increased text size */
   text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);
   margin-bottom: 20px;
}


       /* Back Button */
       .back-btn {
          position: absolute;
          top: 20px;
          left: 20px;
          text-decoration: none;
          font-size: 1.2rem;
          font-weight: bold;
          color: white;
          background: #333;
          padding: 10px 20px;
          border-radius: 8px;
          transition: background 0.3s ease;
       }
```

```
    .back-btn:hover {
        background: #555;
    }

    .table-container {
        display: grid;
        grid-template-columns: repeat(18, 60px);
        gap: 6px;
        justify-content: center;
        margin-top: 20px;
    }

    .element {
        position: relative;
        width: 50px;
        height: 50px;
        border-radius: 8px;
        text-align: center;
        font-weight: bold;
        font-size: 18px;
        color: rgb(253, 253, 253);
        display: flex;
        align-items: center;
        justify-content: center;
        border: 2px solid rgba(255, 255, 255, 0.3);
        transition: transform 0.2s ease-in-out, box-shadow 0.2s ease-in-out;
    }

    .element:hover {
        transform: scale(1.15);
        box-shadow: 0px 4px 12px rgba(0, 0, 0, 0.3);
    }

    .highlight {
        background-color: #60c5ff !important;
    }

    .element-number {
        position: absolute;
        top: 4px;
        left: 6px;
        font-size: 12px;
        font-weight: bold;
        color: rgba(250, 250, 250, 0.8);
    }

    .element-symbol {
        font-size: 24px;
        font-weight: bold;
    }

    .details {
        margin-top: 20px;
        padding: 15px;
        background-color: rgba(0, 0, 0, 0.75);
        border-radius: 10px;
        display: inline-block;
        text-align: left;
        display: flex;
        box-shadow: 0 4px 10px rgba(0, 0, 0, 0.3);
        max-width: 300px;
        font-size: 16px;
        line-height: 1.5;
    }


    /* Dark Matte Theme for Periodic Table */
.nonmetal { background-color: #1b5e20; color: #d4e157; } /* Deep Green with Lime Text */
.noble-gas { background-color: #6d4c41; color: #ffcc80; } /* Dark Brown with Soft Orange */
.alkali-metal { background-color: #880e4f; color: #ff80ab; } /* Dark Magenta with Pink */
```

```
.alkaline-earth-metal { background-color: #b71c1c; color: #ff8a80; } /* Deep Red with Soft Coral */
.metalloid { background-color: #3e2723; color: #ffcc80; } /* Charcoal Brown with Gold */
.halogen { background-color: #004d40; color: #80cbc4; } /* Deep Teal with Aqua */
.transition-metal { background-color: #1a237e; color: #7986cb; } /* Dark Indigo with Lavender */
.post-transition-metal { background-color: #263238; color: #b0bec5; } /* Blue-Gray with Silver */
.lanthanide { background-color: #4a148c; color: #b39ddb; } /* Dark Purple with Lilac */
.actinide { background-color: #212121; color: #eeeeee; } /* Almost Black with Soft White */
.unknown { background-color: #424242; color: #bdbdbd; } /* Neutral Dark Gray with Light Gray */

  </style>
</head>
<body>
  <h1>Periodic Table</h1>
  <div class="table-container" id="periodic-table"></div>
  <div class="details" id="element-details"></div>

  <!-- Back Button -->
  <a href="index.html" class="back-btn">&#8592; Back</a>

  <script>
    const elements = [
    { symbol: "H", name: "Hydrogen", atomicNumber: 1, group: 1, period: 1, weight: 1.008, category: "nonmetal" },
    { symbol: "He", name: "Helium", atomicNumber: 2, group: 18, period: 1, weight: 4.0026, category: "noble gas" },
    { symbol: "Li", name: "Lithium", atomicNumber: 3, group: 1, period: 2, weight: 6.94, category: "alkali metal" },
    { symbol: "Be", name: "Beryllium", atomicNumber: 4, group: 2, period: 2, weight: 9.0122, category: "alkaline earth metal"
},
    { symbol: "B", name: "Boron", atomicNumber: 5, group: 13, period: 2, weight: 10.81, category: "metalloid" },
    { symbol: "C", name: "Carbon", atomicNumber: 6, group: 14, period: 2, weight: 12.011, category: "nonmetal" },
    { symbol: "N", name: "Nitrogen", atomicNumber: 7, group: 15, period: 2, weight: 14.007, category: "nonmetal" },
    { symbol: "O", name: "Oxygen", atomicNumber: 8, group: 16, period: 2, weight: 15.999, category: "nonmetal" },
    { symbol: "F", name: "Fluorine", atomicNumber: 9, group: 17, period: 2, weight: 18.998, category: "nonmetal" },
    { symbol: "Ne", name: "Neon", atomicNumber: 10, group: 18, period: 2, weight: 20.180, category: "noble gas" },
    { symbol: "Na", name: "Sodium", atomicNumber: 11, group: 1, period: 3, weight: 22.990, category: "alkali metal" },
     { symbol: "Mg", name: "Magnesium", atomicNumber: 12, group: 2, period: 3, weight: 24.305, category: "alkaline earth
metal" },
     { symbol: "Al", name: "Aluminum", atomicNumber: 13, group: 13, period: 3, weight: 26.982, category: "post-transition
metal" },
    { symbol: "Si", name: "Silicon", atomicNumber: 14, group: 14, period: 3, weight: 28.085, category: "metalloid" },
    { symbol: "P", name: "Phosphorus", atomicNumber: 15, group: 15, period: 3, weight: 30.974, category: "nonmetal" },
    { symbol: "S", name: "Sulfur", atomicNumber: 16, group: 16, period: 3, weight: 32.06, category: "nonmetal" },
    { symbol: "Cl", name: "Chlorine", atomicNumber: 17, group: 17, period: 3, weight: 35.45, category: "nonmetal" },
    { symbol: "Ar", name: "Argon", atomicNumber: 18, group: 18, period: 3, weight: 39.948, category: "noble gas" },
    { symbol: "K", name: "Potassium", atomicNumber: 19, group: 1, period: 4, weight: 39.098, category: "alkali metal" },
    { symbol: "Ca", name: "Calcium", atomicNumber: 20, group: 2, period: 4, weight: 40.078, category: "alkaline earth metal"
},
    { symbol: "Sc", name: "Scandium", atomicNumber: 21, group: 3, period: 4, weight: 44.956, category: "transition metal" },
    { symbol: "Ti", name: "Titanium", atomicNumber: 22, group: 4, period: 4, weight: 47.867, category: "transition metal" },
    { symbol: "V", name: "Vanadium", atomicNumber: 23, group: 5, period: 4, weight: 50.942, category: "transition metal" },
    { symbol: "Cr", name: "Chromium", atomicNumber: 24, group: 6, period: 4, weight: 51.996, category: "transition metal" },
    { symbol: "Mn", name: "Manganese", atomicNumber: 25, group: 7, period: 4, weight: 54.938, category: "transition metal"
},
    { symbol: "Fe", name: "Iron", atomicNumber: 26, group: 8, period: 4, weight: 55.845, category: "transition metal" },
    { symbol: "Co", name: "Cobalt", atomicNumber: 27, group: 9, period: 4, weight: 58.933, category: "transition metal" },
    { symbol: "Ni", name: "Nickel", atomicNumber: 28, group: 10, period: 4, weight: 58.693, category: "transition metal" },
    { symbol: "Cu", name: "Copper", atomicNumber: 29, group: 11, period: 4, weight: 63.546, category: "transition metal" },
    { symbol: "Zn", name: "Zinc", atomicNumber: 30, group: 12, period: 4, weight: 65.38, category: "transition metal" },
   { symbol: "Ga", name: "Gallium", atomicNumber: 31, group: 13, period: 4, weight: 69.723, category: "post-transition metal"
},
    { symbol: "Ge", name: "Germanium", atomicNumber: 32, group: 14, period: 4, weight: 72.630, category: "metalloid" },
    { symbol: "As", name: "Arsenic", atomicNumber: 33, group: 15, period: 4, weight: 74.922, category: "metalloid" },
    { symbol: "Se", name: "Selenium", atomicNumber: 34, group: 16, period: 4, weight: 78.971, category: "nonmetal" },
    { symbol: "Br", name: "Bromine", atomicNumber: 35, group: 17, period: 4, weight: 79.904, category: "nonmetal" },
    { symbol: "Kr", name: "Krypton", atomicNumber: 36, group: 18, period: 4, weight: 83.798, category: "noble gas" },
    { symbol: "Rb", name: "Rubidium", atomicNumber: 37, group: 1, period: 5, weight: 85.4678, category: "alkali metal" },
    { symbol: "Sr", name: "Strontium", atomicNumber: 38, group: 2, period: 5, weight: 87.62, category: "alkaline earth metal"
},
    { symbol: "Y", name: "Yttrium", atomicNumber: 39, group: 3, period: 5, weight: 88.905, category: "transition metal" },
    { symbol: "Zr", name: "Zirconium", atomicNumber: 40, group: 4, period: 5, weight: 91.224, category: "transition metal" },
    { symbol: "Nb", name: "Niobium", atomicNumber: 41, group: 5, period: 5, weight: 92.90637, category: "transition metal" },
    { symbol: "Mo", name: "Molybdenum", atomicNumber: 42, group: 6, period: 5, weight: 95.95, category: "transition metal"
},
```

```
    { symbol: "Tc", name: "Technetium", atomicNumber: 43, group: 7, period: 5, weight: 98, category: "transition metal" },
    { symbol: "Ru", name: "Ruthenium", atomicNumber: 44, group: 8, period: 5, weight: 101.07, category: "transition metal" },
    { symbol: "Rh", name: "Rhodium", atomicNumber: 45, group: 9, period: 5, weight: 102.90550, category: "transition metal"
},
    { symbol: "Pd", name: "Palladium", atomicNumber: 46, group: 10, period: 5, weight: 106.42, category: "transition metal" },
    { symbol: "Ag", name: "Silver", atomicNumber: 47, group: 11, period: 5, weight: 107.8682, category: "transition metal" },
    { symbol: "Cd", name: "Cadmium", atomicNumber: 48, group: 12, period: 5, weight: 112.414, category: "transition metal"
},
   { symbol: "In", name: "Indium", atomicNumber: 49, group: 13, period: 5, weight: 114.818, category: "post-transition metal"
},
    { symbol: "Sn", name: "Tin", atomicNumber: 50, group: 14, period: 5, weight: 118.710, category: "post-transition metal" },
    { symbol: "Sb", name: "Antimony", atomicNumber: 51, group: 15, period: 5, weight: 121.760, category: "metalloid" },
    { symbol: "Te", name: "Tellurium", atomicNumber: 52, group: 16, period: 5, weight: 127.60, category: "metalloid" },
    { symbol: "I", name: "Iodine", atomicNumber: 53, group: 17, period: 5, weight: 126.90447, category: "nonmetal" },
    { symbol: "Xe", name: "Xenon", atomicNumber: 54, group: 18, period: 5, weight: 131.293, category: "noble gas" },
    { symbol: "Cs", name: "Cesium", atomicNumber: 55, group: 1, period: 6, weight: 132.90545196, category: "alkali metal" },
    { symbol: "Ba", name: "Barium", atomicNumber: 56, group: 2, period: 6, weight: 137.327, category: "alkaline earth metal"
},
    { symbol: "La", name: "Lanthanum", atomicNumber: 57, group: 3, period: 6, weight: 138.904, category: "lanthanide" },
    { symbol: "Ce", name: "Cerium", atomicNumber: 58, group: 3, period: 6, weight: 140.116, category: "lanthanide" },
    { symbol: "Pr", name: "Praseodymium", atomicNumber: 59, group: 3, period: 6, weight: 140.90766, category: "lanthanide"
},
    { symbol: "Nd", name: "Neodymium", atomicNumber: 60, group: 3, period: 6, weight: 144.242, category: "lanthanide" },
    { symbol: "Pm", name: "Promethium", atomicNumber: 61, group: 3, period: 6, weight: 145, category: "lanthanide" },
    { symbol: "Sm", name: "Samarium", atomicNumber: 62, group: 3, period: 6, weight: 150.36, category: "lanthanide" },
    { symbol: "Eu", name: "Europium", atomicNumber: 63, group: 3, period: 6, weight: 151.964, category: "lanthanide" },
    { symbol: "Gd", name: "Gadolinium", atomicNumber: 64, group: 3, period: 6, weight: 157.25, category: "lanthanide" },
    { symbol: "Tb", name: "Terbium", atomicNumber: 65, group: 3, period: 6, weight: 158.92535, category: "lanthanide" },
    { symbol: "Dy", name: "Dysprosium", atomicNumber: 66, group: 3, period: 6, weight: 162.500, category: "lanthanide" },
    { symbol: "Ho", name: "Holmium", atomicNumber: 67, group: 3, period: 6, weight: 164.93033, category: "lanthanide" },
    { symbol: "Er", name: "Erbium", atomicNumber: 68, group: 3, period: 6, weight: 167.259, category: "lanthanide" },
    { symbol: "Tm", name: "Thulium", atomicNumber: 69, group: 3, period: 6, weight: 168.93422, category: "lanthanide" },
    { symbol: "Yb", name: "Ytterbium", atomicNumber: 70, group: 3, period: 6, weight: 173.04, category: "lanthanide" },
    { symbol: "Lu", name: "Lutetium", atomicNumber: 71, group: 3, period: 6, weight: 174.9668, category: "lanthanide" },
    { symbol: "Hf", name: "Hafnium", atomicNumber: 72, group: 4, period: 6, weight: 178.49, category: "transition metal" },
    { symbol: "Ta", name: "Tantalum", atomicNumber: 73, group: 5, period: 6, weight: 180.94788, category: "transition metal"
},
    { symbol: "W", name: "Tungsten", atomicNumber: 74, group: 6, period: 6, weight: 183.84, category: "transition metal" },
    { symbol: "Re", name: "Rhenium", atomicNumber: 75, group: 7, period: 6, weight: 186.207, category: "transition metal" },
    { symbol: "Os", name: "Osmium", atomicNumber: 76, group: 8, period: 6, weight: 190.23, category: "transition metal" },
    { symbol: "Ir", name: "Iridium", atomicNumber: 77, group: 9, period: 6, weight: 192.217, category: "transition metal" },
    { symbol: "Pt", name: "Platinum", atomicNumber: 78, group: 10, period: 6, weight: 195.084, category: "transition metal" },
    { symbol: "Au", name: "Gold", atomicNumber: 79, group: 11, period: 6, weight: 196.966569, category: "transition metal" },
    { symbol: "Hg", name: "Mercury", atomicNumber: 80, group: 12, period: 6, weight: 200.592, category: "transition metal" },
   { symbol: "Tl", name: "Thallium", atomicNumber: 81, group: 13, period: 6, weight: 204.38, category: "post-transition metal"
},
    { symbol: "Pb", name: "Lead", atomicNumber: 82, group: 14, period: 6, weight: 207.2, category: "post-transition metal" },
     { symbol: "Bi", name: "Bismuth", atomicNumber: 83, group: 15, period: 6, weight: 208.98040, category: "post-transition
metal" },
    { symbol: "Po", name: "Polonium", atomicNumber: 84, group: 16, period: 6, weight: 209, category: "post-transition metal"
},
    { symbol: "At", name: "Astatine", atomicNumber: 85, group: 17, period: 6, weight: 210, category: "post-transition metal" },
    { symbol: "Rn", name: "Radon", atomicNumber: 86, group: 18, period: 6, weight: 222, category: "noble gas" },
    { symbol: "Fr", name: "Francium", atomicNumber: 87, group: 1, period: 7, weight: 223, category: "alkali metal" },
    { symbol: "Ra", name: "Radium", atomicNumber: 88, group: 2, period: 7, weight: 226, category: "alkaline earth metal" },
    { symbol: "Ac", name: "Actinium", atomicNumber: 89, group: 3, period: 7, weight: 227, category: "actinide" },
    { symbol: "Th", name: "Thorium", atomicNumber: 90, group: 3, period: 7, weight: 232.0381, category: "actinide" },
    { symbol: "Pa", name: "Protactinium", atomicNumber: 91, group: 3, period: 7, weight: 231.03588, category: "actinide" },
    { symbol: "U", name: "Uranium", atomicNumber: 92, group: 3, period: 7, weight: 238.02891, category: "actinide" },
    { symbol: "Np", name: "Neptunium", atomicNumber: 93, group: 3, period: 7, weight: 237, category: "actinide" },
    { symbol: "Pu", name: "Plutonium", atomicNumber: 94, group: 3, period: 7, weight: 244, category: "actinide" },
    { symbol: "Am", name: "Americium", atomicNumber: 95, group: 3, period: 7, weight: 243, category: "actinide" },
    { symbol: "Cm", name: "Curium", atomicNumber: 96, group: 3, period: 7, weight: 247, category: "actinide" },
    { symbol: "Bk", name: "Berkelium", atomicNumber: 97, group: 3, period: 7, weight: 247, category: "actinide" },
    { symbol: "Cf", name: "Californium", atomicNumber: 98, group: 3, period: 7, weight: 251, category: "actinide" },
    { symbol: "Es", name: "Einsteinium", atomicNumber: 99, group: 3, period: 7, weight: 252, category: "actinide" },
    { symbol: "Fm", name: "Fermium", atomicNumber: 100, group: 3, period: 7, weight: 257, category: "actinide" },
    { symbol: "Md", name: "Mendelevium", atomicNumber: 101, group: 3, period: 7, weight: 258, category: "actinide" },
    { symbol: "No", name: "Nobelium", atomicNumber: 102, group: 3, period: 7, weight: 259, category: "actinide" },
    { symbol: "Lr", name: "Lawrencium", atomicNumber: 103, group: 3, period: 7, weight: 266, category: "actinide" },
```

```
      { symbol: "Rf", name: "Rutherfordium", atomicNumber: 104, group: 4, period: 7, weight: 267, category: "transition metal"
},
      { symbol: "Db", name: "Dubnium", atomicNumber: 105, group: 5, period: 7, weight: 268, category: "transition metal" },
      { symbol: "Sg", name: "Seaborgium", atomicNumber: 106, group: 6, period: 7, weight: 271, category: "transition metal" },
      { symbol: "Bh", name: "Bohrium", atomicNumber: 107, group: 7, period: 7, weight: 272, category: "transition metal" },
      { symbol: "Hs", name: "Hassium", atomicNumber: 108, group: 8, period: 7, weight: 277, category: "transition metal" },
      { symbol: "Mt", name: "Meitnerium", atomicNumber: 109, group: 9, period: 7, weight: 278, category: "unknown" },
      { symbol: "Ds", name: "Darmstadtium", atomicNumber: 110, group: 10, period: 7, weight: 281, category: "unknown" },
      { symbol: "Rg", name: "Roentgenium", atomicNumber: 111, group: 11, period: 7, weight: 282, category: "unknown" },
      { symbol: "Cn", name: "Copernicium", atomicNumber: 112, group: 12, period: 7, weight: 285, category: "unknown" },
      { symbol: "Nh", name: "Nihonium", atomicNumber: 113, group: 13, period: 7, weight: 286, category: "unknown" },
      { symbol: "Fl", name: "Flerovium", atomicNumber: 114, group: 14, period: 7, weight: 289, category: "unknown" },
      { symbol: "Mc", name: "Moscovium", atomicNumber: 115, group: 15, period: 7, weight: 288, category: "unknown" },
      { symbol: "Lv", name: "Livermorium", atomicNumber: 116, group: 16, period: 7, weight: 293, category: "unknown" },
      { symbol: "Ts", name: "Tennessine", atomicNumber: 117, group: 17, period: 7, weight: 294, category: "unknown" },
      { symbol: "Og", name: "Oganesson", atomicNumber: 118, group: 18, period: 7, weight: 294, category: "unknown" }
   ];

   const tableContainer = document.getElementById("periodic-table");
const detailsContainer = document.getElementById("element-details");

for (let i = 0; i < 118; i++) {
   const element = elements.find(e => e.atomicNumber === i + 1);
   const div = document.createElement("div");
   div.classList.add("element");

   if (element) {
      // Assign category class dynamically
      div.classList.add(element.category.replace(/\s+/g, "-"));

      div.innerHTML = `
         <div class="element-number">${element.atomicNumber}</div>
         <div class="element-symbol">${element.symbol}</div>
      `;
      div.setAttribute("title", element.name);

      div.addEventListener("click", () => {
         document.querySelectorAll(".element").forEach(el => el.classList.remove("highlight"));
         div.classList.add("highlight");

         detailsContainer.innerHTML = `
            <h2>${element.name} (${element.symbol})</h2>
            <p>Atomic Number: ${element.atomicNumber}</p>
            <p>Atomic Weight: ${element.weight}</p>
            <p>Group: ${element.group}</p>
            <p>Period: ${element.period}</p>
            <p>Category: ${element.category.replace(/-/g, " ")}</p>
         `;
         detailsContainer.style.display = "block";
      });
   }
   tableContainer.appendChild(div);
}
   </script>
</body>
</html>
```

Module 3
experiment.html –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Select Experiment Type</title>
```

```
<!-- Tailwind CSS CDN -->
<script src="https://cdn.tailwindcss.com"></script>
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;600;700&display=swap" rel="stylesheet">

<style>
  body {
  font-family: 'Poppins', sans-serif;
  background: linear-gradient(to bottom right, #0f172a, #1e293b); /* 🌑 Deep navy to slate */
  overflow: hidden;
  position: relative;
  color: #e2e8f0; /* Light text for contrast */
}

.glass {
  background: rgba(255, 255, 255, 0.05);
  backdrop-filter: blur(12px);
  border: 1px solid rgba(255, 255, 255, 0.12);
  border-radius: 1rem;
  box-shadow: 0 10px 40px rgba(0, 0, 0, 0.5);
}

  /* Back Button */
  .back-btn {
        position: absolute;
        top: 30px;
        left: 20px;
        text-decoration: none;
        font-size: 1.2rem;
        font-weight: bold;
        color: white;
        background: #333;
        padding: 10px 20px;
        border-radius: 8px;
        transition: background 0.3s ease;
    }

    .back-btn:hover {
        background: #555;
    }

.orb {
  position: absolute;
  border-radius: 50%;
  filter: blur(45px);
  opacity: 0.5;
  animation: floatOrb 12s ease-in-out infinite, pulse 4s ease-in-out infinite alternate;
}

.orb1 {
  width: 160px;
  height: 160px;
  background: radial-gradient(circle, #7f5af0, #3a0ca3);
  top: 15%;
  left: 12%;
}

.orb2 {
  width: 130px;
  height: 130px;
  background: radial-gradient(circle, #fb7185, #c9184a);
  top: 65%;
  left: 78%;
}

.orb3 {
  width: 100px;
  height: 100px;
  background: radial-gradient(circle, #38bdf8, #0ea5e9);
  top: 45%;
  left: 48%;
}
```

```
  @keyframes floatOrb {
   0%, 100% { transform: translateY(0px) translateX(0px); }
   50% { transform: translateY(-30px) translateX(20px); }
  }

  @keyframes pulse {
   0% { opacity: 0.4; }
   100% { opacity: 0.8; }
  }
  </style>
</head>
<body class="flex items-center justify-center min-h-screen text-gray-900 relative px-4">

  <!-- Animated Glowing Orbs -->
  <div class="orb orb1"></div>
  <div class="orb orb2"></div>
  <div class="orb orb3"></div>

  <!-- Back Button -->
  <a href="index.html" class="back-btn">&#8592; Back</a>


  <!-- Main Content -->
  <div class="glass px-10 py-12 max-w-xl w-full text-center z-10">
   <h1 class="text-4xl font-bold mb-6 text-white drop-shadow-md" style="text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);">
     Select Experiment Type
    </h1>

   <div class="flex flex-wrap justify-center gap-6">
    <button onclick="location.href='organic.html'"
        class="bg-green-600 hover:bg-green-700 text-white px-6 py-3 rounded-full text-lg font-semibold transition-all duration-300 ease-
in-out button-hover">
     Organic Chemistry
    </button>
    <button onclick="location.href='physical.html'"
         class="bg-yellow-600 hover:bg-yellow-700 text-white px-6 py-3 rounded-full text-lg font-semibold transition-all duration-300
ease-in-out button-hover">
     Physical Chemistry
    </button>
    <button onclick="location.href='inorganic.html'"
        class="bg-blue-600 hover:bg-blue-700 text-white px-6 py-3 rounded-full text-lg font-semibold transition-all duration-300 ease-
in-out button-hover">
     Inorganic Chemistry
    </button>
   </div>
  </div>
</body>
</html>
```

Module 4
organic.html –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Organic Chemistry Lab</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
      font-family: Arial, sans-serif;
    }
```

```
    body {
      display: flex;
      flex-direction: column;
      height: 100vh;
      background-color: #f9f9f9;
    }

    /* Back Button */
    .back-btn {
      position: absolute;
      top: 30px;
      left: 20px;
      text-decoration: none;
      font-size: 1.2rem;
      font-weight: bold;
      color: white;
      background: #333;
      padding: 10px 20px;
      border-radius: 8px;
      transition: background 0.3s ease;
    }

    .back-btn:hover {
      background: #555;
    }

    /* Navbar */
.navbar {
  display: flex;
  justify-content: center; /* Center all items horizontally */
  align-items: center;
  background-color: #004466;
  color: white;
  padding: 15px 20px;
  position: relative; /* Needed for absolute positioning of nav links */
}

.navbar h1 {
  font-size: 3rem; /* Increased text size */
  text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);
  margin-bottom: 15px;
  text-align: center; /* Ensures text is centered */
  flex-grow: 1; /* Allows title to occupy center */
}

/* Navbar Links */
.nav-links {
  position: absolute;
  right: 20px; /* Keeps the navigation links on the right */
}

.nav-links a {
  color: white;
  text-decoration: none;
  margin: 0 10px;
  font-weight: bold;
}

    /* Page Layout */
    .container {
      display: flex;
      flex: 1;
    }

    /* Sidebar */
    .sidebar {
      width: 18%;
```

```
      background-color: #0077b6;
      color: white;
      padding: 15px;
    }
    .sidebar h2 {
      font-size: 18px;
      margin-bottom: 10px;
    }
    .sidebar ul {
      list-style: none;
    }
    .sidebar ul li {
      padding: 10px;
      border-bottom: 1px solid #fff;
      cursor: pointer;
    }
    .sidebar ul li:hover {
      background-color: #48cae4;
    }

    /* Main Content */
    .main-content {
      flex: 1;
      padding: 20px;
      background-color: white;
    }
    .main-content h2 {
      margin-bottom: 10px;
    }
    .interactive-area {
      background-color: #e9ecef;
      height: 70vh;
      border-radius: 10px;
      padding: 15px;
      text-align: center;
      display: flex;
      justify-content: center;
      align-items: center;
      font-size: 18px;
      color: #555;
    }

    /* Right Panel */
    .right-panel {
      width: 25%;
      background-color: #e0fbfc;
      padding: 15px;
    }
    .right-panel h3 {
      margin-bottom: 10px;
    }

    /* Footer */
    .footer {
      background-color: #004466;
      color: white;
      text-align: center;
      padding: 10px;
      font-size: 14px;
    }

  </style>
</head>
<body>

  <!-- Navbar -->
  <div class="navbar">
```

```
    <h1>Organic Chemistry Virtual Lab</h1>
    </div>
  </div>

   <!-- Back Button -->
   <a href="experiment.html" class="back-btn">&#8592; Back</a>

  <!-- Container -->
  <div class="container">

  <!-- Sidebar -->
<div class="sidebar">
  <h2>🗁 Organic Chemistry Topics</h2>
  <ul>
    <li><a href="IUPAC_nomenclature.html" style="color: white; text-decoration: none;">IUPAC Nomenclature</a></li>
        <li><a href="reaction_mechanisms.html" style="color: white; text-decoration: none;">Types of Reaction Mechanisms</a></li>
    <li><a href="kinetics_energy.html" style="color: white; text-decoration: none;">Kinetics & Energy Diagrams</a></li>
    <li><a href="stereochemistry.html" style="color: white; text-decoration: none;">Stereochemistry Simulator</a></li>
    <li><a href="iconic_compounds.html" style="color: white; text-decoration: none;">Iconic Compounds Archive</a></li>
        <li><a href="compound_calculator.html" style="color: white; text-decoration: none;">Chemical Equation Solver</a></li>
    <li><a href="organic_synthesis.html" style="color: white; text-decoration: none;">Organic Synthesis Builder</a></li>
  </ul>
</div>

    <!-- Main Content -->
    <div class="main-content">
      <h2 style="color: #08550a;">🧪 Interactive Chemistry Lab</h2>
      <div class="interactive-area">
        Select a feature from the sidebar to begin exploring!
      </div>
    </div>

    <!-- Right Panel -->
    <div class="right-panel">
      <h3>ℹ Interactive Panel</h3>
      <p>📌 View molecule structures in 3D.</p>
      <p>🔄 Stepwise reaction explanations.</p>
      <p>🤖 AI-Powered Reaction Solver.</p>
      <p>📝 MCQs & Quiz Section.</p>
    </div>

  </div>

</body>
</html>
```

Module 6
IUPAC_nomenclature.html–

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Virtual Chemistry Lab - IUPAC Nomenclature</title>
  <script src="https://embed.molview.org/embed.js"></script>
  <script src="https://cdnjs.cloudflare.com/ajax/libs/jquery/3.6.0/jquery.min.js"></script>
  <style>
```

```
 body {
   font-family: 'Poppins', sans-serif;
   text-align: center;
   background: linear-gradient(to right, #1e3c72, #2a5298);
   color: #fff;
   margin: 0;
   padding: 20px;

   /* Flexbox for Centering */
   display: flex;
   flex-direction: column;
   align-items: center;  /* Centers content horizontally */
   justify-content: center; /* Centers content vertically */
   min-height: 100vh; /* Full height for vertical centering */
}

/* Back Button */
.back-btn {
   position: absolute;
   top: 20px;
   left: 20px;
   text-decoration: none;
   font-size: 1.2rem;
   font-weight: bold;
   color: white;
   background: #333;
   padding: 12px 24px;
   border-radius: 8px;
   transition: background 0.3s ease;
   border: none;
   box-shadow: 2px 2px 10px rgba(0, 0, 0, 0.2);
}

.back-btn:hover {
   background: #555555;
}

/* Main Heading */
h1 {
   color: #f8f9fa;
   font-size: 3rem; /* Increased text size */
   font-weight: 700;
   text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);
   text-align: center; /* Ensures text is centered */
   flex-grow: 1; /* Allows title to occupy center */
   margin-bottom: 20px;
}

/* Subheading */
h2 {
   color: #fcd34d;
   font-size: 2rem;
   font-weight: 600;
   margin: 30px 0 20px;
}

/* Main Container */
.container {
   max-width: 900px;
   width: 90%;
   background: rgba(255, 255, 255, 0.1);
   padding: 25px;
   border-radius: 10px;
   box-shadow: 0px 4px 10px rgba(0, 0, 0, 0.3);
   text-align: center;
   margin: auto; /* Ensures it's centered horizontally */
}
```

```
/* Input Fields */
.input-group {
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 10px;
  margin-bottom: 20px;
  flex-wrap: wrap;
}

input[type="text"], select {
  padding: 12px;
  border: 2px solid #fcd34d;
  border-radius: 6px;
  width: 100%;
  max-width: 350px;
  font-size: 1rem;
  text-align: center;
}

/* Buttons */
button {
  background: #fcd34d;
  color: #1e3c72;
  padding: 12px 20px;
  border: none;
  border-radius: 5px;
  cursor: pointer;
  transition: background-color 0.3s ease;
  font-weight: bold;
  font-size: 1rem;
}

button:hover {
  background: #eab308;
}

/* Tab Navigation */
.tab {
  display: flex;
  justify-content: center;
  gap: 10px;
  background: rgba(255, 255, 255, 0.2);
  border-radius: 5px;
  margin-bottom: 20px;
  padding: 10px;
}

.tab button {
  border: none;
  outline: none;
  cursor: pointer;
  padding: 14px 20px;
  transition: 0.3s;
  background: transparent;
  color: white;
  font-size: 1.5rem;
  font-weight: Bold;
}

.tab button:hover {
  background: rgba(255, 255, 255, 0.3);
}

.tab button.active {
  background: #fcd34d;
```

```
    color: #1e3c72;
    border-radius: 5px 5px 0 0;
}

/* Tab Content */
.tabcontent {
    display: none;
    padding: 20px;
    border-radius: 0 0 5px 5px;
    animation: fadeEffect 1s;
    background: rgba(255, 255, 255, 0.1);
    text-align: left;
}

/* Fade-in effect */
@keyframes fadeEffect {
    from {opacity: 0;}
    to {opacity: 1;}
}

/* Table Styling */
table {
    width: 100%;
    border-collapse: collapse;
    margin: 20px 0;
    background: rgba(255, 255, 255, 0.2);
    border-radius: 10px;
    overflow: hidden;
}

table, th, td {
    border: 1px solid rgba(255, 255, 255, 0.3);
    padding: 12px;
    text-align: left;
    color: white;
}

th {
    background: #fcd34d;
    color: #1e3c72;
}

tr:nth-child(even) {
    background: rgba(255, 255, 255, 0.1);
}

/* Quiz Container */
.quiz-container {
    margin: 20px 0;
    padding: 15px;
    background: rgba(255, 255, 255, 0.2);
    border-radius: 5px;
}

.quiz-feedback {
    margin-top: 15px;
    padding: 10px;
    border-radius: 5px;
    display: none;
}

.correct {
    background: #22c55e;
    color: white;
}

.incorrect {
```

```
  background: #ef4444;
  color: white;
}

  </style>
</head>
<body>
  <div class="container">
    <h1>IUPAC Nomenclature</h1>

     <!-- Back Button -->
  <a href="organic.html" class="back-btn">&#8592; Back</a>

    <div class="tab">
      <button class="tablinks active" onclick="openTab(event, 'MoleculeViewer')">3D Molecule Viewer</button>
      <button class="tablinks" onclick="openTab(event, 'NomenclatureGuide')">Nomenclature Guide</button>
      <button class="tablinks" onclick="openTab(event, 'PracticeExercises')">Practice Exercises</button>
      <button class="tablinks" onclick="openTab(event, 'ReferenceData')">Reference Tables</button>
    </div>

    <div id="MoleculeViewer" class="tabcontent" style="display: block;">
      <h2>Interactive 3D Molecular Structures</h2>
      <iframe id="molview" width="600" height="400" src="https://embed.molview.org/v1/?smiles=C6H6"></iframe>

      <p>Enter a chemical formula, SMILES notation, or select a common molecule:</p>
      <input type="text" id="moleculeInput" placeholder="e.g., C6H6 or CC(=O)O">
      <button onclick="updateMolecule()">Load Structure</button>

      <p>Or select from common compounds:</p>
      <select id="commonMolecules" onchange="loadCommonMolecule()">
        <option value="">-- Select a common molecule --</option>
        <option value="C6H6">Benzene (C6H6)</option>
        <option value="CCO">Ethanol (CH3CH2OH)</option>
        <option value="CC(=O)C">Acetone (CH3COCH3)</option>
        <option value="C">Methane (CH4)</option>
        <option value="CCC">Propane (C3H8)</option>
        <option value="CCCC">Butane (C4H10)</option>
        <option value="CCCCCC">Hexane (C6H14)</option>
        <option value="CCCCO">1-Butanol (C4H9OH)</option>
        <option value="CC(=O)O">Acetic acid (CH3COOH)</option>
        <option value="OC1=CC=CC=C1">Phenol (C6H5OH)</option>
        <option value="CC1=CC=CC=C1">Toluene (C7H8)</option>
        <option value="NC1=CC=CC=C1">Aniline (C6H5NH2)</option>
        <option value="C=O">Formaldehyde (HCHO)</option>
      </select>

      <div id="moleculeDetails" class="card hidden">
        <h3 id="moleculeName">Compound Name</h3>
        <p id="moleculeFormula">Chemical Formula: </p>
        <p id="moleculeIUPAC">IUPAC Name: </p>
        <p id="moleculeCommon">Common Name: </p>
      </div>
    </div>

    <div id="NomenclatureGuide" class="tabcontent">
      <h2>IUPAC Nomenclature Guide</h2>

      <div class="card">
        <h3>Step-by-Step Nomenclature Process</h3>
        <button onclick="toggleSteps()">Show/Hide Detailed Guide</button>

        <div id="steps" class="hidden">
          <div class="step">
            <strong>1. Identify the parent hydrocarbon chain or ring system:</strong>
            <ul>
              <li>For acyclic compounds, find the longest continuous carbon chain</li>
              <li>For cyclic compounds, the ring is usually the parent structure</li>
```

```
            <li>If there are multiple options, choose the chain with the most substituents</li>
          </ul>
        </div>

        <div class="step">
          <strong>2. Number the carbon atoms in the parent chain:</strong>
          <ul>
            <li>Start numbering from the end nearest to a substituent or functional group</li>
            <li>When multiple substituents are present, use the numbering that gives the lowest set of locants</li>
            <li>For rings, start numbering from the carbon attached to a substituent, giving priority to the substituent that comes first alphabetically</li>
          </ul>
        </div>

        <div class="step">
          <strong>3. Identify and name all substituents:</strong>
          <ul>
            <li>Alkyl groups: methyl ($CH_3$-), ethyl ($C_2H_5$-), etc.</li>
            <li>Halogen atoms: fluoro (F-), chloro (Cl-), bromo (Br-), iodo (I-)</li>
            <li>Other groups: nitro ($NO_2$-), amino ($NH_2$-), etc.</li>
          </ul>
        </div>

        <div class="step">
          <strong>4. Identify and name the principal functional group:</strong>
          <ul>
            <li>Functional groups are assigned in order of precedence (see priority table)</li>
            <li>The highest priority group determines the suffix of the name</li>
            <li>Lower priority groups are named as prefixes</li>
          </ul>
        </div>

        <div class="step">
          <strong>5. Assign locants (position numbers):</strong>
          <ul>
            <li>Add numbers to indicate the positions of substituents and functional groups</li>
            <li>Use hyphens to separate numbers from names</li>
            <li>Use commas to separate multiple numbers</li>
          </ul>
        </div>

        <div class="step">
          <strong>6. Arrange substituent names alphabetically:</strong>
          <ul>
            <li>Ignore prefixes like di-, tri-, iso-, etc. when alphabetizing</li>
            <li>But include cyclo-, sec-, tert- when alphabetizing</li>
          </ul>
        </div>

        <div class="step">
          <strong>7. Apply multiplicity prefixes:</strong>
          <ul>
            <li>Use di-, tri-, tetra-, etc. when the same group appears multiple times</li>
            <li>Include numbers for each position</li>
          </ul>
        </div>

        <div class="step">
          <strong>8. Assemble the complete name:</strong>
          <ul>
            <li>Format: (locant-)[prefix][multiplicity prefix][substituent name]-[parent chain]-[functional group suffix]</li>
            <li>Place names and numbers according to IUPAC conventions</li>
          </ul>
        </div>
      </div>
    </div>
```

```
<div class="card">
  <h3>Naming Different Classes of Compounds</h3>

  <h4>Alkanes</h4>
  <p>Saturated hydrocarbons with the general formula C<sub>n</sub>H<sub>2n+2</sub></p>
  <table>
    <tr>
      <th>Number of Carbon Atoms</th>
      <th>Prefix</th>
      <th>Name</th>
      <th>Formula</th>
    </tr>
    <tr>
      <td>1</td>
      <td>meth-</td>
      <td>methane</td>
      <td>CH<sub>4</sub></td>
    </tr>
    <tr>
      <td>2</td>
      <td>eth-</td>
      <td>ethane</td>
      <td>C<sub>2</sub>H<sub>6</sub></td>
    </tr>
    <tr>
      <td>3</td>
      <td>prop-</td>
      <td>propane</td>
      <td>C<sub>3</sub>H<sub>8</sub></td>
    </tr>
    <tr>
      <td>4</td>
      <td>but-</td>
      <td>butane</td>
      <td>C<sub>4</sub>H<sub>10</sub></td>
    </tr>
    <tr>
      <td>5</td>
      <td>pent-</td>
      <td>pentane</td>
      <td>C<sub>5</sub>H<sub>12</sub></td>
    </tr>
    <tr>
      <td>6</td>
      <td>hex-</td>
      <td>hexane</td>
      <td>C<sub>6</sub>H<sub>14</sub></td>
    </tr>
    <tr>
      <td>7</td>
      <td>hept-</td>
      <td>heptane</td>
      <td>C<sub>7</sub>H<sub>16</sub></td>
    </tr>
    <tr>
      <td>8</td>
      <td>oct-</td>
      <td>octane</td>
      <td>C<sub>8</sub>H<sub>18</sub></td>
    </tr>
    <tr>
      <td>9</td>
      <td>non-</td>
      <td>nonane</td>
      <td>C<sub>9</sub>H<sub>20</sub></td>
    </tr>
    <tr>
```

```
            <td>10</td>
            <td>dec-</td>
            <td>decane</td>
            <td>C10H22</td>
          </tr>
        </table>

        <h4>1] Alkenes</h4>
        <p>Unsaturated hydrocarbons containing at least one carbon-carbon double bond (C=C)</p>
        <p>Naming convention: [prefix][number of C atoms]ene</p>
        <div class="example">
          Example: CH2=CH2 is ethene (common name: ethylene)
        </div>


        <h4>2] Alkynes</h4>
        <p>Unsaturated hydrocarbons containing at least one carbon-carbon triple bond (C≡C)</p>
        <p>Naming convention: [prefix][number of C atoms]yne</p>
        <div class="example">
          Example: HC≡CH is ethyne (common name: acetylene)
        </div>

        <h4>3] Alcohols</h4>
        <p>Compounds with the hydroxyl (-OH) functional group</p>
        <p>Naming convention: [prefix][number of C atoms]anol</p>
        <div class="example">
          Example: CH3CH2OH is ethanol
        </div>

        <h4>4] Aldehydes</h4>
        <p>Compounds with the -CHO functional group</p>
        <p>Naming convention: [prefix][number of C atoms]anal</p>
        <div class="example">
          Example: CH3CHO is ethanal (common name: acetaldehyde)
        </div>

        <h4>5] Ketones</h4>
        <p>Compounds with the carbonyl (C=O) group bonded to two carbon atoms</p>
        <p>Naming convention: [prefix][number of C atoms]anone</p>
        <div class="example">
          Example: CH3COCH3 is propanone (common name: acetone)
        </div>

        <h4>6] Carboxylic Acids</h4>
        <p>Compounds with the -COOH functional group</p>
        <p>Naming convention: [prefix][number of C atoms]anoic acid</p>
        <div class="example">
          Example: CH3COOH is ethanoic acid (common name: acetic acid)
        </div>

        <h4>7] Ethers</h4>
        <p>Compounds with an oxygen atom bonded to two carbon atoms (R-O-R')</p>
        <p>Naming convention: [alkyl group 1] [alkyl group 2] ether</p>
        <div class="example">
          Example: CH3OCH3 is dimethyl ether
        </div>

        <h4>8] Amines</h4>
        <p>Compounds derived from ammonia (NH3) where one or more hydrogen atoms are replaced by alkyl groups</p>
        <p>Naming convention: [N-substituents][parent chain]amine</p>
        <div class="example">
          Example: CH3NH2 is methanamine (common name: methylamine)
        </div>
      </div>
    </div>

    <div id="PracticeExercises" class="tabcontent">
```

```
        <h2>Practice IUPAC Naming Exercises</h2>

        <div class="card">
          <h3>Naming Practice</h3>
          <p>Enter the IUPAC name for the displayed molecule:</p>

          <div id="practiceContainer">
                                                    <iframe id="practiceMolView" width="500" height="300"
src="https://embed.molview.org/v1/?smiles=CCO"></iframe>
            <p id="practiceFormula">Current molecule: CH3CH2OH</p>
            <input type="text" id="compoundName" placeholder="Enter IUPAC name">
            <button onclick="validateName()">Check Answer</button>
            <button onclick="getNextPractice()">Next Compound</button>
            <div id="result"></div>
          </div>
        </div>

        <div class="card">
          <h3>Multiple Choice Quiz</h3>
          <div class="quiz-container">
            <p id="quizQuestion">What is the IUPAC name for CH3CH2OH?</p>
            <div class="quiz-options">
              <label><input type="radio" name="quiz" value="option1"> Methanol</label>
              <label><input type="radio" name="quiz" value="option2"> Ethanol</label>
              <label><input type="radio" name="quiz" value="option3"> Propanol</label>
              <label><input type="radio" name="quiz" value="option4"> Ethyl alcohol</label>
            </div>
            <button onclick="checkQuizAnswer()">Submit Answer</button>
            <div id="quizFeedback" class="quiz-feedback"></div>
          </div>
          <button onclick="loadNextQuiz()">Next Question</button>
        </div>
      </div>

      <div id="ReferenceData" class="tabcontent">
        <h2>IUPAC Nomenclature Reference Tables</h2>

        <div class="card">
          <h3>Functional Group Priorities</h3>
          <p>Functional groups listed in order of decreasing priority:</p>
          <table>
            <tr>
              <th>Rank</th>
              <th>Functional Group</th>
              <th>Suffix</th>
              <th>Prefix (when not principal group)</th>
            </tr>
            <tr>
              <td>1</td>
              <td>Carboxylic acid (-COOH)</td>
              <td>-oic acid</td>
              <td>carboxy-</td>
            </tr>
            <tr>
              <td>2</td>
              <td>Ester (-COOR)</td>
              <td>-oate</td>
              <td>alkoxycarbonyl-</td>
            </tr>
            <tr>
              <td>3</td>
              <td>Amide (-CONH2)</td>
              <td>-amide</td>
              <td>carbamoyl-</td>
            </tr>
            <tr>
              <td>4</td>
```

```
            <td>Nitrile (-CN)</td>
            <td>-nitrile</td>
            <td>cyano-</td>
          </tr>
          <tr>
            <td>5</td>
            <td>Aldehyde (-CHO)</td>
            <td>-al</td>
            <td>formyl-</td>
          </tr>
          <tr>
            <td>6</td>
            <td>Ketone (C=O)</td>
            <td>-one</td>
            <td>oxo-</td>
          </tr>
          <tr>
            <td>7</td>
            <td>Alcohol (-OH)</td>
            <td>-ol</td>
            <td>hydroxy-</td>
          </tr>
          <tr>
            <td>8</td>
            <td>Thiol (-SH)</td>
            <td>-thiol</td>
            <td>mercapto-</td>
          </tr>
          <tr>
            <td>9</td>
            <td>Amine (-NH2)</td>
            <td>-amine</td>
            <td>amino-</td>
          </tr>
          <tr>
            <td>10</td>
            <td>Ether (-OR)</td>
            <td>(used as prefix only)</td>
            <td>alkoxy-</td>
          </tr>
        </table>
      </div>

      <div class="card">
        <h3>Common Substituent Groups</h3>
        <table>
          <tr>
            <th>Substituent</th>
            <th>Formula</th>
            <th>Prefix</th>
          </tr>
          <tr>
            <td>Methyl</td>
            <td>-CH3</td>
            <td>methyl-</td>
          </tr>
          <tr>
            <td>Ethyl</td>
            <td>-CH2CH3</td>
            <td>ethyl-</td>
          </tr>
          <tr>
            <td>Propyl</td>
            <td>-CH2CH2CH3</td>
            <td>propyl-</td>
          </tr>
          <tr>
```

```
            <td>Isopropyl</td>
            <td>-CH(CH3)2</td>
            <td>isopropyl- or 1-methylethyl-</td>
          </tr>
          <tr>
            <td>Butyl</td>
            <td>-CH2CH2CH2CH3</td>
            <td>butyl-</td>
          </tr>
          <tr>
            <td>Phenyl</td>
            <td>-C6H5</td>
            <td>phenyl-</td>
          </tr>
          <tr>
            <td>Hydroxyl</td>
            <td>-OH</td>
            <td>hydroxy-</td>
          </tr>
          <tr>
            <td>Amino</td>
            <td>-NH2</td>
            <td>amino-</td>
          </tr>
          <tr>
            <td>Nitro</td>
            <td>-NO2</td>
            <td>nitro-</td>
          </tr>
          <tr>
            <td>Fluoro</td>
            <td>-F</td>
            <td>fluoro-</td>
          </tr>
          <tr>
            <td>Chloro</td>
            <td>-Cl</td>
            <td>chloro-</td>
          </tr>
          <tr>
            <td>Bromo</td>
            <td>-Br</td>
            <td>bromo-</td>
          </tr>
          <tr>
            <td>Iodo</td>
            <td>-I</td>
            <td>iodo-</td>
          </tr>
        </table>
      </div>

      <div class="card">
        <h3>Multiplicity Prefixes</h3>
        <table>
          <tr>
            <th>Number</th>
            <th>Prefix</th>
          </tr>
          <tr>
            <td>2</td>
            <td>di-</td>
          </tr>
          <tr>
            <td>3</td>
            <td>tri-</td>
          </tr>
```

```
            <tr>
              <td>4</td>
              <td>tetra-</td>
            </tr>
            <tr>
              <td>5</td>
              <td>penta-</td>
            </tr>
            <tr>
              <td>6</td>
              <td>hexa-</td>
            </tr>
            <tr>
              <td>7</td>
              <td>hepta-</td>
            </tr>
            <tr>
              <td>8</td>
              <td>octa-</td>
            </tr>
            <tr>
              <td>9</td>
              <td>nona-</td>
            </tr>
            <tr>
              <td>10</td>
              <td>deca-</td>
            </tr>
          </table>
        </div>

        <div class="card">
          <h3>Rules for Locant Assignment</h3>
          <ol>
            <li>The principal functional group gets the lowest possible number</li>
            <li>Multiple occurrences of the same substituent should receive the lowest set of numbers</li>
                <li>When comparing sets of locants, the set that gives the lowest number at the first point of difference is
preferred</li>
            <li>When double or triple bonds are present, they should be given the lowest possible numbers</li>
            <li>Substituents are listed alphabetically, ignoring di-, tri-, etc. prefixes</li>
          </ol>
        </div>
      </div>
    </div>

    <script>
      // Tab control functions
      function openTab(evt, tabName) {
        var i, tabcontent, tablinks;
        tabcontent = document.getElementsByClassName("tabcontent");
        for (i = 0; i < tabcontent.length; i++) {
          tabcontent[i].style.display = "none";
        }
        tablinks = document.getElementsByClassName("tablinks");
        for (i = 0; i < tablinks.length; i++) {
          tablinks[i].className = tablinks[i].className.replace(" active", "");
        }
        document.getElementById(tabName).style.display = "block";
        evt.currentTarget.className += " active";
      }

      // Molecule viewer functions
      function updateMolecule() {
        let input = document.getElementById("moleculeInput").value;
                                                                    document.getElementById("molview").src                =
`https://embed.molview.org/v1/?smiles=${encodeURIComponent(input)}`;
        updateMoleculeDetails(input);
```

```
}

function loadCommonMolecule() {
  let select = document.getElementById("commonMolecules");
  let selectedValue = select.options[select.selectedIndex].value;
  if (selectedValue) {
    document.getElementById("moleculeInput").value = selectedValue;
    updateMolecule();
  }
}

function updateMoleculeDetails(smiles) {
  // This is a simplified version. In a real implementation, you would
  // need a database or API to get the real details for each molecule.
  let details = {
    "C6H6": {
      name: "Benzene",
      formula: "C6H6",
      iupac: "Benzene",
      common: "Benzene"
    },
    "CCO": {
      name: "Ethanol",
      formula: "C2H6O",
      iupac: "Ethanol",
      common: "Ethyl alcohol"
    },
    "CC(=O)C": {
      name: "Acetone",
      formula: "C3H6O",
      iupac: "Propanone",
      common: "Acetone"
    },
    "C": {
      name: "Methane",
      formula: "CH4",
      iupac: "Methane",
      common: "Methane"
    },
    "CCC": {
      name: "Propane",
      formula: "C3H8",
      iupac: "Propane",
      common: "Propane"
    },
    "CCCC": {
      name: "Butane",
      formula: "C4H10",
      iupac: "Butane",
      common: "Butane"
    },
    "CCCCCC": {
      name: "Hexane",
      formula: "C6H14",
      iupac: "Hexane",
      common: "Hexane"
    },
    "CCCCO": {
      name: "1-Butanol",
      formula: "C4H10O",
      iupac: "Butan-1-ol",
      common: "n-Butyl alcohol"
    },
    "CC(=O)O": {
      name: "Acetic acid",
      formula: "C2H4O2",
      iupac: "Ethanoic acid",
```

```
            common: "Acetic acid"
          },
          "OC1=CC=CC=C1": {
            name: "Phenol",
            formula: "C6H6O",
            iupac: "Phenol",
            common: "Phenol"
          },
          "CC1=CC=CC=C1": {
            name: "Toluene",
            formula: "C7H8",
            iupac: "Methylbenzene",
            common: "Toluene"
          },
          "NC1=CC=CC=C1": {
            name: "Aniline",
            formula: "C6H7N",
            iupac: "Phenylamine",
            common: "Aniline"
          },
          "C=O": {
            name: "Formaldehyde",
            formula: "CH2O",
            iupac: "Methanal",
            common: "Formaldehyde"
          }
        };

        let moleculeInfo = details[smiles];
        if (moleculeInfo) {
          document.getElementById("moleculeName").textContent = moleculeInfo.name;
          document.getElementById("moleculeFormula").textContent = "Chemical Formula: " + moleculeInfo.formula;
          document.getElementById("moleculeIUPAC").textContent = "IUPAC Name: " + moleculeInfo.iupac;
          document.getElementById("moleculeCommon").textContent = "Common Name: " + moleculeInfo.common;
          document.getElementById("moleculeDetails").classList.remove("hidden");
        } else {
          document.getElementById("moleculeDetails").classList.add("hidden");
        }
      }

      // Practice exercises functions
      const practiceMolecules = [
        { smiles: "CCO", formula: "CH3CH2OH", name: "ethanol" },
        { smiles: "CC(=O)C", formula: "CH3COCH3", name: "propanone" },
        { smiles: "CCCC", formula: "CH3CH2CH2CH3", name: "butane" },
        { smiles: "C1=CC=CC=C1", formula: "C6H6", name: "benzene" },
        { smiles: "CC(=O)O", formula: "CH3COOH", name: "ethanoic acid" },
        { smiles: "CCCCO", formula: "CH3CH2CH2CH2OH", name: "butan-1-ol" },
        { smiles: "CC(C)C", formula: "(CH3)2CHCH3", name: "2-methylpropane" },
        { smiles: "C=O", formula: "HCHO", name: "methanal" },
        { smiles: "CCN", formula: "CH3CH2NH2", name: "ethanamine" },
        { smiles: "C#C", formula: "HC≡CH", name: "ethyne" }
      ];

      let currentPracticeIndex = 0;

      function loadPracticeMolecule(index) {
        let molecule = practiceMolecules[index];
                                                    document.getElementById("practiceMolView").src =
` https://embed.molview.org/v1/?smiles=${encodeURIComponent(molecule.smiles)}` ;
        document.getElementById("practiceFormula").textContent = "Current molecule: " + molecule.formula;
        document.getElementById("compoundName").value = "";
        document.getElementById("result").textContent = "";
        document.getElementById("result").className = "";
      }

      function validateName() {
```

```
        let userInput = document.getElementById("compoundName").value.toLowerCase().trim();
        let correctName = practiceMolecules[currentPracticeIndex].name.toLowerCase();
        let resultDiv = document.getElementById("result");

        if (userInput === correctName) {
          resultDiv.textContent = "Correct! Well done!";
          resultDiv.className = "correct";
        } else {
          resultDiv.textContent = `Incorrect. The correct name is: ${practiceMolecules[currentPracticeIndex].name}`;
          resultDiv.className = "incorrect";
        }
      }

      function getNextPractice() {
        currentPracticeIndex = (currentPracticeIndex + 1) % practiceMolecules.length;
        loadPracticeMolecule(currentPracticeIndex);
      }

      // Quiz functions
      const quizQuestions = [
        {
          question: "What is the IUPAC name for CH3CH2OH?",
          options: ["Methanol", "Ethanol", "Propanol", "Ethyl alcohol"],
          correct: 1,
          explanation: "The IUPAC name for CH3CH2OH is ethanol, as it has two carbon atoms with an -OH functional group."
        },
        {
          question: "What is the correct name for CH3COCH3?",
          options: ["Ethanal", "Propanal", "Propanone", "Butanone"],
          correct: 2,
          explanation: "CH3COCH3 is propanone (common name: acetone), with three carbon atoms and a carbonyl group in the middle."
        },
        {
          question: "Which is the correct IUPAC name for C6H6?",
          options: ["Cyclohexane", "Benzene", "Hexene", "Hexane"],
          correct: 1,
          explanation: "C6H6 is benzene, an aromatic hydrocarbon with a ring structure."
        },
        {
          question: "What is the IUPAC name for CH3COOH?",
          options: ["Methanoic acid", "Ethanoic acid", "Propanoic acid", "Butanoic acid"],
          correct: 1,
          explanation: "CH3COOH is ethanoic acid (common name: acetic acid), with two carbon atoms and a carboxyl group."
        },
        {
          question: "Which functional group has the highest priority in IUPAC naming?",
          options: ["Alcohol (-OH)", "Ketone (C=O)", "Carboxylic acid (-COOH)", "Amine (-NH2)"],
          correct: 2,
          explanation: "Carboxylic acids have the highest priority, followed by esters, amides, nitriles, aldehydes, then ketones."
        }
      ];

      let currentQuizIndex = 0;

      function loadQuizQuestion(index) {
        let question = quizQuestions[index];
        document.getElementById("quizQuestion").textContent = question.question;

        let options = document.querySelectorAll('.quiz-options label');
        for (let i = 0; i < options.length; i++) {
          options[i].textContent = question.options[i];
          options[i].querySelector('input').value = `option${i+1}`;
          options[i].querySelector('input').checked = false;
        }
```

```
        document.getElementById("quizFeedback").textContent = "";
        document.getElementById("quizFeedback").style.display = "none";
    }

    function checkQuizAnswer() {
        let selectedOption = document.querySelector('input[name="quiz"]:checked');
        let feedbackDiv = document.getElementById("quizFeedback");

        if (!selectedOption) {
            feedbackDiv.textContent = "Please select an answer!";
            feedbackDiv.className = "incorrect";
            feedbackDiv.style.display = "block";
            return;
        }

        let selectedIndex = parseInt(selectedOption.value.replace("option", "")) - 1;
        let correctIndex = quizQuestions[currentQuizIndex].correct;

        if (selectedIndex === correctIndex) {
            feedbackDiv.textContent = "Correct! " + quizQuestions[currentQuizIndex].explanation;
            feedbackDiv.className = "correct";
        } else {
            feedbackDiv.textContent = `Incorrect. The correct answer is: ${quizQuestions[currentQuizIndex].options[correctIndex]}. ${quizQuestions[currentQuizIndex].explanation}`;
            feedbackDiv.className = "incorrect";
        }

        feedbackDiv.style.display = "block";
    }

    function loadNextQuiz() {
        currentQuizIndex = (currentQuizIndex + 1) % quizQuestions.length;
        loadQuizQuestion(currentQuizIndex);
    }

    // Toggle functions
    function toggleSteps() {
        let stepsDiv = document.getElementById("steps");
        stepsDiv.classList.toggle("hidden");
    }

    // Initialize the page
    document.addEventListener("DOMContentLoaded", function() {
        loadPracticeMolecule(0);
        loadQuizQuestion(0);
    });
  </script>
</body>
</html>
```

Module 7
reaction_mechanisms.html–

```
<!DOCTYPE html>
<html lang="en">
<head>
  <script src="https://cdnjs.cloudflare.com/ajax/libs/3Dmol/1.2.0/3Dmol-min.js"></script>
  <script src="https://cdnjs.cloudflare.com/ajax/libs/jquery/3.6.0/jquery.min.js"></script>

  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Reaction Mechanisms</title>
<style>
  * {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  }
  body {
    display: flex;
    flex-direction: column;
    min-height: 100vh;
    background-color: #f5f5f5;
  }

  .navbar {
    text-align: center;
    background: linear-gradient(135deg, #004466, #006699);
    color: white;
    padding: 20px;
    box-shadow: 0 2px 10px rgba(0,0,0,0.2);
  }
  .navbar h1 {
    font-size: 3rem; /* Increased text size */
    text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);
    margin-bottom: 15px;
    text-align: center; /* Ensures text is centered */
    flex-grow: 1; /* Allows title to occupy center */
  }

  .back-btn {
    position: absolute;
    top: 20px;
    left: 20px;
    text-decoration: none;
    font-size: 1.2rem;
    font-weight: bold;
    color: white;
    background: #333;
    padding: 10px 20px;
    border-radius: 8px;
    transition: background 0.3s ease;
  }

  .back-btn:hover {
    background: #555;
  }

  .container {
    display: flex;
    flex: 1;
    gap: 20px;
    padding: 20px;
  }

  .sidebar {
    width: 250px;
    background: white;
    border-radius: 10px;
    padding: 15px;
    box-shadow: 0 2px 10px rgba(0,0,0,0.1);
  }
  .sidebar h2 {
    font-size: 1.2rem;
    margin-bottom: 15px;
    color: #004466;
    padding-bottom: 10px;
```

```
    border-bottom: 2px solid #f0f0f0;
  }
  .sidebar ul { list-style: none; }
  .sidebar ul li {
    padding: 12px;
    margin-bottom: 8px;
    border-radius: 6px;
    cursor: pointer;
    transition: all 0.3s ease;
    font-weight: 500;
    color: #555;
    background-color: #f9f9f9;
  }
  .sidebar ul li:hover {
    background-color: #e0f7ff;
    color: #004466;
    transform: translateX(5px);
  }
  .sidebar ul li.active {
    background-color: #004466;
    color: white;
  }

  .main-content {
    flex: 1;
    background: white;
    border-radius: 10px;
    padding: 20px;
    box-shadow: 0 2px 10px rgba(0,0,0,0.1);
  }
  .main-content h2 {
    margin-bottom: 20px;
    color: #004466;
    font-size: 1.5rem;
  }

  .reaction-steps {
    padding: 20px;
    border-radius: 8px;
    background-color: #f9f9f9;
    width: 100%;
    line-height: 1.6;
  }
  .reaction-steps h3 {
    color: #006699;
    margin-bottom: 10px;
  }
  .reaction-steps ol {
    padding-left: 20px;
  }
  .reaction-steps li {
    margin-bottom: 10px;
  }
  .conditions-box {
    background-color: #e0f7ff;
    padding: 15px;
    border-radius: 8px;
    margin-top: 15px;
    border-left: 4px solid #004466;
  }
  .example-box {
    background-color: #f0f0f0;
    padding: 15px;
    border-radius: 8px;
    margin-top: 15px;
    font-family: monospace;
    overflow-x: auto;
```

```
}

.right-panel {
  width: 300px;
  background: white;
  border-radius: 10px;
  padding: 20px;
  box-shadow: 0 2px 10px rgba(0,0,0,0.1);
}
.right-panel h3 {
  color: #004466;
  margin-bottom: 15px;
  font-size: 1.2rem;
  padding-bottom: 10px;
  border-bottom: 2px solid #f0f0f0;
}
.quiz-box {
  background-color: #f9f9f9;
  padding: 15px;
  border-radius: 8px;
}
.quiz-box p {
  margin-bottom: 15px;
  font-weight: 500;
}
.quiz-box button {
  display: block;
  width: 100%;
  padding: 10px;
  margin-bottom: 10px;
  background-color: white;
  border: 1px solid #ddd;
  border-radius: 5px;
  cursor: pointer;
  transition: all 0.3s;
  text-align: left;
}
.quiz-box button:hover {
  background-color: #e0f7ff;
  border-color: #004466;
}
#quizFeedback {
  margin-top: 15px;
  font-weight: bold;
  min-height: 20px;
}
.correct {
  color: #28a745;
}
.incorrect {
  color: #dc3545;
}

#moleculeViewer {
  width: 100%;
  height: 300px;
  border-radius: 8px;
  margin-bottom: 20px;
  box-shadow: 0 2px 5px rgba(0,0,0,0.1);
  background-color: #f9f9f9;
}

@media (max-width: 1024px) {
  .container {
    flex-direction: column;
  }
  .sidebar, .right-panel {
```

```
        width: 100%;
        margin-bottom: 20px;
      }
    }
  </style>
</head>
<body>
  <div class="navbar">
    <h1>Reaction Mechanisms</h1>
  </div>

  <a href="organic.html" class="back-btn">&amp;#8592; Back</a>

  <div class="container">
    <div class="sidebar">
      <h2>🗁 Reaction Mechanisms</h2>
      <ul id="mechanismList">
        <li data-mechanism="sn1" class="active">SN1 Reaction</li>
        <li data-mechanism="sn2">SN2 Reaction</li>
        <li data-mechanism="e1">E1 Elimination</li>
        <li data-mechanism="e2">E2 Elimination</li>
        <li data-mechanism="electrophilic">Electrophilic Addition</li>
        <li data-mechanism="nucleophilic">Nucleophilic Addition</li>
        <li data-mechanism="free_radical">Free Radical Substitution</li>
        <li data-mechanism="pericyclic">Pericyclic Reaction</li>
        <li data-mechanism="redox">Oxidation and Reduction</li>
        <li data-mechanism="organometallic">Organometallic Reaction</li>
      </ul>
    </div>

    <div class="main-content">
      <h2>🖉 Step-by-Step Reaction Mechanism</h2>

      <div id="moleculeViewer"></div>

      <div class="interactive-area" id="reactionDisplay">
        <h3>SN1 Reaction</h3>
        <p>Select a different reaction from the sidebar to explore its mechanism</p>
      </div>

      <div class="reaction-steps" id="reactionSteps">
        <h3>Mechanism Steps</h3>
        <ol id="mechanismSteps">
          <li>Step 1: Formation of carbocation (Rate-determining step).</li>
          <li>Step 2: Nucleophilic attack on carbocation.</li>
          <li>Step 3: Deprotonation (if needed) to stabilize the product.</li>
        </ol>

        <div class="conditions-box" id="reactionConditions">
          <strong>Reaction Conditions:</strong> Favored by polar protic solvents (e.g., water, ethanol). Works best with
tertiary alkyl halides.
        </div>

        <div class="example-box" id="reactionExample">
          <strong>Example:</strong> (CH3)3CBr + H2O → (CH3)3COH + HBr
        </div>
      </div>
    </div>

    <div class="right-panel">
      <h3>🗹 Quiz Section</h3>
      <div class="quiz-box">
        <p id="quizQuestion">What is the rate-determining step of SN1?</p>
        <div id="quizOptions">
          <button data-index="0">Carbocation formation</button>
          <button data-index="1">Nucleophilic attack</button>
```

```
            <button data-index="2">Leaving group departure</button>
            <button data-index="3">Proton transfer</button>
          </div>
          <p id="quizFeedback"></p>
        </div>

        <h3 style="margin-top: 30px;">💡 Key Concepts</h3>
        <div class="quiz-box" id="keyConcepts">
          <p>• SN1 is a two-step mechanism</p>
          <p>• Rate depends only on substrate concentration</p>
          <p>• Results in racemization at chiral centers</p>
        </div>
      </div>
    </div>

    <script>
      // Data for reactions, molecules, and quizzes
      const REACTIONS_DATA = {
        sn1: {
          name: "SN1 Reaction",
          steps: [
            "Formation of carbocation (Rate-determining step)",
            "Nucleophilic attack on carbocation",
            "Deprotonation (if needed) to stabilize the product"
          ],
          conditions: "Favored by polar protic solvents (e.g., water, ethanol). Works best with tertiary alkyl halides.",
          example: "(CH3)3CBr + H2O → (CH3)3COH + HBr",
          concepts: [
            "SN1 is a two-step mechanism",
            "Rate depends only on substrate concentration",
            "Results in racemization at chiral centers",
            "Favored by stable carbocations"
          ]
        },
        sn2: {
          name: "SN2 Reaction",
          steps: [
            "Nucleophile attacks from the opposite side of the leaving group",
            "Simultaneous departure of the leaving group",
            "Inversion of configuration (stereochemistry changes)"
          ],
          conditions: "Favored by polar aprotic solvents (e.g., DMSO, acetone). Works best with primary alkyl halides.",
          example: "CH3CH2Br + OH⁻ → CH3CH2OH + Br⁻",
          concepts: [
            "SN2 is a single-step mechanism",
            "Rate depends on both substrate and nucleophile concentration",
            "Results in inversion of configuration",
            "Steric hindrance greatly affects the rate"
          ]
        },
        e1: {
          name: "E1 Elimination",
          steps: [
            "Formation of carbocation (Rate-determining step)",
            "Base abstracts a proton, forming an alkene"
          ],
          conditions: "Favored by weak bases and heat. Works best with tertiary alkyl halides.",
          example: "(CH3)3CBr → (CH3)2C=CH2 + HBr",
          concepts: [
            "E1 is a two-step mechanism",
            "Competes with SN1 pathway",
            "Zaitsev's rule predicts major product",
            "Favored by heat"
          ]
        },
        e2: {
          name: "E2 Elimination",
```

```
      steps: [
        "Strong base abstracts a proton",
        "Simultaneous leaving group departure",
        "Formation of alkene (Requires anti-periplanar geometry)"
      ],
      conditions: "Favored by strong bases (e.g., NaOEt, NaH). Works best with primary and secondary alkyl halides.",
      example: "CH3CH2Br + NaOEt → CH2=CH2 + NaBr + EtOH",
      concepts: [
        "E2 is a single-step mechanism",
        "Requires anti-periplanar geometry",
        "Follows Zaitsev's rule generally",
        "Strong bases favor E2 over SN2"
      ]
    },
    electrophilic: {
      name: "Electrophilic Addition",
      steps: [
        "Electrophile (H+) attacks the π-bond of an alkene",
        "Carbocation formation (Rate-determining step)",
        "Nucleophilic attack by the halide or other nucleophile"
      ],
      conditions: "Occurs in acidic medium (e.g., HBr, HCl). Follows Markovnikov's rule.",
      example: "C2H4 + HBr → C2H5Br",
      concepts: [
        "Markovnikov's rule predicts product",
        "Carbocation stability determines major product",
        "Can have rearrangements",
        "Anti-Markovnikov products possible with peroxides"
      ]
    },
    nucleophilic: {
      name: "Nucleophilic Addition",
      steps: [
        "Nucleophile attacks carbonyl carbon",
        "Formation of tetrahedral intermediate",
        "Protonation to stabilize the product"
      ],
      conditions: "Common in aldehydes & ketones. Occurs in both acidic and basic conditions.",
      example: "CH3CHO + HCN → CH3CH(OH)CN",
      concepts: [
        "Carbonyl carbon is electrophilic",
        "Aldehydes are more reactive than ketones",
        "Can be reversible or irreversible",
        "Sterics affect reactivity"
      ]
    },
    free_radical: {
      name: "Free Radical Substitution",
      steps: [
        "Initiation - Homolytic cleavage of Cl2 by UV light",
        "Propagation - Free radicals attack alkanes, forming new radicals",
        "Termination - Radicals combine to form stable products"
      ],
      conditions: "Requires UV light or heat. Occurs in halogenation of alkanes.",
      example: "CH4 + Cl2 (UV) → CH3Cl + HCl",
      concepts: [
        "Three stages: initiation, propagation, termination",
        "Produces mixture of products",
        "Selectivity: 3° > 2° > 1° H",
        "Can be controlled by stoichiometry"
      ]
    },
    pericyclic: {
  name: "Pericyclic Reaction",
  steps: [
    "Concerted movement of electrons through a cyclic transition state",
    "Reaction proceeds without intermediates",
```

```
      "Stereochemistry governed by orbital symmetry (Woodward-Hoffmann rules)"
    ],
    conditions: "Typically requires heat or UV light; occurs under thermal or photochemical activation.",
    example: "1,3-Butadiene + ethene → cyclohexene (Diels–Alder reaction)",
    concepts: [
      "Reaction is stereospecific",
      "No carbocation or radical intermediates",
      "Includes Diels–Alder, electrocyclic, and sigmatropic reactions",
      "Orbital symmetry controls whether reaction is allowed or forbidden"
    ]
  },
  redox: {
    name: "Oxidation and Reduction Reaction",
    steps: [
      "Oxidation: Increase in C–O or decrease in C–H bonds",
      "Reduction: Increase in C–H or decrease in C–O bonds",
      "Involves specific oxidizing or reducing agents"
    ],
    conditions: "Depends on reagents: PCC, KMnO4, NaBH4, or LiAlH4; may require specific solvents or temperatures.",
    example: "CH3CH2OH + PCC → CH3CHO (oxidation of ethanol to acetaldehyde)",
    concepts: [
      "Redox reactions change the oxidation state of carbon atoms",
      "Primary alcohols can be oxidized to aldehydes or acids",
      "Aldehydes and ketones can be reduced to alcohols",
      "Different reagents offer control over how far the reaction goes"
    ]
  },
  organometallic: {
    name: "Organometallic Reaction",
    steps: [
      "Formation of organometallic reagent (e.g., Grignard or organolithium)",
      "Reagent acts as a nucleophile, attacking an electrophilic carbon",
      "Final product is isolated after aqueous or acidic work-up"
    ],
    conditions: "Must be performed under anhydrous conditions, usually in dry ether solvents.",
    example: "CH3MgBr + CH3CHO → CH3CH(OH)CH3 (after acid workup)",
    concepts: [
      "Grignard reagents react with carbonyl compounds to form alcohols",
      "Highly reactive with water or air—strictly dry conditions required",
      "Used in forming C–C bonds in synthesis",
      "Product depends on the type of carbonyl used (aldehyde, ketone, ester, etc.)"
    ]
  }
};


  const MOLECULE_FILES = {
  sn1: "https://files.rcsb.org/ligands/view/BRM.pdb",  // Bromomethane
  sn2: "https://files.rcsb.org/ligands/view/CHL.pdb",  // Chloromethane
  e1: "https://files.rcsb.org/ligands/view/ETO.pdb",   // Ethanol
  e2: "https://files.rcsb.org/ligands/view/ETH.pdb",   // Ethene
  electrophilic: "https://files.rcsb.org/ligands/view/ETH.pdb", // Ethene
  nucleophilic: "https://files.rcsb.org/ligands/view/FOR.pdb",  // Formaldehyde
  free_radical: "https://files.rcsb.org/ligands/view/CH4.pdb",   // Methane
  pericyclic: "https://files.rcsb.org/ligands/view/BIK.pdb",  // Example of butadiene (for Diels-Alder)
  redox: "https://files.rcsb.org/ligands/view/NAH.pdb",  // Example of a carbonyl compound like acetone
  organometallic: "https://files.rcsb.org/ligands/view/MAG.pdb"  // Example of a Grignard reagent
};

  const QUIZ_DATA = {
    sn1: [
      {
        question: "What is the rate-determining step of SN1?",
        options: ["Carbocation formation", "Nucleophilic attack", "Leaving group departure", "Proton transfer"],
        answer: 0
      },
      {
```

```
        question: "Which type of substrate reacts fastest in SN1?",
        options: ["Methyl", "Primary", "Secondary", "Tertiary"],
        answer: 3
      }
    ],
    sn2: [
      {
        question: "What happens to stereochemistry in SN2?",
        options: ["Retention", "Inversion", "Racemization", "No change"],
        answer: 1
      },
      {
        question: "Which solvent favors SN2 reactions?",
        options: ["Water", "Ethanol", "DMSO", "Hexane"],
        answer: 2
      }
    ],
    e1: [
      {
        question: "Which product is favored in E1 elimination?",
        options: ["Most substituted alkene", "Least substituted alkene", "Equal mixture", "Depends on the nucleophile"],
        answer: 0
      }
    ],
    e2: [
      {
        question: "What geometry is required for E2 elimination?",
        options: ["Syn-periplanar", "Anti-periplanar", "Syn-clinal", "Anti-clinal"],
        answer: 1
      }
    ],
    electrophilic: [
      {
        question: "Which rule predicts the major product in electrophilic addition?",
        options: ["Markovnikov's rule", "Zaitsev's rule", "Saytzeff's rule", "Hoffmann's rule"],
        answer: 0
      }
    ],
    nucleophilic: [
      {
        question: "Which is more reactive toward nucleophilic addition?",
        options: ["Aldehydes", "Ketones", "Esters", "Amides"],
        answer: 0
      }
    ],
    free_radical: [
      {
        question: "What initiates free radical substitution?",
        options: ["Heat", "UV light", "Base", "Acid"],
        answer: 1
      }
    ],
    pericyclic: [
  {
    question: "What is a key feature of pericyclic reactions?",
    options: [
      "Free radical intermediates",
      "Cyclic transition states",
      "Carbocation formation",
      "Base-catalyzed mechanism"
    ],
    answer: 1
  }
],
redox: [
  {
    question: "Which reagent is commonly used to reduce aldehydes to alcohols?",
```

```
        options: ["PCC", "KMnO4", "NaBH4", "HCl"],
        answer: 2
      }
    ],
    organometallic: [
      {
        question: "What solvent is typically used in Grignard reactions?",
        options: ["Water", "Ethanol", "Diethyl ether", "Acetic acid"],
        answer: 2
      }
    ]
};

    // App state
    let currentMechanism = 'sn1';
    let currentQuestion = 0;
    let viewer = null;

    // Initialize the page
    document.addEventListener("DOMContentLoaded", function() {
      initializeMoleculeViewer();
      setupEventListeners();
      loadMechanism('sn1');
    });

    // Initialize the 3D molecule viewer with proper error handling
    function initializeMoleculeViewer() {
      try {
        viewer = $3Dmol.createViewer(document.getElementById("moleculeViewer"), {
          backgroundColor: "white"
        });
        console.log("Molecule viewer initialized successfully");
      } catch (error) {
        console.error("Failed to initialize molecule viewer:", error);
        document.getElementById("moleculeViewer").innerHTML =
            "<p style='padding: 20px; color: #666;'>Molecule viewer could not be loaded. Please check your internet connection.</p>";
      }
    }

    // Set up event listeners
    function setupEventListeners() {
      // Set up mechanism selection
      document.querySelectorAll('#mechanismList li').forEach(item => {
        item.addEventListener('click', function() {
          const mechanism = this.getAttribute('data-mechanism');
          loadMechanism(mechanism);

          // Update active class
          document.querySelectorAll('#mechanismList li').forEach(li => {
            li.classList.remove('active');
          });
          this.classList.add('active');
        });
      });
    }

    // Load molecule data with error handling
    function loadMolecule(mechanismType) {
      if (!viewer || !MOLECULE_FILES[mechanismType]) {
        console.log("Viewer or molecule file not available");
        return;
      }

      try {
        viewer.clear();
        $.get(MOLECULE_FILES[mechanismType], function(data) {
```

```
            viewer.addModel(data, "pdb");
            viewer.setStyle({}, {stick: {radius: 0.2}});
            viewer.zoomTo();
            viewer.render();
        }).fail(function(error) {
            console.error("Failed to load molecule data:", error);
            document.getElementById("moleculeViewer").innerHTML =
                "<p style='padding: 20px; color: #666;'>Failed to load molecule data. Please try again later.</p>";
        });
    } catch (error) {
        console.error("Error in molecule loading:", error);
    }
}

// Load mechanism data
function loadMechanism(mechanismType) {
    if (!REACTIONS_DATA[mechanismType]) return;

    currentMechanism = mechanismType;
    currentQuestion = 0;
    const reaction = REACTIONS_DATA[mechanismType];

    // Update reaction display
    document.getElementById('reactionDisplay').innerHTML = `
        <h3>${reaction.name}</h3>
        <p>Select a different reaction from the sidebar to explore its mechanism</p>
    `;

    // Update mechanism steps
    const stepsHTML = reaction.steps.map(step => `<li>${step}</li>`).join('');
    document.getElementById('mechanismSteps').innerHTML = stepsHTML;

    // Update conditions and example
    document.getElementById('reactionConditions').innerHTML =
        `<strong>Reaction Conditions:</strong> ${reaction.conditions}`;
    document.getElementById('reactionExample').innerHTML =
        `<strong>Example:</strong> ${reaction.example}`;

    // Update key concepts
    const conceptsHTML = reaction.concepts.map(concept => `<p>• ${concept}</p>`).join('');
    document.getElementById('keyConcepts').innerHTML = conceptsHTML;

    // Load molecule
    loadMolecule(mechanismType);

    // Load quiz question
    loadQuizQuestion();
}

// Load quiz question
function loadQuizQuestion() {
    const quizzes = QUIZ_DATA[currentMechanism];
    if (!quizzes || quizzes.length === 0) {
        document.getElementById('quizQuestion').textContent = "No quiz available for this mechanism yet.";
        document.getElementById('quizOptions').style.display = "none";
        document.getElementById('quizFeedback').textContent = '';
        return;
    }

    document.getElementById('quizOptions').style.display = "block";
    const quiz = quizzes[currentQuestion];
    document.getElementById('quizQuestion').textContent = quiz.question;

    // Update options
    const optionsContainer = document.getElementById('quizOptions');
    optionsContainer.innerHTML = '';
```

```
        quiz.options.forEach((option, index) => {
          const button = document.createElement('button');
          button.setAttribute('data-index', index);
          button.textContent = option;

          // Add event listener directly to the button
          button.addEventListener('click', function() {
            checkAnswer(index);
          });

          optionsContainer.appendChild(button);
        });

        document.getElementById('quizFeedback').textContent = '';
        document.getElementById('quizFeedback').className = '';
      }

      // Check quiz answer
      function checkAnswer(selectedIndex) {
        const quizzes = QUIZ_DATA[currentMechanism];
        if (!quizzes || quizzes.length === 0) return;

        const quiz = quizzes[currentQuestion];
        const feedback = document.getElementById('quizFeedback');

        if (selectedIndex === quiz.answer) {
          feedback.textContent = "☑ Correct!";
          feedback.className = "correct";
        } else {
          feedback.textContent = "✗ Incorrect! The correct answer is: " + quiz.options[quiz.answer];
          feedback.className = "incorrect";
        }

        // Move to next question after delay
        setTimeout(() => {
          if (quizzes.length > 1) {
            currentQuestion = (currentQuestion + 1) % quizzes.length;
            loadQuizQuestion();
          }
        }, 1500);
      }
    </script>
  </body>
</html>
```

Module 8
kinetics_energy.html–

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Reaction Kinetics & Energy Diagrams</title>
    <meta name="description" content="Interactive Organic Chemistry Concepts" />

    <!-- Include Tailwind CSS via CDN -->
    <script src="https://cdn.tailwindcss.com"></script>

    <!-- Include Chart.js for visualizations -->
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

    <!-- Custom styles -->
```

```
<style>
 :root {
   --primary: #3b82f6;
   --secondary: #10b981;
   --accent: #8b5cf6;
   --background: #f8fafc;
   --foreground: #1e293b;
   --muted: #f1f5f9;
   --muted-foreground: #64748b;
 }

 body {
   font-family: system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif;
   color: var(--foreground);
   background-color: var(--background);
 }

 /* Back Button */
 .back-btn {
      position: absolute;
      top: 15px;
      left: 30px;
      text-decoration: none;
      font-size: 1.2rem;
      font-weight: bold;
      color: white;
      background: #333;
      padding: 10px 20px;
      border-radius: 8px;
      transition: background 0.3s ease;
   }

   .back-btn:hover {
      background: #555;
   }


 .tab-content {
   display: none;
 }

 .tab-content.active {
   display: block;
 }

 .btn {
   padding: 0.5rem 1rem;
   border-radius: 0.375rem;
   font-weight: 500;
   transition: all 0.2s;
 }

 .btn-primary {
   background-color: var(--primary);
   color: white;
 }

 .btn-primary:hover {
   background-color: #2563eb;
 }

 .btn-outline {
   background-color: transparent;
   border: 1px solid #e2e8f0;
 }

 .btn-outline:hover {
   background-color: #f8fafc;
 }

 .card {
```

```
  background-color: white;
  border-radius: 0.5rem;
  box-shadow: 0 1px 3px rgba(0, 0, 0, 0.1);
  border: 1px solid #e2e8f0;
}

.tabs-header button {
  padding: 0.75rem 1rem;
  font-weight: 500;
}

.tabs-header button.active {
  border-bottom: 2px solid var(--primary);
  color: var(--primary);
}

input[type="range"] {
  width: 100%;
}

.slider-container {
  margin: 1rem 0;
}

.slider-label {
  display: flex;
  justify-content: space-between;
  margin-bottom: 0.5rem;
}

.switch {
  position: relative;
  display: inline-block;
  width: 50px;
  height: 24px;
}

.switch input {
  opacity: 0;
  width: 0;
  height: 0;
}

.slider {
  position: absolute;
  cursor: pointer;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  background-color: #ccc;
  transition: .4s;
  border-radius: 24px;
}

.slider:before {
  position: absolute;
  content: "";
  height: 18px;
  width: 18px;
  left: 3px;
  bottom: 3px;
  background-color: white;
  transition: .4s;
  border-radius: 50%;
}

input:checked + .slider {
  background-color: var(--primary);
}
```

```
    input:checked + .slider:before {
      transform: translateX(26px);
    }
  </style>
</head>
<body>

  <!-- Back Button -->
  <a href="organic.html" class="back-btn">&#8592; Back</a>

  <header class="text-white py-6" style="background-color: #004466;">
    <div class="container mx-auto px-4">
      <div class="flex items-center justify-center">
        <div class="flex items-center space-x-2">
          <h1 class="text-3xl font-bold text-center" style="text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);">
            Reaction Kinetics & Energy Diagrams
          </h1>
        </div>
      </div>
    </div>
  </header>


<main class="container mx-auto px-4 py-8">
  <div class="mb-8 text-center">
    <h2 class="text-3xl font-bold mb-2">Visualize Organic Chemistry Concepts</h2>
    <p class="text-gray-600 max-w-2xl mx-auto">
      Explore reaction kinetics and energy diagrams through interactive visualizations. Adjust parameters
      and see how they affect reaction dynamics in real-time.
    </p>
  </div>

  <div class="tabs">
    <div class="tabs-header flex justify-center mb-6 overflow-x-auto">
      <div class="inline-flex bg-gray-100 rounded-lg p-1">
        <button class="tab-btn active px-4 py-2 rounded-md" data-tab="energy-diagrams">Energy Diagrams</button>
        <button class="tab-btn px-4 py-2 rounded-md" data-tab="reaction-rate">Reaction Rate</button>
        <button class="tab-btn px-4 py-2 rounded-md" data-tab="activation-energy">Catalysis</button>
        <button class="tab-btn px-4 py-2 rounded-md" data-tab="mechanism">Mechanisms</button>
        <button class="tab-btn px-4 py-2 rounded-md" data-tab="temperature">Temperature Effect</button>
      </div>
    </div>

    <div class="tab-content active" id="energy-diagrams">
      <div class="card p-6">
        <h2 class="text-xl font-semibold mb-4">Interactive Energy Diagrams</h2>
        <div class="grid md:grid-cols-2 gap-6">
          <div>
            <h3 class="text-lg font-medium mb-4">Reaction Energy Profile</h3>
            <div class="bg-white p-4 rounded-lg border h-[350px]">
              <canvas id="energyDiagramChart"></canvas>
            </div>
          </div>
          <div>
            <h3 class="text-lg font-medium mb-4">Adjust Parameters</h3>
            <div class="space-y-6 bg-gray-100 p-6 rounded-lg">
              <div class="space-y-2">
                <div class="slider-label">
                  <label for="temperature">Temperature (°C):</label>
                  <span id="temperature-value">25</span>
                </div>
                <input type="range" id="temperature" min="0" max="100" step="1" value="25">
                <p class="text-sm text-gray-600 mt-2">
                  Higher temperatures provide more thermal energy to overcome activation barriers.
                </p>
              </div>

              <div class="space-y-2">
                <div class="flex items-center space-x-2">
                  <label class="switch">
                    <input type="checkbox" id="catalyst-toggle">
```

```
                <span class="slider"></span>
              </label>
              <label for="catalyst-toggle">Add Catalyst</label>
            </div>
            <p class="text-sm text-gray-600 mt-2">
              Catalysts provide an alternative reaction pathway with a lower activation energy.
            </p>
          </div>

          <div class="bg-white p-4 rounded border mt-6">
            <h4 class="font-medium mb-2">What's happening?</h4>
            <p class="text-sm" id="reaction-explanation">
              Without a catalyst, the reaction must overcome a higher energy barrier.
              At lower temperatures, fewer molecules have enough energy to react.
            </p>
            <p class="text-sm mt-2" id="activation-energy-display">
              Activation Energy: 30.0 kJ/mol
            </p>
          </div>
        </div>
      </div>
    </div>
    <div class="bg-gray-100 p-4 rounded-lg mt-6">
      <h3 class="text-lg font-medium mb-2">About Energy Diagrams</h3>
      <p class="text-sm">
        Energy diagrams show how the potential energy changes during a chemical reaction. The y-axis represents
        potential energy, while the x-axis represents the reaction coordinate (progress of the reaction). The
        activation energy is the minimum energy required for a reaction to occur, shown as the energy barrier.
      </p>
    </div>
  </div>
</div>

<div class="tab-content" id="reaction-rate">
  <div class="card p-6">
    <h2 class="text-xl font-semibold mb-4">Reaction Rate vs. Concentration</h2>
    <div class="grid md:grid-cols-2 gap-6">
      <div>
        <h3 class="text-lg font-medium mb-4">Reaction Progress</h3>
        <div class="bg-white p-4 rounded-lg border h-[300px]">
          <canvas id="reactionProgressChart"></canvas>
        </div>
      </div>
      <div>
        <h3 class="text-lg font-medium mb-4">Rate vs Concentration</h3>
        <div class="bg-white p-4 rounded-lg border h-[300px]">
          <canvas id="rateVsConcentrationChart"></canvas>
        </div>
      </div>
    </div>
    <div class="bg-gray-100 p-6 rounded-lg mt-6">
      <h3 class="text-lg font-medium mb-4">Adjust Parameters</h3>
      <div class="grid md:grid-cols-2 gap-6">
        <div class="space-y-4">
          <div>
            <label class="block mb-2">Reaction Order</label>
            <select id="reaction-order" class="w-full px-3 py-2 border rounded">
              <option value="zero">Zero Order</option>
              <option value="first" selected>First Order</option>
              <option value="second">Second Order</option>
            </select>
          </div>

          <div class="space-y-2">
            <div class="slider-label">
              <label for="rate-constant">Rate Constant (k):</label>
              <span id="rate-constant-value">0.05</span>
            </div>
            <input type="range" id="rate-constant" min="0.01" max="0.1" step="0.01" value="0.05">
          </div>
        </div>
```

```
          <div class="bg-white p-4 rounded border">
            <h4 class="font-medium mb-2">Rate Law Explanation</h4>
            <div class="equation mb-2" id="rate-law-equation">
              Rate = k[A]
            </div>
            <p class="text-sm mb-2" id="rate-law-explanation">
              In first-order reactions, the rate is directly proportional to reactant concentration.
            </p>
            <p class="text-sm font-medium mt-3">Half-life:</p>
            <p class="text-sm" id="half-life-equation">
              t₁/₂ = 0.693/k
            </p>
          </div>
        </div>
      </div>
      <div class="bg-gray-100 p-4 rounded-lg mt-6">
        <h3 class="text-lg font-medium mb-2">About Reaction Rates</h3>
        <p class="text-sm">
          The rate of a reaction describes how quickly reactants are converted to products. It is influenced by
          concentration of reactants, temperature, presence of catalysts, and the nature of the reactants. The
          reaction order indicates how the rate depends on reactant concentration, as described by the rate law.
        </p>
      </div>
    </div>
  </div>

  <div class="tab-content" id="activation-energy">
    <div class="card p-6">
      <h2 class="text-xl font-semibold mb-4">Activation Energy & Catalysis</h2>
      <div class="flex justify-center space-x-4 mb-4">
        <button class="btn btn-primary pathway-toggle active" data-path="both">Show Both Pathways</button>
        <button class="btn btn-outline pathway-toggle" data-path="uncatalyzed">Uncatalyzed Only</button>
      </div>

      <div class="bg-white p-4 rounded-lg border h-[400px]">
        <canvas id="catalysisChart"></canvas>
      </div>

      <div class="grid md:grid-cols-2 gap-6 mt-6">
        <div class="card">
          <div class="p-6">
            <h3 class="text-lg font-medium mb-4">How Catalysts Work</h3>
            <p class="text-sm mb-4">
              Catalysts provide an alternative reaction pathway with a lower activation energy,
              increasing the reaction rate without being consumed.
            </p>
            <div class="flex items-center justify-center space-x-4 my-6">
              <div class="bg-red-100 text-red-600 p-3 rounded-lg text-center">
                <div class="font-bold">Without Catalyst</div>
                <div class="text-xl">Ea = 50 kJ/mol</div>
              </div>
                    <svg xmlns="http://www.w3.org/2000/svg" width="24" height="24" viewBox="0 0 24 24" fill="none"
stroke="currentColor" stroke-width="2" stroke-linecap="round" stroke-linejoin="round" class="lucide lucide-arrow-right
text-gray-400">
                <path d="M5 12h14"></path>
                <path d="m12 5 7 7-7 7"></path>
              </svg>
              <div class="bg-green-100 text-green-600 p-3 rounded-lg text-center">
                <div class="font-bold">With Catalyst</div>
                <div class="text-xl">Ea = 25 kJ/mol</div>
              </div>
            </div>
            <p class="text-sm mt-4">
              The catalyst forms temporary bonds with reactant molecules, orienting them in a way that
              makes it easier for them to react. This reduces the energy needed to reach the transition state.
            </p>
          </div>
        </div>

        <div class="card">
```

```
          <div class="p-6">
            <h3 class="text-lg font-medium mb-4">Catalytic Mechanisms</h3>
            <ul class="list-disc list-inside space-y-2 text-sm">
              <li>
                <span class="font-medium">Active site binding:</span> Reactants bind to specific sites on the catalyst
              </li>
              <li>
                <span class="font-medium">Bond weakening:</span> Catalysts weaken specific bonds in the reactants
              </li>
              <li>
                <span class="font-medium">Intermediate formation:</span> Forms stable intermediates with lower energy barriers
              </li>
              <li>
                <span class="font-medium">Proper orientation:</span> Positions molecules in optimal configurations for reaction
              </li>
              <li>
                <span class="font-medium">Electron transfer:</span> Facilitates the movement of electrons between molecules
              </li>
            </ul>
            <div class="bg-blue-50 p-3 rounded-lg mt-4">
              <p class="text-sm font-medium text-blue-700">
                Catalytic rate enhancement can be 10⁶ to 10¹² times faster than uncatalyzed reactions!
              </p>
            </div>
          </div>
        </div>
      </div>
    </div>

    <div class="tab-content" id="mechanism">
      <div class="card p-6">
        <h2 class="text-xl font-semibold mb-4">Mechanism Comparison</h2>
        <div class="tabs mechanism-tabs mb-4">
          <div class="flex justify-center space-x-4">
            <button class="btn btn-primary mech-tab-btn active" data-mech="sn">SN1 vs SN2</button>
            <button class="btn btn-outline mech-tab-btn" data-mech="e">E1 vs E2</button>
          </div>
        </div>

        <div class="mech-tab-content active" id="sn-tab">
          <div class="bg-white p-4 rounded-lg border h-[350px]">
            <canvas id="snMechanismChart"></canvas>
          </div>

          <div class="grid md:grid-cols-2 gap-6 mt-6">
            <div class="card">
              <div class="p-6">
                <h3 class="text-lg font-medium mb-3 text-purple-600">SN1 Mechanism</h3>
                <p class="text-sm mb-3">
                  <span class="font-bold">Unimolecular nucleophilic substitution</span> occurs in two steps:
                </p>
                <ol class="list-decimal list-inside text-sm space-y-2">
                  <li>Formation of a carbocation intermediate (slow, rate-determining)</li>
                  <li>Attack by the nucleophile (fast)</li>
                </ol>
                <div class="mt-4 p-3 bg-purple-50 rounded-md">
                  <p class="text-sm font-medium">Favored by:</p>
                  <ul class="list-disc list-inside text-sm mt-1">
                    <li>Tertiary or secondary substrates</li>
                    <li>Polar protic solvents</li>
                    <li>Good leaving groups</li>
                    <li>First-order kinetics: rate = k[substrate]</li>
                  </ul>
                </div>
              </div>
            </div>

            <div class="card">
              <div class="p-6">
                <h3 class="text-lg font-medium mb-3 text-pink-600">SN2 Mechanism</h3>
```

```
          <p class="text-sm mb-3">
            <span class="font-bold">Bimolecular nucleophilic substitution</span> occurs in a single concerted step:
          </p>
          <ul class="list-disc list-inside text-sm">
            <li>Nucleophile attacks from the back side</li>
            <li>Inversion of stereochemistry (Walden inversion)</li>
          </ul>
          <div class="mt-4 p-3 bg-pink-50 rounded-md">
            <p class="text-sm font-medium">Favored by:</p>
            <ul class="list-disc list-inside text-sm mt-1">
              <li>Primary or methyl substrates</li>
              <li>Polar aprotic solvents</li>
              <li>Strong nucleophiles</li>
              <li>Second-order kinetics: rate = k[substrate][nucleophile]</li>
            </ul>
          </div>
        </div>
      </div>
    </div>

    <div class="mech-tab-content hidden" id="e-tab">
      <div class="bg-white p-4 rounded-lg border h-[350px]">
        <canvas id="eMechanismChart"></canvas>
      </div>

      <div class="grid md:grid-cols-2 gap-6 mt-6">
        <div class="card">
          <div class="p-6">
            <h3 class="text-lg font-medium mb-3 text-blue-600">E1 Mechanism</h3>
            <p class="text-sm mb-3">
              <span class="font-bold">Unimolecular elimination</span> occurs in two steps:
            </p>
            <ol class="list-decimal list-inside text-sm space-y-2">
              <li>Formation of a carbocation intermediate (slow)</li>
              <li>Deprotonation by a base (fast)</li>
            </ol>
            <div class="mt-4 p-3 bg-blue-50 rounded-md">
              <p class="text-sm font-medium">Favored by:</p>
              <ul class="list-disc list-inside text-sm mt-1">
                <li>Tertiary substrates</li>
                <li>Polar protic solvents</li>
                <li>Weak bases</li>
                <li>Higher temperatures</li>
                <li>First-order kinetics: rate = k[substrate]</li>
              </ul>
            </div>
          </div>
        </div>

        <div class="card">
          <div class="p-6">
            <h3 class="text-lg font-medium mb-3 text-green-600">E2 Mechanism</h3>
            <p class="text-sm mb-3">
              <span class="font-bold">Bimolecular elimination</span> occurs in a single concerted step:
            </p>
            <ul class="list-disc list-inside text-sm">
              <li>Anti-periplanar arrangement required</li>
              <li>Base removes a proton as the leaving group departs</li>
            </ul>
            <div class="mt-4 p-3 bg-green-50 rounded-md">
              <p class="text-sm font-medium">Favored by:</p>
              <ul class="list-disc list-inside text-sm mt-1">
                <li>Secondary or tertiary substrates</li>
                <li>Strong, bulky bases</li>
                <li>Good leaving groups</li>
                <li>Higher temperatures</li>
                <li>Second-order kinetics: rate = k[substrate][base]</li>
              </ul>
            </div>
          </div>
```

```
          </div>
        </div>
      </div>

      <div class="bg-gray-100 p-4 rounded-lg mt-6">
        <h3 class="text-lg font-medium mb-2">Mechanism Selection Factors</h3>
        <p class="text-sm">
          The reaction pathway (SN1, SN2, E1, or E2) is determined by several factors including
          substrate structure, nucleophile/base strength, solvent, temperature, and steric effects.
          Understanding energy profiles helps predict which mechanism will dominate under given conditions.
        </p>
      </div>
    </div>
  </div>

  <div class="tab-content" id="temperature">
    <div class="card p-6">
      <h2 class="text-xl font-semibold mb-4">Temperature Dependence (Arrhenius Equation)</h2>
      <div class="grid md:grid-cols-2 gap-6">
        <div>
          <h3 class="text-lg font-medium mb-4">Arrhenius Equation</h3>

          <div class="bg-white p-6 rounded-lg border text-center">
            <div class="equation mb-4 text-2xl">k = A·e<sup>-Ea/RT</sup></div>
            <div class="grid grid-cols-3 gap-4 text-sm">
              <div>
                <p class="font-medium">k</p>
                <p>Rate Constant</p>
              </div>
              <div>
                <p class="font-medium">A</p>
                <p>Pre-exponential Factor</p>
              </div>
              <div>
                <p class="font-medium">Ea</p>
                <p>Activation Energy</p>
              </div>
            </div>
            <div class="grid grid-cols-2 gap-4 mt-4 text-sm">
              <div>
                <p class="font-medium">R</p>
                <p>Gas Constant (8.314 J/mol·K)</p>
              </div>
              <div>
                <p class="font-medium">T</p>
                <p>Temperature (Kelvin)</p>
              </div>
            </div>
          </div>

          <div class="mt-6 bg-gray-100 p-4 rounded-lg">
            <h4 class="font-medium mb-2">About the Arrhenius Equation</h4>
            <p class="text-sm">
              The Arrhenius equation describes the temperature dependence of reaction rates. It shows that
              reaction rates increase exponentially with temperature, and that reactions with higher activation
              energies are more sensitive to temperature changes.
            </p>
          </div>

          <div class="mt-6">
            <h4 class="font-medium mb-4">Results from Arrhenius Plot</h4>
            <div class="bg-blue-50 p-4 rounded-lg border border-blue-100">
              <p class="text-sm mb-2">
                <span class="font-medium">Calculated Activation Energy:</span> <span id="calculated-ea">50.00</span> kJ/mol
              </p>
              <p class="text-sm mb-2">
                <span class="font-medium">Current Rate Constant (k):</span> <span id="current-rate">2.21 × 10^-9</span>
              </p>
              <p class="text-sm">
                <span class="font-medium">Temperature Effect:</span> Increasing temperature by 10°C typically
                increases reaction rate by a factor of 2-4 for reactions with activation energies in the range of
```

```
              50-60 kJ/mol.
            </p>
          </div>
        </div>
      </div>

      <div>
        <div class="space-y-6">
          <div>
            <h3 class="text-lg font-medium mb-4">Rate vs Temperature</h3>
            <div class="bg-white p-4 rounded-lg border h-[200px]">
              <canvas id="rateVsTempChart"></canvas>
            </div>
          </div>

          <div>
            <h3 class="text-lg font-medium mb-4">Arrhenius Plot (ln(k) vs 1/T)</h3>
            <div class="bg-white p-4 rounded-lg border h-[200px]">
              <canvas id="arrheniusPlot"></canvas>
            </div>
          </div>

          <div class="bg-gray-100 p-4 rounded-lg">
            <h4 class="font-medium mb-3">Adjust Parameters</h4>
            <div class="space-y-4">
              <div class="space-y-2">
                <div class="slider-label">
                  <label for="temp-slider">Temperature (K):</label>
                  <span id="temp-slider-value">298</span>
                </div>
                <input type="range" id="temp-slider" min="273" max="373" step="1" value="298">
              </div>

              <div class="space-y-2">
                <div class="slider-label">
                  <label for="ea-slider">Activation Energy (kJ/mol):</label>
                  <span id="ea-slider-value">50</span>
                </div>
                <input type="range" id="ea-slider" min="20" max="100" step="1" value="50">
              </div>

              <div class="space-y-2">
                <div class="slider-label">
                  <label for="a-slider">Pre-exponential Factor:</label>
                  <span id="a-slider-value">1.00 × 10^10</span>
                </div>
                <input type="range" id="a-slider" min="7" max="13" step="0.1" value="10">
                <p class="text-xs text-gray-500">
                  Sliding changes the exponent (10^x)
                </p>
              </div>
            </div>
          </div>
        </div>
      </div>
    </div>
  </div>
</main>



<!-- Include the GPTE script tag for use with Lovable -->
<script src="https://cdn.gpteng.co/gptengineer.js" type="module"></script>

<!-- JavaScript for the interactive features -->
<script>
  document.addEventListener('DOMContentLoaded', function() {
    // Tab switching functionality
    const tabBtns = document.querySelectorAll('.tab-btn');
```

```
    const tabContents = document.querySelectorAll('.tab-content');

    tabBtns.forEach(btn => {
      btn.addEventListener('click', () => {
        const tabId = btn.getAttribute('data-tab');

        // Remove active class from all buttons and tabs
        tabBtns.forEach(b => b.classList.remove('active'));
        tabContents.forEach(c => c.classList.remove('active'));

        // Add active class to current button and tab
        btn.classList.add('active');
        document.getElementById(tabId).classList.add('active');

        // Initialize or resize charts when tab is shown
        initializeCharts();
      });
    });

    // Mechanism tab switching
    const mechTabBtns = document.querySelectorAll('.mech-tab-btn');
    const mechTabContents = document.querySelectorAll('.mech-tab-content');

    mechTabBtns.forEach(btn => {
      btn.addEventListener('click', () => {
        const mechId = btn.getAttribute('data-mech');

        // Remove active class from all buttons and tabs
        mechTabBtns.forEach(b => {
          b.classList.remove('active');
          b.classList.remove('btn-primary');
          b.classList.add('btn-outline');
        });
        mechTabContents.forEach(c => c.classList.add('hidden'));

        // Add active class to current button and tab
        btn.classList.add('active');
        btn.classList.add('btn-primary');
        btn.classList.remove('btn-outline');

        document.getElementById(mechId + '-tab').classList.remove('hidden');
        document.getElementById(mechId + '-tab').classList.add('active');

        // Update the mechanism charts
        updateMechanismCharts();
      });
    });

    // Pathway toggle for catalysis
    const pathwayBtns = document.querySelectorAll('.pathway-toggle');
    let showCatalyst = true;

    pathwayBtns.forEach(btn => {
      btn.addEventListener('click', () => {
        const path = btn.getAttribute('data-path');

        pathwayBtns.forEach(b => {
          b.classList.remove('active');
          b.classList.remove('btn-primary');
          b.classList.add('btn-outline');
        });

        btn.classList.add('active');
        btn.classList.add('btn-primary');
        btn.classList.remove('btn-outline');

        showCatalyst = path === 'both';
        updateCatalysisChart();
      });
    });
```

```
// Energy diagram parameters
const tempSlider = document.getElementById('temperature');
const tempValue = document.getElementById('temperature-value');
const catalystToggle = document.getElementById('catalyst-toggle');
const reactionExplanation = document.getElementById('reaction-explanation');
const activationEnergyDisplay = document.getElementById('activation-energy-display');

tempSlider.addEventListener('input', () => {
  tempValue.textContent = tempSlider.value;
  updateEnergyDiagram();
});

catalystToggle.addEventListener('change', () => {
  updateEnergyDiagram();
});

// Reaction rate parameters
const orderSelect = document.getElementById('reaction-order');
const rateConstantSlider = document.getElementById('rate-constant');
const rateConstantValue = document.getElementById('rate-constant-value');
const rateLawEquation = document.getElementById('rate-law-equation');
const rateLawExplanation = document.getElementById('rate-law-explanation');
const halfLifeEquation = document.getElementById('half-life-equation');

orderSelect.addEventListener('change', () => {
  updateRateLaws();
  updateReactionRateCharts();
});

rateConstantSlider.addEventListener('input', () => {
  rateConstantValue.textContent = parseFloat(rateConstantSlider.value).toFixed(3);
  updateReactionRateCharts();
});

// Temperature dependence parameters
const tempDepSlider = document.getElementById('temp-slider');
const tempDepValue = document.getElementById('temp-slider-value');
const eaSlider = document.getElementById('ea-slider');
const eaValue = document.getElementById('ea-slider-value');
const aSlider = document.getElementById('a-slider');
const aValue = document.getElementById('a-slider-value');

tempDepSlider.addEventListener('input', () => {
  tempDepValue.textContent = tempDepSlider.value;
  updateTemperatureCharts();
});

eaSlider.addEventListener('input', () => {
  eaValue.textContent = eaSlider.value;
  updateTemperatureCharts();
});

aSlider.addEventListener('input', () => {
  const value = Math.pow(10, parseFloat(aSlider.value));
  aValue.textContent = formatScientific(value);
  updateTemperatureCharts();
});

// Chart objects
let energyDiagramChart;
let reactionProgressChart;
let rateVsConcentrationChart;
let catalysisChart;
let snMechanismChart;
let eMechanismChart;
let rateVsTempChart;
let arrheniusPlot;

// Chart initialization function
function initializeCharts() {
  initEnergyDiagram();
```

```
  initReactionRateCharts();
  initCatalysisChart();
  initMechanismCharts();
  initTemperatureCharts();
}

// Energy Diagram Chart
function initEnergyDiagram() {
  const ctx = document.getElementById('energyDiagramChart').getContext('2d');

  if (energyDiagramChart) {
    energyDiagramChart.destroy();
  }

  energyDiagramChart = new Chart(ctx, {
    type: 'line',
    data: {
      labels: ['Reactants', 'Transition State', 'Products'],
      datasets: [{
        label: 'Energy Profile',
        data: [0, 30, -5],
        backgroundColor: 'rgba(59, 130, 246, 0.2)',
        borderColor: 'rgba(59, 130, 246, 1)',
        borderWidth: 3,
        tension: 0.4,
        fill: true
      }]
    },
    options: {
      responsive: true,
      maintainAspectRatio: false,
      scales: {
        y: {
          title: {
            display: true,
            text: 'Energy (kJ/mol)'
          },
          min: -10,
          max: 40
        },
        x: {
          title: {
            display: true,
            text: 'Reaction Coordinate'
          }
        }
      },
      plugins: {
        tooltip: {
          callbacks: {
            label: function(context) {
              return `Energy: ${context.raw} kJ/mol`;
            }
          }
        }
      }
    }
  });

  updateEnergyDiagram();
}

// Update Energy Diagram based on parameters
function updateEnergyDiagram() {
  const temperature = parseInt(tempSlider.value);
  const catalystPresent = catalystToggle.checked;

  // Base activation energy
  let activationEnergy = 30;

  // Temperature effect: higher temperature slightly lowers the relative perception of the barrier
```

```
    const tempFactor = 1 - (temperature - 25) / 200; // Normalized effect

    // Catalyst effect: significantly lowers activation energy
    const catalystFactor = catalystPresent ? 0.6 : 1;

    // Adjusted activation energy
    const adjustedActivationEnergy = activationEnergy * tempFactor * catalystFactor;

    // Generate data points
    const data = [];
    data.push(0); // Reactants
    data.push(adjustedActivationEnergy); // Transition state

    // Intermediate (if applicable)
    if (adjustedActivationEnergy > 15) {
      energyDiagramChart.data.labels = ['Reactants', 'Transition State', 'Intermediate', 'Products'];
      data.push(adjustedActivationEnergy * 0.7); // Intermediate
      data.push(-5); // Products
    } else {
      energyDiagramChart.data.labels = ['Reactants', 'Transition State', 'Products'];
      data.push(-5); // Products
    }

    energyDiagramChart.data.datasets[0].data = data;
    energyDiagramChart.options.scales.y.max = Math.max(40, adjustedActivationEnergy + 10);
    energyDiagramChart.update();

    // Update explanation and display
    activationEnergyDisplay.textContent = `Activation Energy: ${adjustedActivationEnergy.toFixed(1)} kJ/mol`;

    let explanation = '';
    if (catalystPresent) {
      explanation += "The catalyst is lowering the activation energy by providing an alternative reaction pathway. ";
    } else {
      explanation += "Without a catalyst, the reaction must overcome a higher energy barrier. ";
    }

    if (temperature > 50) {
      explanation += "Higher temperature provides more molecules with sufficient energy to overcome the barrier.";
    } else {
      explanation += "At lower temperatures, fewer molecules have enough energy to react.";
    }

    reactionExplanation.textContent = explanation;
  }

  // Reaction Rate Charts
  function initReactionRateCharts() {
    const progressCtx = document.getElementById('reactionProgressChart').getContext('2d');
    const rateCtx = document.getElementById('rateVsConcentrationChart').getContext('2d');

    if (reactionProgressChart) {
      reactionProgressChart.destroy();
    }

    if (rateVsConcentrationChart) {
      rateVsConcentrationChart.destroy();
    }

    reactionProgressChart = new Chart(progressCtx, {
      type: 'line',
      data: {
        labels: [], // Will be filled with time points
        datasets: [
          {
            label: '[A] (mol/L)',
            borderColor: 'rgba(59, 130, 246, 1)',
            backgroundColor: 'rgba(59, 130, 246, 0.2)',
            data: [],
            borderWidth: 2,
            fill: false
```

```
        },
        {
          label: 'Reaction Rate (scaled)',
          borderColor: 'rgba(16, 185, 129, 1)',
          backgroundColor: 'rgba(16, 185, 129, 0.2)',
          data: [],
          borderWidth: 2,
          borderDash: [5, 5],
          fill: false
        }
      ]
    },
    options: {
      responsive: true,
      maintainAspectRatio: false,
      scales: {
        y: {
          beginAtZero: true,
          title: {
            display: true,
            text: 'Concentration (M)'
          },
          max: 1
        },
        x: {
          title: {
            display: true,
            text: 'Time'
          }
        }
      }
    }
  });

  rateVsConcentrationChart = new Chart(rateCtx, {
    type: 'line',
    data: {
      labels: [], // Will be filled with concentration points
      datasets: [
        {
          label: 'Rate',
          borderColor: 'rgba(139, 92, 246, 1)',
          backgroundColor: 'rgba(139, 92, 246, 0.2)',
          data: [],
          borderWidth: 3,
          fill: false
        }
      ]
    },
    options: {
      responsive: true,
      maintainAspectRatio: false,
      scales: {
        y: {
          beginAtZero: true,
          title: {
            display: true,
            text: 'Rate'
          }
        },
        x: {
          title: {
            display: true,
            text: 'Concentration (M)'
          },
          max: 1
        }
      }
    }
  });
```

```
        updateRateLaws();
        updateReactionRateCharts();
      }

      // Update rate law displays based on reaction order
      function updateRateLaws() {
        const reactionOrder = orderSelect.value;

        switch (reactionOrder) {
          case 'zero':
            rateLawEquation.textContent = 'Rate = k';
            rateLawExplanation.textContent = 'In zero-order reactions, the rate is independent of reactant concentration.';
            halfLifeEquation.textContent = 't₁/₂ = [A]₀/(2k)';
            break;
          case 'first':
            rateLawEquation.textContent = 'Rate = k[A]';
            rateLawExplanation.textContent = 'In first-order reactions, the rate is directly proportional to reactant concentration.';
            halfLifeEquation.textContent = 't₁/₂ = 0.693/k';
            break;
          case 'second':
            rateLawEquation.textContent = 'Rate = k[A]²';
              rateLawExplanation.textContent = 'In second-order reactions, the rate is proportional to the square of reactant
concentration.';
            halfLifeEquation.textContent = 't₁/₂ = 1/(k[A]₀)';
            break;
        }
      }

      // Update reaction rate charts based on parameters
      function updateReactionRateCharts() {
        const reactionOrder = orderSelect.value;
        const rateConstant = parseFloat(rateConstantSlider.value);
        const initialConcentration = 1.0;

        // Generate concentration data
        const concentrations = [];
        const rates = [];
        const times = [];

        for (let time = 0; time <= 50; time += 2) {
          let concentration;
          let rate;

          // Calculate concentration based on reaction order
          switch (reactionOrder) {
            case 'zero':
              // [A] = [A]₀ - kt
              concentration = Math.max(0, initialConcentration - rateConstant * time);
              rate = rateConstant; // independent of concentration
              break;
            case 'first':
              // [A] = [A]₀ e^(-kt)
              concentration = initialConcentration * Math.exp(-rateConstant * time);
              rate = rateConstant * concentration;
              break;
            case 'second':
              // 1/[A] = 1/[A]₀ + kt
              concentration = 1 / (1/initialConcentration + rateConstant * time);
              rate = rateConstant * concentration * concentration;
              break;
            default:
              concentration = initialConcentration;
              rate = 0;
          }

          times.push(time);
          concentrations.push(parseFloat(concentration.toFixed(4)));
          rates.push(parseFloat((rate * 100).toFixed(4))); // Scale rate for better visualization
        }

        // Update progress chart
```

```
    reactionProgressChart.data.labels = times;
    reactionProgressChart.data.datasets[0].data = concentrations;
    reactionProgressChart.data.datasets[1].data = rates;
    reactionProgressChart.update();

    // Generate rate vs concentration data
    const concPoints = [];
    const ratePoints = [];
    const labels = [];

    for (let conc = 0; conc <= 1; conc += 0.05) {
      let rate = 0;

      // Calculate rate based on reaction order
      switch (reactionOrder) {
        case 'zero':
          rate = rateConstant;
          break;
        case 'first':
          rate = rateConstant * conc;
          break;
        case 'second':
          rate = rateConstant * conc * conc;
          break;
        default:
          rate = 0;
      }

      labels.push(parseFloat(conc.toFixed(2)));
      concPoints.push(parseFloat(conc.toFixed(2)));
      ratePoints.push(parseFloat((rate * 100).toFixed(4))); // Scale rate for better visualization
    }

    // Update rate vs concentration chart
    rateVsConcentrationChart.data.labels = labels;
    rateVsConcentrationChart.data.datasets[0].data = ratePoints;
    rateVsConcentrationChart.update();
  }

  // Catalysis Chart
  function initCatalysisChart() {
    const ctx = document.getElementById('catalysisChart').getContext('2d');

    if (catalysisChart) {
      catalysisChart.destroy();
    }

    catalysisChart = new Chart(ctx, {
      type: 'line',
      data: {
        labels: [], // Will be filled with reaction coordinate points
        datasets: [
          {
            label: 'Uncatalyzed Path',
            borderColor: 'rgba(244, 63, 94, 1)',
            backgroundColor: 'rgba(244, 63, 94, 0.1)',
            data: [],
            borderWidth: 3,
            fill: true
          },
          {
            label: 'Catalyzed Path',
            borderColor: 'rgba(16, 185, 129, 1)',
            backgroundColor: 'rgba(16, 185, 129, 0.1)',
            data: [],
            borderWidth: 3,
            fill: true
          }
        ]
      },
      options: {
```

```
        responsive: true,
        maintainAspectRatio: false,
        scales: {
          y: {
            title: {
              display: true,
              text: 'Energy (kJ/mol)'
            },
            min: -20,
            max: 60
          },
          x: {
            title: {
              display: true,
              text: 'Reaction Coordinate'
            }
          }
        },
        plugins: {
          annotation: {
            annotations: {
              uncatalyzedEa: {
                type: 'line',
                mode: 'vertical',
                scaleID: 'x',
                value: 50,
                borderColor: 'rgba(244, 63, 94, 0.7)',
                borderWidth: 2,
                borderDash: [5, 5],
                label: {
                  content: 'Ea (uncatalyzed)',
                  enabled: true,
                  position: 'center'
                }
              },
              catalyzedEa: {
                type: 'line',
                mode: 'vertical',
                scaleID: 'x',
                value: 25,
                borderColor: 'rgba(16, 185, 129, 0.7)',
                borderWidth: 2,
                borderDash: [5, 5],
                label: {
                  content: 'Ea (catalyzed)',
                  enabled: true,
                  position: 'center'
                }
              }
            }
          }
        }
      });

      updateCatalysisChart();
    }

    // Update catalysis chart
    function updateCatalysisChart() {
      const uncatalyzedEa = 50; // Activation energy without catalyst
      const catalyzedEa = 25; // Activation energy with catalyst

      const xPoints = [];
      const uncatalyzedPoints = [];
      const catalyzedPoints = [];

      // Generate points along reaction coordinate (0-100)
      for (let x = 0; x <= 100; x++) {
        xPoints.push(x);
```

```
    // Uncatalyzed reaction curve
    if (x < 50) {
      // Rising part of the curve (from reactants to transition state)
      uncatalyzedPoints.push(((x / 50) ** 2) * uncatalyzedEa);
    } else {
      // Falling part of the curve (from transition state to products)
      uncatalyzedPoints.push(uncatalyzedEa - (((x - 50) / 50) ** 1.5) * (uncatalyzedEa + 15));
    }

    // Catalyzed reaction curve (more complex path through different intermediates)
    if (x < 25) {
      // First rising part
      catalyzedPoints.push(((x / 25) ** 2) * catalyzedEa);
    } else if (x < 40) {
      // First falling part to intermediate
      catalyzedPoints.push(catalyzedEa - (((x - 25) / 15) ** 1.5) * 12);
    } else if (x < 60) {
      // Second rising part from intermediate
      catalyzedPoints.push(catalyzedEa - 12 + ((x - 40) / 20) * 10);
    } else {
      // Final falling part to products
      catalyzedPoints.push(catalyzedEa - 2 - (((x - 60) / 40) ** 1.5) * (catalyzedEa + 13));
    }
  }

  catalysisChart.data.labels = xPoints;
  catalysisChart.data.datasets[0].data = uncatalyzedPoints;

  if (showCatalyst) {
    catalysisChart.data.datasets[1].data = catalyzedPoints;
    catalysisChart.data.datasets[1].hidden = false;
  } else {
    catalysisChart.data.datasets[1].hidden = true;
  }

  catalysisChart.update();
}

// Mechanism Comparison Charts
function initMechanismCharts() {
  // SN1 vs SN2 chart
  const snCtx = document.getElementById('snMechanismChart').getContext('2d');

  if (snMechanismChart) {
    snMechanismChart.destroy();
  }

  // Data for SN1 reaction energy profile
  const sn1Data = [
    { x: 0, y: 0, label: "Reactants" },
    { x: 30, y: 45, label: "TS1" },
    { x: 60, y: 30, label: "Carbocation Intermediate" },
    { x: 80, y: 40, label: "TS2" },
    { x: 100, y: -10, label: "Products" }
  ];

  // Data for SN2 reaction energy profile
  const sn2Data = [
    { x: 0, y: 0, label: "Reactants" },
    { x: 50, y: 35, label: "Transition State" },
    { x: 100, y: -5, label: "Products" }
  ];

  snMechanismChart = new Chart(snCtx, {
    type: 'line',
    data: {
      labels: ['0%', '25%', '50%', '75%', '100%'],
      datasets: [
        {
          label: 'SN1',
          data: [0, 45, 30, 40, -10],
```

```
        borderColor: 'rgba(139, 92, 246, 1)',
        backgroundColor: 'rgba(139, 92, 246, 0.1)',
        borderWidth: 3,
        fill: false,
        tension: 0.4,
        pointRadius: 5
      },
      {
        label: 'SN2',
        data: [0, 20, 35, 15, -5],
        borderColor: 'rgba(236, 72, 153, 1)',
        backgroundColor: 'rgba(236, 72, 153, 0.1)',
        borderWidth: 3,
        borderDash: [5, 5],
        fill: false,
        tension: 0.4,
        pointRadius: 5
      }
    ]
  },
  options: {
    responsive: true,
    maintainAspectRatio: false,
    scales: {
      y: {
        title: {
          display: true,
          text: 'Energy (kJ/mol)'
        },
        min: -20,
        max: 50
      },
      x: {
        title: {
          display: true,
          text: 'Reaction Coordinate'
        }
      }
    },
    plugins: {
      tooltip: {
        callbacks: {
          label: function(context) {
            const datasetLabel = context.dataset.label;
            const value = context.raw;
            return `${datasetLabel} Energy: ${value} kJ/mol`;
          }
        }
      }
    }
  }
});

// E1 vs E2 chart
const eCtx = document.getElementById('eMechanismChart').getContext('2d');

if (eMechanismChart) {
  eMechanismChart.destroy();
}

// Data for E1 reaction energy profile
const e1Data = [
  { x: 0, y: 0, label: "Reactants" },
  { x: 30, y: 42, label: "TS1" },
  { x: 60, y: 25, label: "Carbocation Intermediate" },
  { x: 80, y: 38, label: "TS2" },
  { x: 100, y: -15, label: "Products" }
];

// Data for E2 reaction energy profile
const e2Data = [
```

```
    { x: 0, y: 0, label: "Reactants" },
    { x: 50, y: 40, label: "Transition State" },
    { x: 100, y: -20, label: "Products" }
  ];

  eMechanismChart = new Chart(eCtx, {
    type: 'line',
    data: {
      labels: ['0%', '25%', '50%', '75%', '100%'],
      datasets: [
        {
          label: 'E1',
          data: [0, 42, 25, 38, -15],
          borderColor: 'rgba(14, 165, 233, 1)',
          backgroundColor: 'rgba(14, 165, 233, 0.1)',
          borderWidth: 3,
          fill: false,
          tension: 0.4,
          pointRadius: 5
        },
        {
          label: 'E2',
          data: [0, 20, 40, 20, -20],
          borderColor: 'rgba(16, 185, 129, 1)',
          backgroundColor: 'rgba(16, 185, 129, 0.1)',
          borderWidth: 3,
          borderDash: [5, 5],
          fill: false,
          tension: 0.4,
          pointRadius: 5
        }
      ]
    },
    options: {
      responsive: true,
      maintainAspectRatio: false,
      scales: {
        y: {
          title: {
            display: true,
            text: 'Energy (kJ/mol)'
          },
          min: -25,
          max: 50
        },
        x: {
          title: {
            display: true,
            text: 'Reaction Coordinate'
          }
        }
      },
      plugins: {
        tooltip: {
          callbacks: {
            label: function(context) {
              const datasetLabel = context.dataset.label;
              const value = context.raw;
              return `${datasetLabel} Energy: ${value} kJ/mol`;
            }
          }
        }
      }
    }
  });
}

// Update mechanism charts
function updateMechanismCharts() {
  if (snMechanismChart) snMechanismChart.update();
  if (eMechanismChart) eMechanismChart.update();
```

```
}

// Temperature Dependence Charts
function initTemperatureCharts() {
 const rateVsTempCtx = document.getElementById('rateVsTempChart').getContext('2d');
 const arrheniusCtx = document.getElementById('arrheniusPlot').getContext('2d');

 if (rateVsTempChart) {
  rateVsTempChart.destroy();
 }

 if (arrheniusPlot) {
  arrheniusPlot.destroy();
 }

 rateVsTempChart = new Chart(rateVsTempCtx, {
  type: 'line',
  data: {
   labels: [], // Will be filled with temperature points
   datasets: [{
    label: 'Rate',
    data: [],
    borderColor: 'rgba(59, 130, 246, 1)',
    backgroundColor: 'rgba(59, 130, 246, 0.1)',
    borderWidth: 2,
    fill: false
   }]
  },
  options: {
   responsive: true,
   maintainAspectRatio: false,
   scales: {
    y: {
     type: 'logarithmic',
     title: {
      display: true,
      text: 'Rate'
     }
    },
    x: {
     title: {
      display: true,
      text: 'Temperature (K)'
     }
    }
   }
  }
 });

 arrheniusPlot = new Chart(arrheniusCtx, {
  type: 'line',
  data: {
   labels: [], // Will be filled with 1/T points
   datasets: [{
    label: 'ln(k)',
    data: [],
    borderColor: 'rgba(139, 92, 246, 1)',
    backgroundColor: 'rgba(139, 92, 246, 0.1)',
    borderWidth: 2,
    fill: false
   }]
  },
  options: {
   responsive: true,
   maintainAspectRatio: false,
   scales: {
    y: {
     title: {
      display: true,
      text: 'ln(k)'
     }
```

```
        },
        x: {
          title: {
            display: true,
            text: '1000/T (K⁻¹)'
          }
        }
      }
    }
  });

  updateTemperatureCharts();
}

// Update temperature dependence charts
function updateTemperatureCharts() {
  const R = 8.314; // Gas constant in J/(mol·K)
  const temperature = parseInt(tempDepSlider.value);
  const activationEnergy = parseInt(eaSlider.value); // kJ/mol
  const preExpFactor = Math.pow(10, parseFloat(aSlider.value));

  // Calculate reaction rate using Arrhenius equation
  const calculateRate = (temp, ea, a) => {
    const eaInJ = ea * 1000; // Convert kJ to J
    return a * Math.exp(-eaInJ / (R * temp));
  };

  // Calculate current rate
  const rate = calculateRate(temperature, activationEnergy, preExpFactor);
  document.getElementById('current-rate').textContent = formatScientific(rate);

  // Generate data for rate vs temperature graph
  const temps = [];
  const rates = [];
  const logRates = [];
  const inverseTemps = [];

  for (let temp = 273; temp <= 373; temp += 5) {
    const rate = calculateRate(temp, activationEnergy, preExpFactor);
    temps.push(temp);
    rates.push(rate);
    logRates.push(Math.log(rate));
    inverseTemps.push((1 / temp) * 1000); // 1/T in 1/K (multiplied by 1000 for better visualization)
  }

  // Update rate vs temperature chart
  rateVsTempChart.data.labels = temps;
  rateVsTempChart.data.datasets[0].data = rates;

  // Add reference line for current temperature
  const currentTempIndex = temps.indexOf(parseInt(temperature));
  if (currentTempIndex >= 0) {
    rateVsTempChart.options.plugins = rateVsTempChart.options.plugins || {};
    rateVsTempChart.options.plugins.annotation = {
      annotations: {
        currentTemp: {
          type: 'line',
          mode: 'vertical',
          scaleID: 'x',
          value: temperature,
          borderColor: 'red',
          borderWidth: 2,
          borderDash: [5, 5],
          label: {
            content: 'Current T',
            enabled: true,
            position: 'top'
          }
        }
      }
    };
```

```
        }

        rateVsTempChart.update();

        // Update Arrhenius plot
        arrheniusPlot.data.labels = inverseTemps;
        arrheniusPlot.data.datasets[0].data = logRates;

        // Calculate activation energy from slope of Arrhenius plot (simple linear regression)
        const n = inverseTemps.length;
        let sumX = 0, sumY = 0, sumXY = 0, sumX2 = 0;

        for (let i = 0; i < n; i++) {
          sumX += inverseTemps[i];
          sumY += logRates[i];
          sumXY += inverseTemps[i] * logRates[i];
          sumX2 += inverseTemps[i] * inverseTemps[i];
        }

        const slope = (n * sumXY - sumX * sumY) / (n * sumX2 - sumX * sumX);

        // Ea = -slope * R
        // The slope is multiplied by 1000 because inverse_temp was multiplied by 1000
        const calculatedEa = -slope * R / 1000;

        document.getElementById('calculated-ea').textContent = calculatedEa.toFixed(2);

        arrheniusPlot.update();
      }

      // Format large numbers in scientific notation
      function formatScientific(value) {
        if (value === 0) return '0';

        const exponent = Math.floor(Math.log10(Math.abs(value)));
        const mantissa = value / Math.pow(10, exponent);

        if (Math.abs(exponent) < 3) {
          return value.toFixed(2);
        }

        return `${mantissa.toFixed(2)} × 10^${exponent}`;
      }

      // Initialize all charts on page load
      initializeCharts();

      // Handle window resize to redraw charts
      window.addEventListener('resize', initializeCharts);
    });
  </script>
 </body>
</html>
```

Module 8
stereochemistry.html–

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Stereochemistry Virtual Lab</title>
  <style>
```

```
:root {
    --primary: #3498db;
    --secondary: #2ecc71;
    --accent: #e74c3c;
    --light: #f5f5f5;
    --dark: #2c3e50;
    --shadow: 0 4px 8px rgba(0,0,0,0.1);
}

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
}

body {
    background-color: var(--light);
    color: var(--dark);
    line-height: 1.6;
}

header {
    background-color: #004466;
    color: white;
    padding: 1.5rem;
    text-align: center;
    box-shadow: var(--shadow);
}

h1 {
    margin-bottom: 0.5rem;
    font-size: 3rem; /* Increased text size */
    text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);
}


 /* Back Button */
  .back-btn {
    position: absolute;
    top: 30px;
    left: 20px;
    text-decoration: none;
    font-size: 1.2rem;
    font-weight: bold;
    color: white;
    background: #333;
    padding: 10px 20px;
    border-radius: 8px;
    transition: background 0.3s ease;
}

.back-btn:hover {
    background: #555;
}

nav {
    background-color: white;
    padding: 1rem;
    box-shadow: var(--shadow);
    position: sticky;
    top: 0;
    z-index: 100;
}

.nav-links {
    display: flex;
    justify-content: center;
    list-style: none;
    gap: 1.5rem;
    flex-wrap: wrap;
```

```
}

.nav-links li a {
    text-decoration: none;
    color: var(--dark);
    font-weight: 600;
    font-size: 1.1rem;
    padding: 0.5rem 1rem;
    border-radius: 4px;
    transition: all 0.3s ease;
}

.nav-links li a:hover,
.nav-links li a.active {
    background-color: var(--primary);
    color: white;
}

main {
    max-width: 1200px;
    margin: 2rem auto;
    padding: 0 1rem;
}

.tool-container {
    background-color: white;
    border-radius: 8px;
    box-shadow: var(--shadow);
    padding: 2rem;
    margin-bottom: 2rem;
}

.tool-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 1.5rem;
    border-bottom: 2px solid var(--light);
    padding-bottom: 1rem;
}

.tool-header h2 {
    color: var(--primary);
    font-size: 1.8rem;
}

.tool-controls {
    display: flex;
    gap: 1rem;
    margin-bottom: 1.5rem;
    flex-wrap: wrap;
}

.tool-content {
    display: flex;
    gap: 2rem;
    flex-wrap: wrap;
}

.visualization {
    flex: 1;
    min-width: 300px;
    height: 400px;
    background-color: var(--light);
    border-radius: 8px;
    display: flex;
    justify-content: center;
    align-items: center;
    position: relative;
}
```

```
.controls-panel {
    flex: 1;
    min-width: 300px;
}

button, select, input {
    padding: 0.75rem 1rem;
    border-radius: 4px;
    border: 1px solid #ddd;
    background-color: white;
    font-size: 1rem;
    cursor: pointer;
    transition: all 0.3s ease;
}

button {
    background-color: var(--primary);
    color: white;
    font-weight: 600;
    border: none;
}

button:hover {
    background-color: #2980b9;
    transform: translateY(-2px);
}

select:focus, input:focus {
    outline: none;
    border-color: var(--primary);
    box-shadow: 0 0 0 2px rgba(52, 152, 219, 0.3);
}

.control-group {
    margin-bottom: 1.5rem;
}

.control-group label {
    display: block;
    margin-bottom: 0.5rem;
    font-weight: 600;
    color: var(--dark);
}

.feedback {
    margin-top: 1.5rem;
    padding: 1rem;
    border-radius: 4px;
    background-color: var(--light);
}

.feedback.correct {
    background-color: rgba(46, 204, 113, 0.2);
    border-left: 4px solid var(--secondary);
}

.feedback.incorrect {
    background-color: rgba(231, 76, 60, 0.2);
    border-left: 4px solid var(--accent);
}

.molecule-canvas {
    width: 100%;
    height: 100%;
    border-radius: 8px;
}

.btn-group {
    display: flex;
    gap: 0.5rem;
    flex-wrap: wrap;
```

```
}

.btn-secondary {
    background-color: var(--secondary);
}

.btn-secondary:hover {
    background-color: #27ae60;
}

.btn-accent {
    background-color: var(--accent);
}

.btn-accent:hover {
    background-color: #c0392b;
}

.tab-container {
    margin-bottom: 1.5rem;
}

.tabs {
    display: flex;
    gap: 0.5rem;
    margin-bottom: 1rem;
}

.tab {
    padding: 0.5rem 1rem;
    border-radius: 4px 4px 0 0;
    background-color: var(--light);
    cursor: pointer;
    border: 1px solid #ddd;
    border-bottom: none;
}

.tab.active {
    background-color: white;
    border-bottom: 2px solid var(--primary);
    font-weight: 600;
}

.tab-content {
    display: none;
    padding: 1rem;
    border: 1px solid #ddd;
    border-radius: 0 4px 4px 4px;
}

.tab-content.active {
    display: block;
}

.split-view {
    display: flex;
    gap: 1rem;
    flex-wrap: wrap;
}

.split-panel {
    flex: 1;
    min-width: 250px;
    border-radius: 8px;
    overflow: hidden;
}

.molecule-gallery {
    display: grid;
    grid-template-columns: repeat(auto-fill, minmax(150px, 1fr));
    gap: 1rem;
```

```
    margin-top: 1.5rem;
}

.molecule-card {
    background-color: white;
    border-radius: 8px;
    padding: 1rem;
    box-shadow: var(--shadow);
    cursor: pointer;
    transition: all 0.3s ease;
    text-align: center;
}

.molecule-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 8px 16px rgba(0,0,0,0.1);
}

.molecule-thumbnail {
    width: 100%;
    height: 100px;
    background-color: var(--light);
    margin-bottom: 0.5rem;
    border-radius: 4px;
    display: flex;
    justify-content: center;
    align-items: center;
}

.progress-bar {
    height: 8px;
    background-color: var(--light);
    border-radius: 4px;
    margin: 1rem 0;
    overflow: hidden;
}

.progress {
    height: 100%;
    background-color: var(--primary);
    width: 0%;
    transition: width 0.5s ease;
}

.step-indicator {
    display: flex;
    justify-content: space-between;
    margin-bottom: 1.5rem;
}

.step {
    width: 30px;
    height: 30px;
    border-radius: 50%;
    background-color: var(--light);
    display: flex;
    justify-content: center;
    align-items: center;
    font-weight: 600;
    position: relative;
}

.step.active {
    background-color: var(--primary);
    color: white;
}

.step.completed {
    background-color: var(--secondary);
    color: white;
}
```

```
.step-line {
    flex: 1;
    height: 2px;
    background-color: var(--light);
    align-self: center;
    margin: 0 0.5rem;
}

.step-line.active {
    background-color: var(--primary);
}

.info-card {
    background-color: rgba(52, 152, 219, 0.1);
    border-left: 4px solid var(--primary);
    padding: 1rem;
    margin-bottom: 1.5rem;
    border-radius: 0 4px 4px 0;
}

.info-card h3 {
    margin-bottom: 0.5rem;
    color: var(--primary);
}

@media (max-width: 768px) {
    .tool-content {
        flex-direction: column;
    }

    .visualization {
        height: 300px;
    }
}

/* Specific component styles */
.slider-container {
    margin: 1.5rem 0;
}

.slider-container label {
    display: block;
    margin-bottom: 0.5rem;
    font-weight: 600;
}

.slider {
    width: 100%;
}

.slider-value {
    display: inline-block;
    margin-left: 1rem;
    font-weight: 600;
}

.toggle-switch {
    position: relative;
    display: inline-block;
    width: 60px;
    height: 34px;
}

.toggle-switch input {
    opacity: 0;
    width: 0;
    height: 0;
}

.toggle-slider {
```

```
    position: absolute;
    cursor: pointer;
    top: 0;
    left: 0;
    right: 0;
    bottom: 0;
    background-color: #ccc;
    transition: .4s;
    border-radius: 34px;
}

.toggle-slider:before {
    position: absolute;
    content: "";
    height: 26px;
    width: 26px;
    left: 4px;
    bottom: 4px;
    background-color: white;
    transition: .4s;
    border-radius: 50%;
}

input:checked + .toggle-slider {
    background-color: var(--primary);
}

input:focus + .toggle-slider {
    box-shadow: 0 0 1px var(--primary);
}

input:checked + .toggle-slider:before {
    transform: translateX(26px);
}

.toggle-label {
    display: flex;
    align-items: center;
    gap: 0.5rem;
}

.toggle-label span {
    font-weight: 600;
}

.molecule-display {
    position: relative;
    width: 100%;
    height: 100%;
}

.control-panel {
    padding: 1rem;
    background-color: var(--light);
    border-radius: 8px;
    margin-bottom: 1.5rem;
}

/* Loading indicator */
.loading {
    position: absolute;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    background-color: rgba(255, 255, 255, 0.8);
    display: flex;
    justify-content: center;
    align-items: center;
    z-index: 10;
    border-radius: 8px;
```

```
        }

        .spinner {
            width: 40px;
            height: 40px;
            border: 4px solid rgba(52, 152, 219, 0.3);
            border-radius: 50%;
            border-top: 4px solid var(--primary);
            animation: spin 1s linear infinite;
        }

        @keyframes spin {
            0% { transform: rotate(0deg); }
            100% { transform: rotate(360deg); }
        }
    </style>
</head>
<body>

    <script src="https://3Dmol.org/build/3Dmol-min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/@chemdoodle/web/chemdoodle.min.js"></script>
<script>
    // Initialize 3Dmol viewers
    document.querySelectorAll('.molecule-display').forEach(container => {
        let viewer = $3Dmol.createViewer(container, {
            backgroundColor: "white"
        });
        container.viewer = viewer; // Store viewer reference
    });

    // Function to load molecule into a viewer
    function loadMolecule(elementId, pdbData) {
        const viewer = document.getElementById(elementId).viewer;
        viewer.clear();
        viewer.addModel(pdbData, "pdb");
        viewer.setStyle({}, { stick: {} });
        viewer.zoomTo();
        viewer.render();
    }

    // Sample PDB data for molecules
    const moleculeData = {
        'glucose-l': `
HETATM    1  C1  UNL     1       0.000   0.000   0.000  1.00  0.00           C
HETATM    2  C2  UNL     1       1.200   0.000   0.000  1.00  0.00           C
HETATM    3  C3  UNL     1       1.800   1.200   0.000  1.00  0.00           C
HETATM    4  O1  UNL     1      -0.800   1.200   0.000  1.00  0.00           O
HETATM    5  O2  UNL     1       2.400  -0.800   0.000  1.00  0.00           O
HETATM    6  O3  UNL     1       3.000   1.800   0.000  1.00  0.00           O
END
        `,
        'glucose-d': `
HETATM    1  C1  UNL     1       0.000   0.000   0.000  1.00  0.00           C
HETATM    2  C2  UNL     1       1.200   0.000   0.000  1.00  0.00           C
HETATM    3  C3  UNL     1       1.800  -1.200   0.000  1.00  0.00           C
HETATM    4  O1  UNL     1      -0.800  -1.200   0.000  1.00  0.00           O
HETATM    5  O2  UNL     1       2.400   0.800   0.000  1.00  0.00           O
HETATM    6  O3  UNL     1       3.000  -1.800   0.000  1.00  0.00           O
END
        `,
        'glucose-racemic': `
HETATM    1  C1  UNL     1       0.000   0.000   0.000  1.00  0.00           C
HETATM    2  C2  UNL     1       1.200   0.000   0.000  1.00  0.00           C
HETATM    3  C3  UNL     1       1.200   1.200   0.000  1.00  0.00           C
HETATM    4  C4  UNL     1       0.000   1.200   0.000  1.00  0.00           C
HETATM    5  O1  UNL     1       0.600  -1.200   0.000  1.00  0.00           O
HETATM    6  O2  UNL     1       1.800   2.400   0.000  1.00  0.00           O
END
        `
        // Add more compounds here as needed
    };
```

```
    // Load molecule on compound selection
    document.getElementById('optical-compound-select').addEventListener('change', function (e) {
        const compound = e.target.value;
        const pdbData = moleculeData[compound];
        if (pdbData) {
            loadMolecule('optical-display', pdbData);
        }
    });

    // Auto-load a default compound on page load
    window.addEventListener('DOMContentLoaded', () => {
        const defaultCompound = 'glucose-l';
        document.getElementById('optical-compound-select').value = defaultCompound;
        document.getElementById('optical-compound-select').dispatchEvent(new Event('change'));
    });
</script>

    <header>
        <h1>Stereochemistry Virtual Laboratory</h1>
        <p1>Explore 3D molecular structures, stereoisomerism, and chirality</p1>
    </header>

    <!-- Back Button -->
    <a href="organic.html" class="back-btn">&#8592; Back</a>

    <nav>
        <ul class="nav-links">
            <li><a href="#rs-configuration" class="active">R/S Configuration</a></li>
            <li><a href="#stereoisomer-builder">Stereoisomer Builder</a></li>
            <li><a href="#optical-activity">Optical Activity</a></li>
            <li><a href="#isomer-counter">Isomer Counter</a></li>
            <li><a href="#cip-visualizer">CIP Rules</a></li>
        </ul>
    </nav>

    <main>

        <!-- Optimized R/S Configuration Tool Section -->
<section id="rs-configuration" class="tool-container">
    <div class="tool-header">
      <h2>Interactive R/S Configuration Tool</h2>
      <div class="btn-group">
        <button class="btn-secondary" id="rs-help">Tutorial</button>
        <button id="rs-reset">Reset</button>
      </div>
    </div>

    <div class="info-card">
      <h3>Understanding R/S Configuration</h3>
      <p>
        Learn to assign R/S configurations to chiral centers using the Cahn-Ingold-Prelog priority rules. Select a molecule, assign priorities, orient it correctly, and determine the configuration interactively.
      </p>
    </div>

    <div class="tool-controls">
      <select id="rs-molecule-select" aria-label="Select Molecule">
        <option value="">-- Choose a molecule --</option>
        <option value="2-butanol">2-Butanol</option>
        <option value="lactic-acid">Lactic Acid</option>
        <option value="alanine">Alanine</option>
        <option value="2-chlorobutane">2-Chlorobutane</option>
        <option value="carvone">Carvone</option>
        <option value="custom">Custom Structure</option>
      </select>
      <div class="btn-group">
        <button id="rs-rotate">Rotate View</button>
        <button id="rs-check" class="btn-secondary">Check Configuration</button>
      </div>
    </div>
```

```
  <div class="tool-content">
   <div class="visualization">
    <div class="molecule-display" id="rs-molecule-display" aria-label="3D Molecule Viewer">
     <div class="loading" id="rs-loading" hidden>
      <div class="spinner"></div>
     </div>
    </div>
   </div>

   <div class="controls-panel">
    <div class="control-group">
     <label for="rs-priority-controls">Assign Priorities</label>
     <div id="rs-priority-controls">
      <p>Please select a molecule to begin priority assignment.</p>
     </div>
    </div>

    <div class="control-group">
     <label>Specify Configuration</label>
     <div class="btn-group">
      <button id="rs-r-config" class="config-btn">R</button>
      <button id="rs-s-config" class="config-btn">S</button>
     </div>
    </div>

    <div class="feedback" id="rs-feedback">
     <p>Feedback on your configuration will appear here.</p>
    </div>
   </div>
  </div>
 </section>

    <!-- Stereoisomer Builder -->
    <section id="stereoisomer-builder" class="tool-container">
      <div class="tool-header">
        <h2>Stereoisomer Builder</h2>
        <div class="btn-group">
          <button class="btn-secondary" id="builder-help">Tutorial</button>
          <button id="builder-reset">Reset</button>
        </div>
      </div>

      <div class="info-card">
        <h3>Exploring Stereoisomers</h3>
        <p>Stereoisomers are compounds with the same connectivity but different spatial arrangements. This tool allows you to build and visualize all possible stereoisomers of a molecule, and identify relationships between them (enantiomers, diastereomers, and meso compounds).</p>
      </div>

      <div class="tool-controls">
        <select id="builder-molecule-select">
          <option value="">Select a molecule</option>
          <option value="2-3-butanediol">2,3-Butanediol</option>
          <option value="tartaric-acid">Tartaric Acid</option>
          <option value="glucose">Glucose</option>
          <option value="threose">Threose</option>
          <option value="custom">Custom Structure</option>
        </select>
        <button id="builder-generate" class="btn-secondary">Generate Stereoisomers</button>
        <select id="builder-view-mode">
          <option value="gallery">Gallery View</option>
          <option value="comparison">Comparison View</option>
        </select>
      </div>

      <div class="tab-container">
        <div class="tabs">
          <div class="tab active" data-tab="all-isomers">All Isomers</div>
          <div class="tab" data-tab="enantiomer-pairs">Enantiomer Pairs</div>
          <div class="tab" data-tab="diastereomers">Diastereomers</div>
```

```
            <div class="tab" data-tab="meso-compounds">Meso Compounds</div>
          </div>

          <div class="tab-content active" id="all-isomers">
            <div class="molecule-gallery" id="isomer-gallery">
              <p>Select a molecule to generate stereoisomers</p>
            </div>
          </div>

          <div class="tab-content" id="enantiomer-pairs">
            <div class="split-view" id="enantiomer-view">
              <p>Select a molecule to view enantiomer pairs</p>
            </div>
          </div>

          <div class="tab-content" id="diastereomers">
            <div class="split-view" id="diastereomer-view">
              <p>Select a molecule to view diastereomers</p>
            </div>
          </div>

          <div class="tab-content" id="meso-compounds">
            <div class="molecule-gallery" id="meso-gallery">
              <p>Select a molecule to identify meso compounds</p>
            </div>
          </div>
        </div>

        <div class="control-panel">
          <div class="control-group">
            <label>Manipulate Stereocenters</label>
            <div id="stereocenter-controls">
              <p>Select a molecule to manipulate stereocenters</p>
            </div>
          </div>
        </div>
      </section>

      <!-- Optical Activity Simulator -->
      <section id="optical-activity" class="tool-container">
        <div class="tool-header">
          <h2>Optical Activity Simulator</h2>
          <div class="btn-group">
            <button class="btn-secondary" id="optical-help">Tutorial</button>
            <button id="optical-reset">Reset</button>
          </div>
        </div>

        <div class="info-card">
          <h3>Understanding Optical Activity</h3>
          <p>Optical activity is the ability of chiral compounds to rotate plane-polarized light. This simulator demonstrates how different stereoisomers and their mixtures affect polarized light based on concentration and path length.</p>
        </div>

        <div class="tool-controls">
          <select id="optical-compound-select">
            <option value="">Select a compound</option>
            <option value="glucose-d">D-Glucose</option>
            <option value="glucose-l">L-Glucose</option>
            <option value="glucose-racemic">Racemic Glucose</option>
            <option value="lactic-acid-d">D-Lactic Acid</option>
            <option value="lactic-acid-l">L-Lactic Acid</option>
            <option value="lactic-acid-racemic">Racemic Lactic Acid</option>
            <option value="custom">Custom Compound</option>
          </select>
          <button id="optical-start">Start Simulation</button>
          <button id="optical-pause">Pause</button>
        </div>

        <div class="tool-content">
          <div class="visualization">
```

```
            <div class="molecule-display" id="optical-display">
              <div class="loading">
                <div class="spinner"></div>
              </div>
            </div>
          </div>

          <div class="controls-panel">
            <div class="control-group">
              <label>Specific Rotation [α]</label>
              <input type="number" id="specific-rotation" value="52.7" step="0.1">
              <small>Degrees · (dm·g/cm³)⁻¹</small>
            </div>

            <div class="slider-container">
              <label>Concentration (g/cm³)</label>
              <input type="range" class="slider" id="concentration" min="0.01" max="1" step="0.01" value="0.1">
              <span class="slider-value" id="concentration-value">0.1 g/cm³</span>
            </div>

            <div class="slider-container">
              <label>Path Length (dm)</label>
              <input type="range" class="slider" id="path-length" min="0.1" max="5" step="0.1" value="1">
              <span class="slider-value" id="path-length-value">1.0 dm</span>
            </div>

            <div class="control-group">
              <label>Observed Rotation</label>
              <div id="observed-rotation">0.0°</div>
            </div>

            <div class="toggle-label">
              <span>Show Formula</span>
              <label class="toggle-switch">
                <input type="checkbox" id="show-formula">
                <span class="toggle-slider"></span>
              </label>
            </div>
          </div>
        </div>

        <div class="info-card" id="rotation-formula" style="display: none;">
          <h3>Calculation of Optical Rotation</h3>
          <p>The observed rotation (α) is calculated using the formula:</p>
          <p><strong>α = [α] × l × c</strong></p>
          <p>Where [α] is the specific rotation, l is the path length in decimeters, and c is the concentration in g/cm³.</p>
        </div>
      </section>

      <!-- Isomer Counter Tool -->
      <section id="isomer-counter" class="tool-container">
        <div class="tool-header">
          <h2>Isomer Counter Tool</h2>
          <div class="btn-group">
            <button class="btn-secondary" id="counter-help">Tutorial</button>
            <button id="counter-reset">Reset</button>
          </div>
        </div>

        <div class="info-card">
          <h3>Counting Stereoisomers</h3>
            <p>This tool helps you identify stereocenters in a molecule and calculate the theoretical number of stereoisomers. It accounts for meso compounds and symmetry, and displays all possible stereoisomers in an interactive gallery.</p>
        </div>

        <div class="tool-controls">
          <select id="counter-molecule-select">
            <option value="">Select a molecule</option>
            <option value="2-3-butanediol">2,3-Butanediol</option>
            <option value="2-3-dichlorobutane">2,3-Dichlorobutane</option>
            <option value="tartaric-acid">Tartaric Acid</option>
```

```
            <option value="glucose">Glucose</option>
            <option value="custom">Custom Structure</option>
          </select>
          <button id="counter-analyze" class="btn-secondary">Analyze Structure</button>
        </div>

        <div class="tool-content">
          <div class="visualization">
            <div class="molecule-display" id="counter-display">
              <div class="loading">
                <div class="spinner"></div>
              </div>
            </div>
          </div>

          <div class="controls-panel">
            <div class="control-group">
              <label>Structure Analysis</label>
              <div id="counter-results">
                <p>Select a molecule to analyze</p>
              </div>
            </div>

            <div class="control-group">
              <label>Theoretical Maximum</label>
              <div id="theoretical-max">-</div>
            </div>

            <div class="control-group">
              <label>Actual Number (accounting for meso)</label>
              <div id="actual-count">-</div>
            </div>

            <div class="toggle-label">
              <span>Highlight Stereocenters</span>
              <label class="toggle-switch">
                <input type="checkbox" id="highlight-stereocenters" checked>
                <span class="toggle-slider"></span>
              </label>
            </div>
          </div>
        </div>

        <div class="molecule-gallery" id="stereoisomer-gallery">
          <p>Analyze a structure to view all stereoisomers</p>
        </div>
      </section>

      <!-- CIP Rule Visualizer -->
      <section id="cip-visualizer" class="tool-container">
        <div class="tool-header">
          <h2>Step-by-Step CIP Rule Visualizer</h2>
          <div class="btn-group">
            <button class="btn-secondary" id="cip-help">Tutorial</button>
            <button id="cip-reset">Reset</button>
          </div>
        </div>

        <div class="info-card">
          <h3>Cahn-Ingold-Prelog Priority Rules</h3>
          <p>The CIP system allows systematic naming of stereoisomers. This tool guides you through applying CIP rules step-by-step to determine substituent priorities and ultimately assign R or S configuration.</p>
        </div>

        <div class="tool-controls">
          <select id="cip-molecule-select">
            <option value="">Select a molecule</option>
            <option value="2-butanol">2-Butanol</option>
            <option value="lactic-acid">Lactic Acid</option>
            <option value="alanine">Alanine</option>
            <option value="cysteine">Cysteine</option>
```

```
            <option value="2-bromobutane">2-Bromobutane</option>
            <option value="custom">Custom Structure</option>
        </select>
        <button id="start-tutorial" class="btn-secondary">Start Tutorial</button>
    </div>

    <div class="step-indicator">
        <div class="step active" id="step1">1</div>
        <div class="step-line"></div>
        <div class="step" id="step2">2</div>
        <div class="step-line"></div>
        <div class="step" id="step3">3</div>
        <div class="step-line"></div>
        <div class="step" id="step4">4</div>
        <div class="step-line"></div>
        <div class="step" id="step5">5</div>
    </div>

    <div class="tool-content">
        <div class="visualization">
            <div class="molecule-display" id="cip-display">
                <div class="loading">
                    <div class="spinner"></div>
                </div>
            </div>
        </div>

        <div class="controls-panel" id="cip-tutorial-panel">
            <div class="step-content active" id="step1-content">
                <h3>Step 1: Identify the Stereocenter</h3>
                <p>A stereocenter is an atom (usually carbon) with four different substituents.</p>
                <div class="control-group">
                    <button id="step1-next" class="btn-secondary">Next Step</button>
                </div>
            </div>

            <div class="step-content" id="step2-content">
                <h3>Step 2: Identify the Four Substituents</h3>
                <p>Find the four different groups attached to the stereocenter.</p>
                <div id="substituent-list"></div>
                <div class="control-group">
                    <button id="step2-prev">Previous</button>
                    <button id="step2-next" class="btn-secondary">Next Step</button>
                </div>
            </div>

            <div class="step-content" id="step3-content">
                <h3>Step 3: Rank the Substituents</h3>
                <p>Assign priorities based on atomic number: higher atomic number = higher priority.</p>
                <div class="control-group">
                    <button id="step3-prev">Previous</button>
                    <button id="step3-next" class="btn-secondary">Next Step</button>
                </div>
            </div>

            <div class="step-content" id="step4-content">
                <h3>Step 4: Orient the Molecule</h3>
                <p>Position the lowest priority group (usually 4) pointing away from you.</p>
                <div class="control-group">
                    <button id="step4-prev">Previous</button>
                    <button id="step4-next" class="btn-secondary">Next Step</button>
                </div>
            </div>

            <div class="step-content" id="step5-content">
                <h3>Step 5: Determine Configuration</h3>
                <p>If the order of 1 → 2 → 3 is clockwise, it's R. If it's counterclockwise, it's S.</p>
                <div class="result-box" id="cip-result">
                    <strong>Result:</strong> <span id="configuration-result">Pending</span>
                </div>
                <div class="control-group">
```

```
                    <button id="step5-prev">Previous</button>
                    <button id="restart-tutorial" class="btn-secondary">Restart</button>
                  </div>
                </div>
              </div> <!-- end controls-panel -->
            </div> <!-- end tool-content -->
        </section> <!-- end cip-visualizer -->

        <!-- Scripts -->
        <script src="scripts/main.js"></script>
        <!-- Add any other script files as needed -->

    </body>
</html>
```

Module 9
iconic_compounds.html–

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Advanced 3D Molecular Structure Viewer</title>
    <script src="https://cdnjs.cloudflare.com/ajax/libs/3Dmol/2.0.3/3Dmol-min.js"></script>
    <script src="https://cdnjs.cloudflare.com/ajax/libs/jquery/3.6.4/jquery.min.js"></script>
    <style>
          body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    text-align: center;
    margin: 0;
    padding: 0;
    background: linear-gradient(135deg, #1b2430, #587898);
     color: #e0e6ed; /* optional: soft off-white text for readability */
    line-height: 1.6;
}

/* Back Button */
.back-btn {
          position: absolute;
          top: 60px;
          left: 20px;
          text-decoration: none;
          font-size: 1.2rem;
          font-weight: bold;
          color: white;
          background: #333;
          padding: 10px 20px;
          border-radius: 8px;
          transition: background 0.3s ease;
       }

       .back-btn:hover {
          background: #555;
       }

.page-wrapper {
    max-width: 1200px;
    margin: 0 auto;
    padding: 30px 20px;
}

/* Title Styling */
 h1 {
    color: #ffffff;
```

```
    margin-bottom: 8px;
    font-size: 3.5rem;
    font-weight: 700;
    text-shadow: 3px 3px 6px rgba(241, 87, 87, 0.934);

}

p.subtitle {
    color: #ffffff;
    margin-top: 0;
    margin-bottom: 30px;
    font-size: 2rem;
    text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);

}

#viewer-container {
    width: 800px;
    height: 500px;
    margin: 0 auto 30px;
    position: relative;
    box-shadow: 0 8px 30px rgba(0, 0, 0, 0.12);
    border-radius: 12px;
    overflow: hidden;
    background-color: #fff;
    transition: transform 0.3s ease, box-shadow 0.3s ease;
}

#viewer-container:hover {
    transform: translateY(-5px);
    box-shadow: 0 12px 36px rgba(0, 0, 0, 0.16);
}

#viewer {
    width: 100%;
    height: 100%;
    border: none;
    background-color: #fff;
}

#controls {
    margin: 30px auto;
    padding: 25px;
    background-color: #fff;
    border-radius: 12px;
    box-shadow: 0 4px 20px rgba(0, 0, 0, 0.08);
    max-width: 800px;
}

.btn-group {
    margin-bottom: 24px;
}

.btn-group h3 {
    margin-bottom: 16px;
    color: #1a365d;
    font-size: 1.25rem;
    font-weight: 600;
    position: relative;
    padding-bottom: 10px;
}

.btn-group h3::after {
    content: '';
    position: absolute;
    bottom: 0;
    left: 50%;
```

```
    transform: translateX(-50%);
    width: 60px;
    height: 3px;
    background-color: #3b82f6;
    border-radius: 2px;
}

button {
    margin: 6px;
    padding: 12px 20px;
    font-size: 15px;
    font-weight: 500;
    cursor: pointer;
    border: none;
    border-radius: 8px;
    background-color: #3b82f6;
    color: white;
    transition: all 0.25s ease;
    box-shadow: 0 2px 8px rgba(59, 130, 246, 0.3);
}

button:hover {
    background-color: #2563eb;
    transform: translateY(-3px);
    box-shadow: 0 4px 12px rgba(37, 99, 235, 0.4);
}

button:active {
    transform: translateY(-1px);
}

button.primary {
    background-color: #10b981;
    box-shadow: 0 2px 8px rgba(16, 185, 129, 0.3);
}

button.primary:hover {
    background-color: #059669;
    box-shadow: 0 4px 12px rgba(5, 150, 105, 0.4);
}

button.secondary {
    background-color: #8b5cf6;
    box-shadow: 0 2px 8px rgba(139, 92, 246, 0.3);
}

button.secondary:hover {
    background-color: #7c3aed;
    box-shadow: 0 4px 12px rgba(124, 58, 237, 0.4);
}

.style-options {
    display: flex;
    justify-content: center;
    flex-wrap: wrap;
    gap: 8px;
    margin-top: 16px;
}

.style-options button {
    padding: 10px 16px;
    font-size: 14px;
    background-color: #f3f4f6;
    color: #4b5563;
    box-shadow: 0 2px 5px rgba(0, 0, 0, 0.05);
}
```

```
.style-options button:hover {
    background-color: #e5e7eb;
    color: #1f2937;
    box-shadow: 0 3px 8px rgba(0, 0, 0, 0.08);
}

.style-options button.active {
    background-color: #dbeafe;
    color: #1e40af;
    border-left: 3px solid #3b82f6;
}

.info-panel {
    background-color: #f8fafc;
    border-radius: 12px;
    padding: 25px;
    margin-top: 30px;
    text-align: left;
    max-width: 800px;
    margin-left: auto;
    margin-right: auto;
    box-shadow: 0 4px 16px rgba(0, 0, 0, 0.06);
    border-top: 4px solid #3b82f6;
}

.info-panel h3 {
    margin-top: 0;
    color: #1a365d;
    font-size: 1.25rem;
    font-weight: 600;
    margin-bottom: 16px;
}

.info-panel p {
    margin-bottom: 12px;
    color: #475569;
    line-height: 1.7;
}

.hidden {
    display: none;
}

/* Responsive adjustments */
@media (max-width: 850px) {
    #viewer-container {
        width: 100%;
        height: 400px;
    }

    button {
        padding: 10px 16px;
        font-size: 14px;
    }
}

@media (max-width: 600px) {
    h1 {
        font-size: 2rem;
    }

    .page-wrapper {
        padding: 20px 15px;
    }

    #viewer-container {
        height: 350px;
```

```
    }

    #controls, .info-panel {
        padding: 20px;
    }

    .btn-group {
        margin-bottom: 18px;
    }
}

/* Light/dark mode toggle (optional) */
.mode-toggle {
    position: fixed;
    top: 20px;
    right: 20px;
    padding: 10px;
    background-color: #fff;
    border-radius: 50%;
    box-shadow: 0 2px 10px rgba(0, 0, 0, 0.1);
    cursor: pointer;
    z-index: 100;
    width: 40px;
    height: 40px;
    display: flex;
    align-items: center;
    justify-content: center;
    transition: all 0.3s ease;
}

.mode-toggle:hover {
    transform: rotate(30deg);
}

/* Animation effects */
@keyframes fadeIn {
    from { opacity: 0; transform: translateY(20px); }
    to { opacity: 1; transform: translateY(0); }
}

.fade-in {
    animation: fadeIn 0.6s ease forwards;
}

.btn-group, .info-panel {
    animation: fadeIn 0.6s ease forwards;
}

/* Custom scrollbar */
::-webkit-scrollbar {
    width: 10px;
}

::-webkit-scrollbar-track {
    background: #f1f5f9;
}

::-webkit-scrollbar-thumb {
    background: #cbd5e1;
    border-radius: 5px;
}

::-webkit-scrollbar-thumb:hover {
    background: #94a3b8;
}
    </style>
</head>
```

```
<body>


    <!-- Back Button -->
    <a href="organic.html" class="back-btn">&#8592; Back</a>

    <h1>3D Molecular Structure Viewer</h1>
    <p class="subtitle">Explore organic molecules with real-time 3D visualization</p>

    <div id="viewer-container">
        <div id="viewer"></div>
    </div>

    <div id="controls">
        <div class="btn-group">
            <h3>Select Molecule</h3>
            <button onclick="loadMolecule('methane')" class="primary">Methane (CH₄)</button>
            <button onclick="loadMolecule('ethanol')" class="primary">Ethanol (C₂H₅OH)</button>
            <button onclick="loadMolecule('benzene')" class="primary">Benzene (C₆H₆)</button>
            <button onclick="loadMolecule('aspirin')" class="primary">Aspirin (C₉H₈O₄)</button>
            <button onclick="loadMolecule('water')" class="primary">Water (H2O)</button>
            <button onclick="loadMolecule('carbonDioxide')" class="primary">CarbonDioxide (CO₂)</button>
            <button onclick="loadMolecule('glucose')" class="primary">Glucose (C₆H₁₂O₆)</button>
            <button onclick="loadMolecule('acetone')" class="primary">Acetone (C₃H₆O)</button>




        </div>

        <div class="btn-group">
            <h3>Display Options</h3>
            <button onclick="setStyle('stick')">Stick</button>
            <button onclick="setStyle('sphere')">Space-filling</button>
            <button onclick="setStyle('line')">Line</button>
        </div>

        <div class="btn-group">
            <h3>View Controls</h3>
            <button onclick="toggleLabels()" class="secondary">Toggle Labels</button>
            <button onclick="toggleSpin()" class="secondary">Toggle Spin</button>
            <button onclick="resetView()" class="secondary">Reset View</button>
        </div>
    </div>

    <div id="molecule-info" class="info-panel hidden">
        <h3 id="info-title">Molecule Information</h3>
        <p id="info-formula"><strong>Formula:</strong> <span id="formula-text"></span></p>
        <p id="info-details"></p>
    </div>

    <script>
        // Initialize the viewer
        let viewer;
        let currentMolecule = '';
        let isSpinning = false;
        let showLabels = false;

        $(document).ready(function() {
            const element = $("#viewer")[0];
            viewer = $3Dmol.createViewer(element, {
                backgroundColor: "white"
            });
            loadMolecule('methane'); // Load methane by default
        });

        const molecules = {
```

```
  methane: {
    pdb: `HEADER    METHANE
ATOM      1  C   MET     1       0.000   0.000   0.000  1.00  0.00           C
ATOM      2  H1  MET     1       0.889   0.000   0.629  1.00  0.00           H
ATOM      3  H2  MET     1      -0.889   0.000   0.629  1.00  0.00           H
ATOM      4  H3  MET     1       0.000   0.889  -0.629  1.00  0.00           H
ATOM      5  H4  MET     1       0.000  -0.889  -0.629  1.00  0.00           H
END`,
    formula: "CH₄",
    description: "Methane is the simplest alkane and the main component of natural gas. It consists of one carbon atom bonded to four hydrogen atoms in a tetrahedral arrangement."
  },
  ethanol: {
    pdb: `HEADER    ETHANOL
ATOM      1  C1  ETH     1      -1.153  -0.022   0.000  1.00  0.00           C
ATOM      2  C2  ETH     1       0.363  -0.022   0.000  1.00  0.00           C
ATOM      3  O   ETH     1       0.853   1.300   0.000  1.00  0.00           O
ATOM      4  H1  ETH     1      -1.534  -1.050   0.000  1.00  0.00           H
ATOM      5  H2  ETH     1      -1.534   0.488   0.890  1.00  0.00           H
ATOM      6  H3  ETH     1      -1.534   0.488  -0.890  1.00  0.00           H
ATOM      7  H4  ETH     1       0.744  -0.532  -0.890  1.00  0.00           H
ATOM      8  H5  ETH     1       0.744  -0.532   0.890  1.00  0.00           H
ATOM      9  H6  ETH     1       1.823   1.300   0.000  1.00  0.00           H
END`,
    formula: "C₂H₅OH",
    description: "Ethanol is a simple alcohol composed of a two-carbon chain with a hydroxyl group (-OH) attached. It's the alcohol found in alcoholic beverages and is also used as a solvent and fuel."
  },
  benzene: {
    pdb: `HEADER    BENZENE
ATOM      1  C1  BNZ     1       0.000   1.398   0.000  1.00  0.00           C
ATOM      2  C2  BNZ     1       1.211   0.699   0.000  1.00  0.00           C
ATOM      3  C3  BNZ     1       1.211  -0.699   0.000  1.00  0.00           C
ATOM      4  C4  BNZ     1       0.000  -1.398   0.000  1.00  0.00           C
ATOM      5  C5  BNZ     1      -1.211  -0.699   0.000  1.00  0.00           C
ATOM      6  C6  BNZ     1      -1.211   0.699   0.000  1.00  0.00           C
ATOM      7  H1  BNZ     1       0.000   2.484   0.000  1.00  0.00           H
ATOM      8  H2  BNZ     1       2.151   1.242   0.000  1.00  0.00           H
ATOM      9  H3  BNZ     1       2.151  -1.242   0.000  1.00  0.00           H
ATOM     10  H4  BNZ     1       0.000  -2.484   0.000  1.00  0.00           H
ATOM     11  H5  BNZ     1      -2.151  -1.242   0.000  1.00  0.00           H
ATOM     12  H6  BNZ     1      -2.151   1.242   0.000  1.00  0.00           H
END`,
    formula: "C₆H₆",
    description: "Benzene is an aromatic hydrocarbon with a ring of six carbon atoms and six hydrogen atoms. Its special structure with delocalized electrons gives it unique chemical properties and stability."
  },
  aspirin: {
    pdb: `HEADER    ASPIRIN
ATOM      1  C1  ASP     1      -1.334   0.496   0.008  1.00  0.00           C
ATOM      2  C2  ASP     1      -2.237   1.544   0.102  1.00  0.00           C
ATOM      3  C3  ASP     1      -3.605   1.313   0.122  1.00  0.00           C
ATOM      4  C4  ASP     1      -4.073   0.010   0.046  1.00  0.00           C
ATOM      5  C5  ASP     1      -3.175  -1.045  -0.049  1.00  0.00           C
ATOM      6  C6  ASP     1      -1.806  -0.812  -0.070  1.00  0.00           C
ATOM      7  C7  ASP     1       0.131   0.736  -0.012  1.00  0.00           C
ATOM      8  C8  ASP     1       2.449  -0.060  -0.003  1.00  0.00           C
ATOM      9  C9  ASP     1       2.954  -1.475  -0.120  1.00  0.00           C
ATOM     10  O1  ASP     1       0.639   1.830  -0.089  1.00  0.00           O
ATOM     11  O2  ASP     1       0.837  -0.410   0.082  1.00  0.00           O
ATOM     12  O3  ASP     1       3.126   0.925   0.084  1.00  0.00           O
ATOM     13  O4  ASP     1      -0.960  -1.851  -0.165  1.00  0.00           O
ATOM     14  H1  ASP     1      -1.867   2.560   0.160  1.00  0.00           H
ATOM     15  H2  ASP     1      -4.305   2.134   0.197  1.00  0.00           H
ATOM     16  H3  ASP     1      -5.139  -0.186   0.060  1.00  0.00           H
ATOM     17  H4  ASP     1      -3.547  -2.059  -0.108  1.00  0.00           H
ATOM     18  H5  ASP     1       2.610  -2.065   0.731  1.00  0.00           H
ATOM     19  H6  ASP     1       4.043  -1.493  -0.125  1.00  0.00           H
```

```
ATOM     20  H7  ASP     1       2.612  -1.950  -1.043  1.00  0.00           H
ATOM     21  H8  ASP     1      -1.437  -2.672  -0.211  1.00  0.00           H
END`,
        formula: "C9H8O4",
       description: "Aspirin (acetylsalicylic acid) is a medication used to reduce pain, fever, and inflammation. It consists of a benzene ring with carboxylic acid and acetoxy groups attached."
    },
    water: {
        pdb: `HEADER    WATER
ATOM      1  O   H2O     1       0.000   0.000   0.000  1.00  0.00           O
ATOM      2  H1  H2O     1       0.758   0.000   0.504  1.00  0.00           H
ATOM      3  H2  H2O     1      -0.758   0.000   0.504  1.00  0.00           H
END`,
        formula: "H2O",
        description: "Water is a simple molecule consisting of two hydrogen atoms bonded to one oxygen atom. It is essential for life and serves as a solvent for many biochemical reactions."
    },
    carbonDioxide: {
        pdb: `HEADER    CARBON DIOXIDE
ATOM      1  C   CO2     1       0.000   0.000   0.000  1.00  0.00           C
ATOM      2  O1  CO2     1       1.160   0.000   0.000  1.00  0.00           O
ATOM      3  O2  CO2     1      -1.160   0.000   0.000  1.00  0.00           O
END`,
        formula: "CO2",
         description: "Carbon dioxide is a colorless, odorless gas that is a byproduct of respiration and combustion. It is vital for photosynthesis in plants but is also a major greenhouse gas."
    },
    glucose: {
        pdb: `HEADER    GLUCOSE
ATOM      1  C1  GLU     1       0.000   0.000   0.000  1.00  0.00           C
ATOM      2  C2  GLU     1       1.212   0.000   0.000  1.00  0.00           C
ATOM      3  C3  GLU     1       1.212   1.212   0.000  1.00  0.00           C
ATOM      4  C4  GLU     1       0.000   1.212   0.000  1.00  0.00           C
ATOM      5  C5  GLU     1      -1.212   0.000   0.000  1.00  0.00           C
ATOM      6  C6  GLU     1      -1.212  -1.212   0.000  1.00  0.00           C
ATOM      7  O1  GLU     1       2.121   0.000   0.000  1.00  0.00           O
ATOM      8  O2  GLU     1      -2.121   0.000   0.000  1.00  0.0            O
END`,
        formula: "C6H12O6",
        description: "Glucose is a simple sugar and an essential carbohydrate in metabolism. It is a primary source of energy for cells and is found in many foods."
    },
    acetone: {
        pdb: `HEADER    ACETONE
ATOM      1  C1  ACT     1       0.000   0.000   0.000  1.00  0.00           C
ATOM      2  C2  ACT     1       1.206   0.000   0.000  1.00  0.00           C
ATOM      3  O   ACT     1       2.412   0.000   0.000  1.00  0.00           O
ATOM      4  H1  ACT     1      -0.758   0.000   0.629  1.00  0.00           H
ATOM      5  H2  ACT     1      -0.758   0.000  -0.629  1.00  0.00           H
ATOM      6  H3  ACT     1       1.206   1.206   0.000  1.00  0.00           H
END`,
        formula: "C3H6O",
        description: "Acetone is a solvent used in various industrial processes and as a cleaning agent. It is the simplest ketone with a distinctive odor and is widely used in cosmetics and household products."
    }
};

    // Function to load a molecule
    function loadMolecule(type) {
        if (!molecules[type]) {
            console.error("Molecule not found:", type);
            return;
        }

        currentMolecule = type;
        viewer.clear();

        const moleculeData = molecules[type];
```

```
        viewer.addModel(moleculeData.pdb, "pdb");

        // Apply default style
        setStyle('stick');

        // Center and zoom
        viewer.zoomTo();

        // Update info panel
        $("#molecule-info").removeClass("hidden");
        $("#info-title").text(type.charAt(0).toUpperCase() + type.slice(1));
        $("#formula-text").text(moleculeData.formula);
        $("#info-details").text(moleculeData.description);

        // Add labels if enabled
        if (showLabels) {
            addLabels();
        }

        // Start spinning if enabled
        if (isSpinning) {
            viewer.spin(true);
        }
    }

    // Function to set the display style
    function setStyle(style) {
        viewer.setStyle({}, {});  // Clear previous styles

        switch(style) {
            case 'stick':
                viewer.setStyle({}, {stick: {radius: 0.2, colorscheme: 'Jmol', singleBonds: true}});
                break;
            case 'sphere':
                viewer.setStyle({}, {sphere: {scale: 0.5, colorscheme: 'Jmol'}});
                break;
            case 'line':
                viewer.setStyle({}, {line: {colorscheme: 'Jmol'}});
                break;
        }

        // Add labels if enabled
        if (showLabels) {
            addLabels();
        }

        viewer.render();
    }

    // Function to toggle spinning
    function toggleSpin() {
        isSpinning = !isSpinning;
        viewer.spin(isSpinning);
    }

    // Function to toggle atom labels
    function toggleLabels() {
        showLabels = !showLabels;

        if (showLabels) {
            addLabels();
        } else {
            viewer.removeAllLabels();
        }

        viewer.render();
    }

    // Function to add atom labels
```

```
        function addLabels() {
            viewer.removeAllLabels();

            const atoms = viewer.getModel().selectedAtoms({});
            for (let i = 0; i < atoms.length; i++) {
                const atom = atoms[i];
                viewer.addLabel(atom.atom.elem, {
                    position: atom,
                    backgroundColor: 'rgba(255, 255, 255, 0.5)',
                    fontColor: 'black',
                    fontSize: 12
                });
            }
        }

        // Function to reset the view
        function resetView() {
            viewer.rotate(0, 0, 0, 0, true);
            viewer.spin(false);
            isSpinning = false;
            viewer.zoomTo();
        }
    </script>
</body>
</html>
```

Module 10

Physical.html–

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Virtual Chemistry Laboratory</title>
    <script src="https://cdn.jsdelivr.net/npm/three@0.132.2/build/three.min.js"></script>
    <script src="https://cdn.jsdelivr.net/npm/three@0.132.2/examples/js/controls/OrbitControls.js"></script>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        /* Title Styling */
        h1 {
            font-size: 3.5rem;
            color: white;
            text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);
            font-weight: 700;
            margin-bottom: 10px;
        }

        p {
            font-size: 1.2rem;
            margin-bottom: 20px;
        }

        body {
            font-family: Arial, sans-serif;
            line-height: 1.6;
            background: linear-gradient(135deg, #496b7d, #17559c);
        }
        .container {
            width: 90%;
            max-width: 1200px;
```

```
    margin: 0 auto;
    padding: 20px;
}
.tab-container {
    display: flex;
    background-color: #333;
}
.tab-button {
    flex: 1;
    padding: 10px;
    text-align: center;
    color: white;
    cursor: pointer;
    transition: background-color 0.3s;
}
.tab-button:hover {
    background-color: #555;
}
.tab-button.active {
    background-color: #4CAF50;
}
.tab-content {
    display: none;
    padding: 20px;
    background-color: white;
    min-height: 500px;
}
.tab-content.active {
    display: block;
}
.element-cell {
    border: 1px solid #ddd;
    padding: 5px;
    text-align: center;
    cursor: pointer;
    transition: transform 0.2s;
}
.element-cell:hover {
    transform: scale(1.1);
    z-index: 10;
}
/* Crystal Structure Styles */
#crystal-viewer {
    width: 100%;
    height: 400px;
    background-color: #f0f0f0;
}
/* Back Button Style */
 .back-btn {
    position: absolute;
    top: 20px;
    left: 20px;
    text-decoration: none;
    font-size: 1.2rem;
    font-weight: bold;
    color: white;
    background: #333;
    padding: 10px 20px;
    border-radius: 8px;
    transition: background 0.3s ease;
}

.back-btn:hover {
    background: #555;
}
/* Electrochemistry Styles */
.feature-intro {
    margin-bottom: 20px;
    padding: 15px;
    background-color: #f8f8f8;
    border-left: 4px solid #4CAF50;
```

```
        }
        .feature-intro ul {
            margin-left: 20px;
        }
        .calculator-form {
            margin-bottom: 25px;
            padding: 15px;
            border: 1px solid #ddd;
            border-radius: 5px;
        }
        .calculator-form h3 {
            margin-bottom: 10px;
            color: #4CAF50;
        }
        .form-group {
            margin-bottom: 15px;
        }
        .form-group label {
            display: block;
            margin-bottom: 5px;
            font-weight: bold;
        }
        .form-group input {
            width: 100%;
            padding: 8px;
            border: 1px solid #ddd;
            border-radius: 4px;
        }
        .btn-calculate {
            padding: 8px 16px;
            background-color: #4CAF50;
            color: white;
            border: none;
            border-radius: 4px;
            cursor: pointer;
        }
        .btn-calculate:hover {
            background-color: #45a049;
        }
        .result-box {
            margin-top: 10px;
            padding: 10px;
            border-left: 3px solid #4CAF50;
            background-color: #f9f9f9;
            display: none;
        }
        /* Responsive Design */
        @media (max-width: 768px) {
            .tab-container {
                flex-direction: column;
            }
        }
    </style>
</head>
<body>
    <div class="container">

        <!-- Back Button -->
    <a href="experiment.html" class="back-btn">&#8592; Back</a>

        <h1 style="text-align: center; margin-bottom: 30px;">Virtual Chemistry Laboratory</h1>

        <div class="tab-container">
            <div class="tab-button" onclick="openTab('electrochemistry')">Electrochemistry</div>
            <div class="tab-button" onclick="openTab('crystal-structures')">Crystal Structures</div>
            <div class="tab-button" onclick="openTab('quantum-chemistry')">Quantum Chemistry</div>
        </div>

        <!-- Electrochemistry Tab -->
        <div id="electrochemistry-tab" class="tab-content">
            <div class="feature-intro">
```

```
        <h2>Electrochemistry Calculator</h2>
            <p>An <strong>electrochemistry calculator</strong> is a tool used to perform various calculations related to electrochemical reactions, including:</p>
        <ul>
                <li><strong>Nernst Equation Calculations</strong> – Determines the electrode potential at non-standard conditions.</li>
            <li><strong>Cell Potential (E°cell)</strong> – Calculates the standard electrode potential of a galvanic cell.</li>
             <li><strong>Faraday's Law of Electrolysis</strong> – Computes the mass of substances deposited or liberated at an electrode during electrolysis.</li>
            <li><strong>Electrolyte Conductivity & Resistance</strong> – Finds conductivity based on ion concentration.</li>
            <li><strong>Gibbs Free Energy (ΔG) & Equilibrium Constant (K)</strong> – Relates cell potential to spontaneity of reactions.</li>
        </ul>
    </div>

    <!-- Nernst Equation Calculator -->
    <div class="calculator-form">
        <h3>Nernst Equation Calculator</h3>
        <div class="form-group">
            <label>Standard Cell Potential (E°):</label>
            <input type="number" id="standard-potential" step="0.01" placeholder="Enter value in volts">
        </div>
        <div class="form-group">
            <label>Temperature (K):</label>
            <input type="number" id="temperature" value="298.15" placeholder="Default: 298.15 K">
        </div>
        <div class="form-group">
            <label>Number of electrons transferred (n):</label>
            <input type="number" id="electrons" value="2" min="1" placeholder="Default: 2">
        </div>
        <div class="form-group">
            <label>Reaction Quotient (Q):</label>
            <input type="number" id="reaction-quotient" step="0.01" placeholder="Enter value">
        </div>
        <button class="btn-calculate" onclick="calculateNernst()">Calculate Cell Potential</button>
        <div id="nernst-result" class="result-box"></div>
    </div>

    <!-- Cell Potential Calculator -->
    <div class="calculator-form">
        <h3>Cell Potential (E°<sub>cell</sub>) Calculator</h3>
        <div class="form-group">
            <label>Cathode Potential (E°<sub>cathode</sub>):</label>
            <input type="number" id="cathode-potential" step="0.01" placeholder="Enter value in volts">
        </div>
        <div class="form-group">
            <label>Anode Potential (E°<sub>anode</sub>):</label>
            <input type="number" id="anode-potential" step="0.01" placeholder="Enter value in volts">
        </div>
        <button class="btn-calculate" onclick="calculateCellPotential()">Calculate Cell Potential</button>
        <div id="cell-potential-result" class="result-box"></div>
    </div>

    <!-- Faraday's Law Calculator -->
    <div class="calculator-form">
        <h3>Faraday's Law of Electrolysis</h3>
        <div class="form-group">
            <label>Current (I) in Amperes:</label>
            <input type="number" id="current" step="0.01" placeholder="Enter current">
        </div>
        <div class="form-group">
            <label>Time (t) in seconds:</label>
            <input type="number" id="time" placeholder="Enter time">
        </div>
        <div class="form-group">
            <label>Molar Mass (M) in g/mol:</label>
            <input type="number" id="molar-mass" step="0.01" placeholder="Enter molar mass">
        </div>
        <div class="form-group">
            <label>Number of electrons (n):</label>
            <input type="number" id="faraday-electrons" value="1" min="1" placeholder="Default: 1">
```

```
            </div>
            <button class="btn-calculate" onclick="calculateFaraday()">Calculate Mass Deposited</button>
            <div id="faraday-result" class="result-box"></div>
          </div>

          <!-- Electrolyte Conductivity Calculator -->
          <div class="calculator-form">
            <h3>Electrolyte Conductivity</h3>
            <div class="form-group">
              <label>Concentration (C) in mol/L:</label>
              <input type="number" id="concentration" step="0.001" placeholder="Enter concentration">
            </div>
            <div class="form-group">
              <label>Molar Ionic Conductivity (Λ<sub>m</sub>) in S·cm²/mol:</label>
              <input type="number" id="molar-conductivity" step="0.1" placeholder="Enter value">
            </div>
            <button class="btn-calculate" onclick="calculateConductivity()">Calculate Conductivity</button>
            <div id="conductivity-result" class="result-box"></div>
          </div>

          <!-- Gibbs Free Energy Calculator -->
          <div class="calculator-form">
            <h3>Gibbs Free Energy (ΔG) & Equilibrium Constant (K)</h3>
            <div class="form-group">
              <label>Cell Potential (E°) in volts:</label>
              <input type="number" id="gibbs-potential" step="0.01" placeholder="Enter cell potential">
            </div>
            <div class="form-group">
              <label>Number of electrons (n):</label>
              <input type="number" id="gibbs-electrons" value="2" min="1" placeholder="Default: 2">
            </div>
            <div class="form-group">
              <label>Temperature (K):</label>
              <input type="number" id="gibbs-temperature" value="298.15" placeholder="Default: 298.15 K">
            </div>
            <button class="btn-calculate" onclick="calculateGibbs()">Calculate ΔG and K</button>
            <div id="gibbs-result" class="result-box"></div>
          </div>
        </div>

        <!-- Crystal Structures Tab -->
<div id="crystal-structures-tab" class="tab-content">
  <div class="feature-intro">
    <h2>Crystal Structures Calculator</h2>
    <p>A <strong>crystal structures calculator</strong> helps analyze and compute various properties of crystalline materials, including:</p>
    <ul>
      <li><strong>Unit Cell Parameters</strong> – Determines lattice constants (a, b, c) and interaxial angles (α, β, γ).</li>
      <li><strong>Atomic Packing Factor (APF)</strong> – Calculates the packing efficiency of atoms in different crystal structures.</li>
      <li><strong>Bravais Lattice Classification</strong> – Identifies the type of unit cell (Cubic, Tetragonal, Hexagonal, etc.).</li>
      <li><strong>X-ray Diffraction (Bragg's Law)</strong> – Computes interplanar spacing using diffraction angles.</li>
      <li><strong>Density of Crystals</strong> – Finds the theoretical density of a crystal based on atomic mass and volume.</li>
    </ul>
  </div>


  <!-- Crystal Structures Visualization -->
  <div class="feature-visualization">
    <h2 style="color: #4CAF50;">Crystal Structures Visualization</h2>
    <div>
      <label for="crystal-type">Crystal Type:</label>
      <select id="crystal-type">
        <option value="fcc">Face-Centered Cubic (FCC)</option>
        <option value="bcc">Body-Centered Cubic (BCC)</option>
        <option value="hcp">Hexagonal Close-Packed (HCP)</option>
      </select>
    </div>
    <div id="crystal-viewer"></div>
    <div id="structure-details"></div>
```

```
    </div>
</div>



<!-- Quantum Chemistry Tab -->
<div id="quantum-chemistry-tab" class="tab-content">
    <div class="feature-intro">
        <h2>Quantum Chemistry Calculator</h2>
        <p>A <strong>quantum chemistry calculator</strong> helps analyze and compute various quantum mechanical properties of molecules, including:</p>
        <ul>
            <li><strong>Schrödinger Equation Solver</strong> – Computes wavefunctions and energy levels for simple quantum systems.</li>
            <li><strong>Atomic & Molecular Orbitals</strong> – Visualizes electron orbitals for different elements and molecules.</li>
            <li><strong>Electron Density & Probability Distributions</strong> – Calculates and represents electron density maps.</li>
        </ul>
    </div>

    <!-- Schrödinger Equation Solver -->
    <div class="feature-box">
        <h3>🔢 Schrödinger Equation Solver</h3>
        <label>System Type:</label>
        <select id="quantum-system">
            <option value="particle-in-box">Particle in a Box</option>
            <option value="harmonic-oscillator">Harmonic Oscillator</option>
            <option value="hydrogen-atom">Hydrogen Atom</option>
        </select>
        <label>Energy Level (n):</label>
        <input type="number" id="energy-level" min="1" value="1">
        <button onclick="solveSchrodinger()">Calculate Wavefunction</button>
        <div id="schrodinger-result" class="result-box"></div>
    </div>

    <!-- 3D Molecular Orbital Visualization -->
    <div class="feature-box">
        <h3>⚛ 3D Molecular Orbital Visualization</h3>
        <label>Select an orbital:</label>
        <select id="orbital-type">
            <option value="s">s-Orbital</option>
            <option value="p">p-Orbital</option>
            <option value="d">d-Orbital</option>
            <option value="f">f-Orbital</option>
        </select>
        <button onclick="loadOrbital()">View Orbital</button>
        <div id="orbital-visualization"></div>
    </div>

    <!-- Electron Density Maps -->
    <div class="feature-box">
        <h3>📊 Electron Density Maps</h3>
        <p>Visualizing where electrons are most likely to be found in an atom.</p>
        <button onclick="showElectronDensity()">View Electron Density</button>
        <div id="density-map"></div>
    </div>
</div>

<!-- Add this to your main JavaScript section -->
<script>
// Existing functions remain the same
function loadOrbital() {
    let type = document.getElementById("orbital-type").value;
    let orbitalDiv = document.getElementById("orbital-visualization");
    orbitalDiv.innerHTML = `<p>Rendering 3D ${type}-Orbital...</p>`;

    let color = type === "s" ? "#FF5733" :
                type === "p" ? "#3498DB" :
                type === "d" ? "#2ECC71" : "#9B59B6";

    orbitalDiv.innerHTML += `
```

```
        <div style="width:150px; height:150px; background:${color};
            border-radius:50%; margin:20px auto;
            box-shadow:0 0 20px ${color};">
        </div>
        <p>This is a simplified representation of a ${type}-orbital.</p>`;
}

function showElectronDensity() {
    let densityDiv = document.getElementById("density-map");
    densityDiv.innerHTML = `<p>Generating Electron Density Map...</p>`;

    densityDiv.innerHTML += `
        <div style="width:200px; height:200px; background:radial-gradient(circle, rgba(128,0,128,0.8) 0%, rgba(128,0,128,0.1)
70%);
            border-radius:50%; margin:20px auto;
            animation:pulse 1.5s infinite;">
        </div>
        <p>This visualization represents electron probability density.</p>`;
}

function solveSchrodinger() {
    const systemType = document.getElementById("quantum-system").value;
    const n = parseInt(document.getElementById("energy-level").value);
    let result = document.getElementById("schrodinger-result");
    result.style.display = "block";

    let energy, wavefunction;

    switch(systemType) {
        case "particle-in-box":
            energy = `E_${n} = (n²π²ħ²)/(2mL²) = ${(n*n).toFixed(0)} × constant`;
            wavefunction = `ψ_${n}(x) = √(2/L) sin(nπx/L)`;
            break;
        case "harmonic-oscillator":
            energy = `E_${n} = ħω(n + 1/2) = ${(n + 0.5).toFixed(1)} × ħω`;
            wavefunction = `ψ_${n}(x) = (α/π)^(1/4) (1/√(2^n n!)) H_${n}(√α x) e^(-αx²/2)`;
            break;
        case "hydrogen-atom":
            energy = `E_${n} = -13.6 eV / n² = ${(-13.6/(n*n)).toFixed(2)} eV`;
            wavefunction = `ψ_${n,0,0}(r) ∝ e^(-r/na₀)`;
            break;
    }

    result.innerHTML = `
        <h4>Results for ${systemType.replace('-', ' ').toUpperCase()}:</h4>
        <p>Energy Level: ${energy}</p>
        <p>Wavefunction: ${wavefunction}</p>`;
}
</script>

<!-- Styles -->
<style>
.feature-box {
    margin-bottom: 20px;
    padding: 15px;
    border: 2px solid #4CAF50;
    border-radius: 8px;
    background: #f9f9f9;
}

button {
    margin-top: 10px;
    padding: 10px;
    background: #4CAF50;
    color: white;
    border: none;
    border-radius: 5px;
}

button:hover {
    background: #45a049;
```

```
}

.result-box {
    margin-top: 15px;
    padding: 15px;
    background-color: #f0f8ff;
}
</style>

    <script>
        // Tab Management
        function openTab(tabName) {
            // Hide all tab contents
            const tabContents = document.querySelectorAll('.tab-content');
            tabContents.forEach(content => content.classList.remove('active'));

            // Remove active from all tab buttons
            const tabButtons = document.querySelectorAll('.tab-button');
            tabButtons.forEach(button => button.classList.remove('active'));

            // Show selected tab content
            document.getElementById(`${tabName}-tab`).classList.add('active');

            // Highlight selected tab button
            event.currentTarget.classList.add('active');

            // Reinitialize specific tab contents
            if (tabName === 'crystal-structures') initCrystalStructures();
        }

        // Initialize the first tab when page loads
        document.addEventListener('DOMContentLoaded', function() {
            // Set first tab as active by default
            document.querySelector('.tab-button').classList.add('active');
            document.getElementById('electrochemistry-tab').classList.add('active');
        });

        // Electrochemistry Calculations

        // 1. Nernst Equation
        function calculateNernst() {
            const E0 = parseFloat(document.getElementById('standard-potential').value);
            const T = parseFloat(document.getElementById('temperature').value);
            const n = parseFloat(document.getElementById('electrons').value);
            const Q = parseFloat(document.getElementById('reaction-quotient').value);

            if (isNaN(E0) || isNaN(T) || isNaN(Q) || isNaN(n)) {
                document.getElementById('nernst-result').style.display = 'block';
                document.getElementById('nernst-result').innerHTML = `<p>Please enter all values.</p>`;
                return;
            }

            // Nernst Equation: E = E0 - (RT/nF) * ln(Q)
            const R = 8.314; // Gas constant (J/mol·K)
            const F = 96485; // Faraday constant (C/mol)

            const cellPotential = E0 - ((R * T) / (n * F)) * Math.log(Q);

            document.getElementById('nernst-result').style.display = 'block';
            document.getElementById('nernst-result').innerHTML = `
                <p>Calculated Cell Potential: <strong>${cellPotential.toFixed(4)} V</strong></p>
                <p>At ${T} K with reaction quotient Q = ${Q}</p>
            `;
        }

        // 2. Cell Potential
        function calculateCellPotential() {
            const cathodePotential = parseFloat(document.getElementById('cathode-potential').value);
            const anodePotential = parseFloat(document.getElementById('anode-potential').value);

            if (isNaN(cathodePotential) || isNaN(anodePotential)) {
```

```
        document.getElementById('cell-potential-result').style.display = 'block';
        document.getElementById('cell-potential-result').innerHTML = `<p>Please enter both electrode potentials.</p>`;
        return;
    }

    // Cell potential = E(cathode) - E(anode)
    const cellPotential = cathodePotential - anodePotential;
    const spontaneity = cellPotential > 0 ? 'Spontaneous' : 'Non-spontaneous';

    document.getElementById('cell-potential-result').style.display = 'block';
    document.getElementById('cell-potential-result').innerHTML = `
        <p>Standard Cell Potential (E°<sub>cell</sub>): <strong>${cellPotential.toFixed(4)} V</strong></p>
        <p>Reaction is <strong>${spontaneity}</strong> under standard conditions.</p>
    `;
}

// 3. Faraday's Law
function calculateFaraday() {
    const current = parseFloat(document.getElementById('current').value);
    const time = parseFloat(document.getElementById('time').value);
    const molarMass = parseFloat(document.getElementById('molar-mass').value);
    const n = parseFloat(document.getElementById('faraday-electrons').value);

    if (isNaN(current) || isNaN(time) || isNaN(molarMass) || isNaN(n)) {
        document.getElementById('faraday-result').style.display = 'block';
        document.getElementById('faraday-result').innerHTML = `<p>Please enter all values.</p>`;
        return;
    }

    // Faraday's Law: m = (M * I * t) / (n * F)
    const F = 96485; // Faraday constant (C/mol)
    const massDeposited = (molarMass * current * time) / (n * F);

    document.getElementById('faraday-result').style.display = 'block';
    document.getElementById('faraday-result').innerHTML = `
        <p>Mass Deposited: <strong>${massDeposited.toFixed(4)} g</strong></p>
        <p>For a current of ${current} A applied for ${time} seconds</p>
    `;
}

// 4. Electrolyte Conductivity
function calculateConductivity() {
    const concentration = parseFloat(document.getElementById('concentration').value);
    const molarConductivity = parseFloat(document.getElementById('molar-conductivity').value);

    if (isNaN(concentration) || isNaN(molarConductivity)) {
        document.getElementById('conductivity-result').style.display = 'block';
        document.getElementById('conductivity-result').innerHTML = `<p>Please enter all values.</p>`;
        return;
    }

    // Conductivity (κ) = C * Λm
    const conductivity = concentration * molarConductivity;
    const resistivity = 1 / conductivity;

    document.getElementById('conductivity-result').style.display = 'block';
    document.getElementById('conductivity-result').innerHTML = `
        <p>Electrolyte Conductivity (κ): <strong>${conductivity.toFixed(4)} S/cm</strong></p>
        <p>Resistivity (ρ): <strong>${resistivity.toFixed(4)} Ω·cm</strong></p>
    `;
}

// 5. Gibbs Free Energy & Equilibrium Constant
function calculateGibbs() {
    const cellPotential = parseFloat(document.getElementById('gibbs-potential').value);
    const n = parseFloat(document.getElementById('gibbs-electrons').value);
    const T = parseFloat(document.getElementById('gibbs-temperature').value);

    if (isNaN(cellPotential) || isNaN(n) || isNaN(T)) {
        document.getElementById('gibbs-result').style.display = 'block';
        document.getElementById('gibbs-result').innerHTML = `<p>Please enter all values.</p>`;
```

```
        return;
    }

    // Constants
    const F = 96485; // Faraday constant (C/mol)
    const R = 8.314; // Gas constant (J/mol·K)

    // Gibbs Free Energy: ΔG = -nFE
    const deltaG = -n * F * cellPotential;

    // Equilibrium Constant: K = exp(-ΔG/RT)
    const K = Math.exp(-deltaG / (R * T));

    document.getElementById('gibbs-result').style.display = 'block';
    document.getElementById('gibbs-result').innerHTML = `
        <p>Gibbs Free Energy (ΔG): <strong>${deltaG.toFixed(2)} J/mol</strong></p>
        <p>Equilibrium Constant (K): <strong>${K.toExponential(4)}</strong></p>
        <p>Reaction is <strong>${deltaG < 0 ? 'spontaneous' : 'non-spontaneous'}</strong> at ${T} K</p>
    `;
}

// Crystal Structures 3D Visualization
function initCrystalStructures() {
    const viewer = document.getElementById('crystal-viewer');
    const structureDetails = document.getElementById('structure-details');
    const crystalTypeSelect = document.getElementById('crystal-type');

    // Clear previous scene
    viewer.innerHTML = '';

    // Three.js setup
    const scene = new THREE.Scene();
    const camera = new THREE.PerspectiveCamera(75, viewer.offsetWidth / 400, 0.1, 1000);
    const renderer = new THREE.WebGLRenderer();
    renderer.setSize(viewer.offsetWidth, 400);
    viewer.appendChild(renderer.domElement);

    // Crystal structure data
    const structures = {
        'fcc': {
            name: 'Face-Centered Cubic',
            positions: [
                [0,0,0], [0,1,0], [1,0,0], [1,1,0],
                [0,0,1], [0,1,1], [1,0,1], [1,1,1],
                [0.5, 0.5, 0], [0.5, 0, 0.5], [0, 0.5, 0.5],
                [0.5, 0.5, 1], [0.5, 1, 0.5], [1, 0.5, 0.5]
            ],
            details: 'Common in metals like copper, silver, gold'
        },
        'bcc': {
            name: 'Body-Centered Cubic',
            positions: [
                [0,0,0], [1,0,0], [0,1,0], [0,0,1],
                [1,1,0], [1,0,1], [0,1,1], [1,1,1],
                [0.5,0.5,0.5]
            ],
            details: 'Found in metals like iron, chromium, tungsten'
        },
        'hcp': {
            name: 'Hexagonal Close-Packed',
            positions: [
                [0, 0, 0], [2/3, 1/3, 0], [1/3, 2/3, 0],
                [0, 0, 1/2], [2/3, 1/3, 1/2], [1/3, 2/3, 1/2]
            ],
            details: 'Common in titanium, cobalt, and magnesium'
        }
    };

    // Visualization function
    function createCrystalStructure(type) {
        // Clear previous scene
```

```
            while(scene.children.length > 0) {
                scene.remove(scene.children[0]);
            }

            const structure = structures[type];
            const geometry = new THREE.SphereGeometry(0.2, 32, 32);
            const material = new THREE.MeshPhongMaterial({color: 0x4488ff});

            // Add atoms
            structure.positions.forEach(pos => {
                const mesh = new THREE.Mesh(geometry, material);
                mesh.position.set(pos[0], pos[1], pos[2]);
                scene.add(mesh);
            });

            // Add lighting
            const light = new THREE.PointLight(0xffffff, 1);
            light.position.set(5, 5, 5);
            scene.add(light);
            scene.add(new THREE.AmbientLight(0x404040));

            // Position camera
            camera.position.z = 5;

            // Render
            function animate() {
                requestAnimationFrame(animate);
                scene.rotation.y += 0.01;
                renderer.render(scene, camera);
            }
            animate();

            // Update details
            structureDetails.innerHTML = `
                <h3>${structure.name}</h3>
                <p>${structure.details}</p>
            `;
        }

        // Initial structure
        createCrystalStructure('fcc');

        // Change structure on select
        crystalTypeSelect.addEventListener('change', (e) => {
            createCrystalStructure(e.target.value);
        });
    }
    </script>
</body>
</html>
```

Module 11
inorganic.html–

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Inorganic Chemistry Lab</title>
    <style>
        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            font-family: Arial, sans-serif;
        }
```

```
body {
    display: flex;
    flex-direction: column;
    height: 100vh;
    background-color: #ffecaf;
}

/* Back Button */
.back-btn {
    position: absolute;
    top: 30px;
    left: 20px;
    text-decoration: none;
    font-size: 1.2rem;
    font-weight: bold;
    color: white;
    background: #333;
    padding: 10px 20px;
    border-radius: 8px;
    transition: background 0.3s ease;
}

.back-btn:hover {
    background: #555;
}

/* Navbar */
.navbar {
    display: flex;
    justify-content: center; /* Center title */
    align-items: center;
    background-color: #004466;
    color: white;
    padding: 15px 20px;
    position: relative; /* Allows absolute positioning for links */
}

.navbar h1 {
    font-size: 3rem;
    text-shadow: 2px 2px 4px rgba(241, 87, 87, 0.934);
    margin-bottom: 15px;
    text-align: center;
    flex-grow: 1;
}

/* Page Layout */
.container {
    display: flex;
    flex: 1;
}

/* Sidebar */
.sidebar {
    width: 18%;
    background-color: #0077b6;
    color: white;
    padding: 15px;
}
.sidebar h2 {
    font-size: 18px;
    margin-bottom: 10px;
}
.sidebar ul {
    list-style: none;
}
.sidebar ul li {
    padding: 10px;
    border-bottom: 1px solid #fff;
    cursor: pointer;
}
.sidebar ul li:hover {
```

```
            background-color: #48cae4;
        }

        /* Main Content */
        .main-content {
            flex: 1;
            padding: 20px;
            background-color: white;
        }
        .main-content h2 {
            margin-bottom: 10px;
        }
        .interactive-area {
            background-color: #e9ecef;
            height: 70vh;
            border-radius: 10px;
            padding: 15px;
            text-align: center;
            display: flex;
            justify-content: center;
            align-items: center;
            font-size: 18px;
            color: #555;
        }

        /* Right Panel */
        .right-panel {
            width: 25%;
            background-color: #e0fbfc;
            padding: 15px;
        }
        .right-panel h3 {
            margin-bottom: 10px;
        }

        /* Footer */
        .footer {
            background-color: #004466;
            color: white;
            text-align: center;
            padding: 10px;
            font-size: 14px;
        }

    </style>
</head>
<body>

    <!-- Navbar -->
    <div class="navbar">
        <h1>Inorganic Chemistry Virtual Lab</h1>
    </div>

    <!-- Back Button -->
    <a href="experiment.html" class="back-btn">&#8592; Back</a>

    <!-- Container -->
    <div class="container">

        <!-- Sidebar -->
        <div class="sidebar">
            <h2>🗁 Inorganic Chemistry Topics</h2>
            <ul>
                <li>Periodic Table & Trends</li>
                <li>Coordination Chemistry</li>
                <li>Crystal Field Theory</li>
                <li>Acids, Bases & Salts</li>
                <li>Qualitative Analysis</li>
                <li>Metal Extraction & Metallurgy</li>
                <li>Redox Reactions & Electrochemistry</li>
            </ul>
```

```
    </div>

    <!-- Main Content -->
    <div class="main-content">
      <h2 style="color: #08550a;">🧪 Interactive Inorganic Chemistry Lab</h2>
      <div class="interactive-area">
        Select a feature from the sidebar to begin exploring!
      </div>
    </div>

    <!-- Right Panel -->
    <div class="right-panel">
      <h3>i Interactive Panel</h3>
      <p>⚛️ Periodic Trends Visualizer.</p>
      <p>🔬 Stepwise Coordination Complex Formation.</p>
      <p> ⚡ Electrochemical Cell Simulations.</p>
      <p>📝 MCQs & Quiz Section.</p>
    </div>

  </div>
</body>
</html>
```